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LINKU Stereo Vision System Report

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Abstract

This report presents the conceptual design, optical characterization, experimental validation, and maturation roadmap of the LINKU stereo vision system, developed to support visual monitoring and three-dimensional reconstruction of a tethered satellite configuration in Low Earth Orbit. The system is intended to complement conventional telemetry by providing direct geometric information about tether deployment, spatial configuration, curvature, and post-deployment dynamic behavior.

Orbital tether observation presents a non-conventional computer vision problem. The tether is expected to appear as a thin, low-texture structure observed against a predominantly dark background under highly directional solar illumination. At mission-relevant distances of up to approximately 100 m, a millimeter-scale tether progressively transitions from a geometrically resolvable object to a sub-pixel radiometric feature. Consequently, the reconstruction problem cannot be addressed reliably using standard texture-based photogrammetry alone.

To evaluate this problem, the study combines optical observability analysis, controlled laboratory experimentation, and comparative assessment of three reconstruction approaches: conventional photogrammetry, geometry-driven stereo reconstruction, and exploratory AI-assisted processing. The analysis quantified the geometric and radiometric limits of representative COTS camera configurations, while the experimental campaign reproduced key orbital-like visual constraints, including low environmental texture, dark backgrounds, directional illumination, and thin tether-like targets.

The results show that conventional photogrammetry is unsuitable for isolated tether reconstruction under low-texture conditions due to its dependence on stable visual features and background texture. In contrast, the proposed geometry-driven framework successfully reconstructed planar and volumetric tether configurations using centerline extraction, geometric continuity, stereo correspondence, and constrained triangulation. AI-based methods were found to be more appropriate as supporting tools for segmentation, topology interpretation, and ambiguity resolution rather than as primary scientific reconstruction engines.

Overall, this work demonstrates the feasibility of optical tether reconstruction under representative orbital-like conditions and establishes a physically interpretable reconstruction strategy for future tether monitoring missions. The report also defines the main maturation steps required to evolve the current laboratory prototype toward a flight-capable visual observation system.

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Introduction

The LINKU mission involves the operation of a tethered multi-satellite configuration in Low Earth Orbit, where visual information can provide valuable support for deployment verification, anomaly detection, and post-deployment dynamic characterization. A central element of this capability is a dedicated stereo vision system designed to observe the tether during deployment and to support geometric reconstruction after the tethered configuration stabilizes.

Unlike conventional spacecraft telemetry, which provides indirect or localized measurements, a stereo camera system can provide direct visual information about the physical shape and spatial evolution of the tether. This capability is relevant because the tether may experience curvature, oscillations, partial out-of-plane motion, rebounds, slack conditions, or entanglement during different mission stages. Visual reconstruction can therefore complement GNSS, inertial, and tether-tension measurements by adding spatial context to the interpretation of tether dynamics.

However, observing a tether in orbit presents significant optical and computational challenges. Conventional terrestrial photogrammetry assumes textured surfaces, diffuse illumination, and visually rich environments that support feature extraction and image matching. These assumptions are not valid for the LINKU observation scenario, where the tether is thin, nearly textureless, and expected to be observed against a low-radiance background under highly directional illumination. As a result, the problem must be treated as a geometric and radiometric detection problem rather than as a standard texture-based reconstruction task.

This report addresses that problem through a progressive validation strategy. First, the operational role of the stereo vision system is defined in relation to the LINKU mission. Then, the optical and radiometric limits of tether observation are analyzed to determine the conditions under which the tether can be geometrically resolved or detected as a sub-pixel radiometric feature. Based on these constraints, the report presents an experimental and processing framework that compares conventional photogrammetry, a geometry-driven stereo reconstruction approach, and exploratory AI-assisted methods.

The document is organized as follows:

- Chapter 1: Defines the operational concept of the stereo vision system and its role within the LINKU mission.
- Chapter 2: Reviews previous tethered missions and the state of the art in reconstruction methods for slender structures.
- Chapter 3: Develops the theoretical and optical analysis required to evaluate tether observability.
- Chapter 4: Presents the experimental and processing methodology.
- Chapter 5: Describes the hardware configuration and laboratory validation environment.
- Chapter 6: Presents the experimental results and comparative discussion of the proposed reconstruction strategies.
- Chapter 7: Outlines the future work and system maturation roadmap.
- Chapter 8: Summarizes the principal conclusions and contributions of the study.

Objectives of the Stereo Vision System and validation scope

The stereo vision system is intended to support the LINKU mission by providing direct visual and geometric information about the tethered satellite configuration. Its primary mission-level objectives are:

- To verify the nominal deployment of the tether and identify possible anomalies during the separation process between satellites.
- To characterize the three-dimensional spatial geometry of the tether during post-deployment operations.
- To support the analysis of tether oscillations, curvature, and deformation through stereo geometric reconstruction.
- To provide visual geometric information complementary to GNSS, inertial, and tether-tension measurements.

In addition to these mission-level objectives, this report focuses on the experimental validation of the optical observation and reconstruction strategy. Specifically, it evaluates whether a thin tether-like structure can be detected and reconstructed under representative orbital-like visual conditions, including low background texture, directional illumination, high radiometric contrast, and sub-pixel target visibility.

Therefore, **the scope of this document is not to qualify a final flight camera system, but to establish the feasibility, limitations, and maturation path of the proposed stereo vision architecture**. The validation emphasizes three aspects: optical observability, comparison of reconstruction approaches, and experimental demonstration of a geometry-driven stereo reconstruction framework.

Chapter 1

Stereo vision system operational concept

This chapter defines the operational concept of the LINKU stereo vision system and its role within the mission architecture. The system is intended to provide direct visual information about the deployment state, spatial geometry, and dynamic behavior of the tethered satellite configuration throughout the mission.

Unlike conventional telemetry sensors, which provide localized or indirect measurements, the stereo vision system is designed to extract geometric information about the tether's physical shape and spatial evolution. This capability is particularly relevant during deployment and post-deployment operations, where the tether may experience oscillations, curvature, transient instabilities, or partial out-of-plane motion.

The stereo vision system is therefore conceived as a complementary observation instrument that supports both operational monitoring and post-processing geometric analysis. Its functionality is closely linked with other onboard measurements, including GNSS data, inertial sensing, and tether-tension measurements, which together provide a more complete characterization of the tethered system dynamics.

1.1 Mission operational stages and observational objectives

The stereo vision system operates during two principal operational stages of the LINKU mission: the tether deployment stage and the post-deployment characterization stage.

First operational stage: Tether deployment

The deployment stage represents the most critical operational period for visual observation. During this stage, SAT-B and SAT-C transition from an initially coupled configuration, as shown in Figure 1.1, to a totally separated tethered configuration, as illustrated in Figure 1.2. The deployment begins when SAT-B ejects SAT-C, initiating the progressive extension of the tether between both bodies. The stereo vision system continuously monitors this separation process to observe deployment dynamics and identify potential anomalies. The tether has a maximum designed length of approximately 100 m, which defines the upper limit of the expected separation range during nominal deployment.

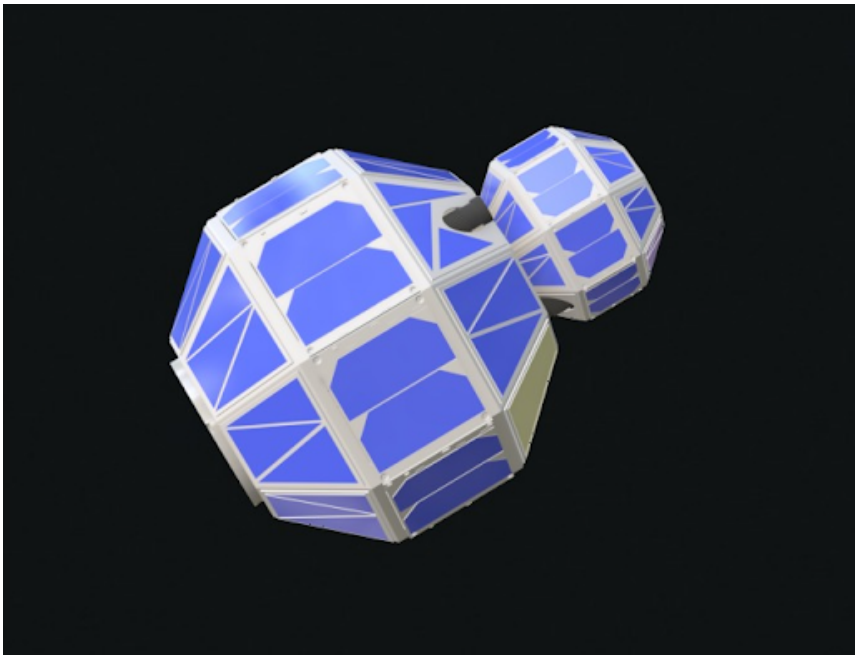


Figure 1.1: Conceptual visualization of the first operational stage of the LINKU tether mission sequence between SAT-B and SAT-C. The figure depicts the initial pre-deployment configuration, in which SAT-C remains mechanically coupled to SAT-B prior to release. The subsequent mass separation initiates tether deployment, resulting in the progressive extension of the tether between the two masses. This stage represents the critical operational period for verifying nominal tether deployment and identifying potential anomalies through visual monitoring.

The primary role of the stereo vision system during deployment is to provide visual confirmation of the nominal extension sequence and support the identification of possible anomalies throughout the separation process. These anomalies may include:

- Tether entanglement
- Asymmetric deployment
- Rebounds
- Unexpected oscillatory behavior
- Partial deployment
- Sudden changes in tether geometry

At this stage, maintaining continuous target centering within the image plane is inherently challenging due to spacecraft attitude variations immediately following separation. Because the cameras are rigidly mounted to the spacecraft structure, the apparent position of the tether and companion satellite may drift within the field of view during transient rotational motion.

Additionally, maintaining fully stabilized attitude control throughout the deployment sequence may not be feasible due to constraints on spacecraft power, control authority, and volume. Consequently, the optical system must tolerate moderate angular deviations and transient motion while preserving sufficient situational awareness of the deployment region.

For this reason, the optical configuration prioritizes wide-field observation capability during deployment operations, increasing the probability of maintaining partial or complete visibility of the tether and companion satellite despite residual spacecraft motion and pointing uncertainties.

Second operational stage: Post-deployment characterization

After deployment stabilization, the tethered system transitions to a characterization stage focused on observing and analyzing tether dynamics under orbital conditions.

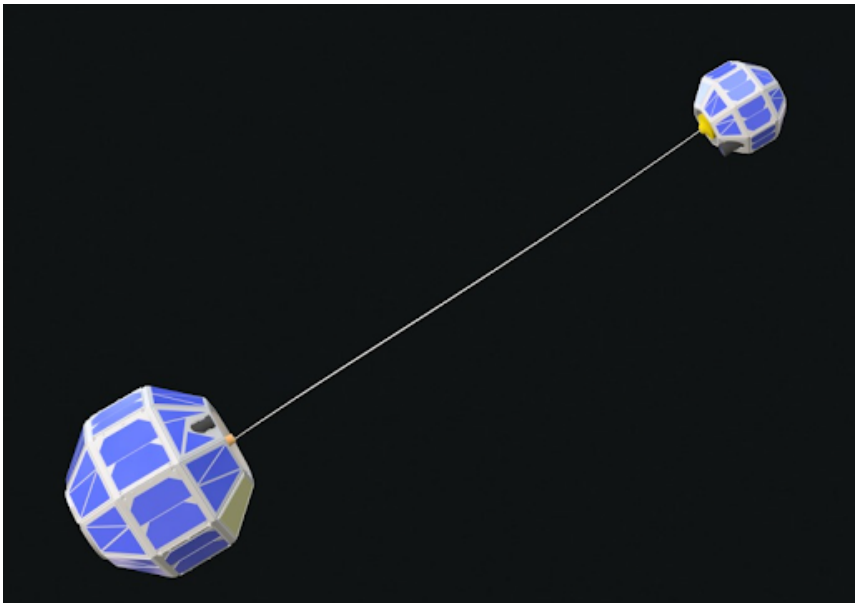


Figure 1.2: Conceptual visualization of the second operational stage of the LINKU tether mission sequence between SAT-B and SAT-C. The figure illustrates the post-deployment configuration, where the tethered system reaches its nominal geometry and tends to align with the local radial (Nadir–Zenith) direction under gravity-gradient effects. This stage enables the characterization of tether geometry, dynamic behavior, and structural stability through visual observation and reconstruction techniques.

During this operational stage, the tethered system is expected to progressively evolve toward a gravity-gradient stabilized configuration, as shown in Figure 1.2. Under this regime, the tether axis and the connected satellite masses tend to align approximately along the local radial direction of the orbit, with the system oscillating around the Nadir-Zenith axis that points toward and away from the Earth's center. However, residual oscillations, perturbations, and out-of-plane motion may still occur due to orbital disturbances and deployment transients.

The observational priorities during this stage include:

- Characterization of tether curvature and spatial orientation
- Analysis of oscillatory motion
- Estimation of deformation behavior
- Generation of geometric datasets for dynamic model validation

Unlike the deployment stage, this operational regime prioritizes geometric reconstruction accuracy over wide-angle situational awareness. Consequently, image processing focuses on extracting the tether geometry from stereo imagery under highly constrained visual conditions.

In addition to tether observation, the stereo vision system architecture also enables exploratory evaluation of secondary proximity-awareness capabilities. In particular, the mission considers the possibility of reconstructing the relative geometry of the companion SAT-C mass from SAT-B stereo observations, providing preliminary information that may support future rendezvous or proximity-operation concepts without interfering with the primary tether-observation objectives.

1.2 Instrument justification and data contextualization

The stereo vision system is integrated into the LINKU mission to provide direct geometric observation of the tethered configuration and to contextualize measurements obtained from other onboard instruments. Conventional telemetry measurements provide only partial information regarding tether behavior. For example:

- GNSS measurements provide relative separation vectors between spacecraft but do not describe the physical shape of the tether between them.
- Tether-tension measurements provide scalar force information at attachment points but cannot directly determine whether the tether remains straight, curved, oscillating, or partially entangled.
- Inertial measurements provide spacecraft attitude and rotational dynamics but do not directly capture the spatial geometry of the tether itself.

The stereo vision system complements these measurements by providing visual geometric information regarding the actual tether configuration throughout the mission.

The resulting data has two principal operational functions:

Qualitative operational awareness

The visual observations provide direct situational awareness during deployment and post-deployment operations, enabling mission operators to identify anomalies or unexpected tether behavior through image inspection and geometric interpretation.

Quantitative geometric reconstruction

Stereo image processing enables the extraction of geometric information from the observed tether, including:

- Three-dimensional spatial coordinates
- Curvature estimation
- Oscillation characterization
- Relative geometric evolution over time

These datasets are intended to support the validation of tether dynamic models and to complement the interpretation of GNSS, inertial, and tether-tension measurements obtained during flight operations.

Chapter 2

Background and State of the Art

This chapter reviews previous tethered satellite missions and the current state of the art in optical reconstruction methods for slender structures. The objective is to establish the technological and methodological context that motivates the development of the LINKU stereo vision system.

The chapter first analyzes the observational limitations encountered in previous tethered missions, particularly regarding the lack of direct geometric visualization of the deployed tether. Subsequently, the chapter reviews the theoretical foundations of conventional photogrammetry, the limitations of texture-based reconstruction under orbital conditions, and alternative approaches based on geometric extraction and emerging AI-assisted methods.

Through this analysis, the chapter identifies the principal technological gap addressed by the LINKU project, the absence of a dedicated stereo reconstruction approach specifically adapted to thin, textureless tether structures observed under high-contrast orbital illumination.

2.1 Lessons from previous tethered missions

Several tethered satellite missions have demonstrated the scientific and technological relevance of tether systems in orbit, including applications related to electrodynamic propulsion, gravity-gradient stabilization, orbital dynamics characterization, and formation control. However, despite these advances, direct visual reconstruction of deployed tether geometry has remained a major unresolved challenge.

Historically, most tethered missions relied primarily on indirect telemetry measurements or limited visual inspection capabilities rather than on dedicated geometric reconstruction systems. Consequently, the actual spatial configuration of the tether during deployment and post-deployment operations could not always be directly characterized, particularly under dynamic or anomalous conditions.

In the context of nano- and micro-satellites, the STARS-II mission developed by Kagawa University demonstrated the deployment concept of an electrodynamic tether between a Mother Satellite (MS) and a Daughter Satellite (DS), as illustrated in Figure 2.1. [1]. This mission validated the feasibility of tether deployment within a nanosatellite architecture and demonstrated the operational relevance of tethered systems for future orbital applications.

However, despite the relevance of this architecture, STARS-II did not incorporate a dedicated visual reconstruction capability to directly estimate the three-dimensional geometry of the tether during flight operations. Due to operational limitations and the absence of dedicated optical instrumentation, deployment analysis relied primarily on indirect observations derived from orbital behavior and telemetry interpretation.

While these measurements provided valuable mission-level information, they did not allow di-

rect observation of tether curvature, oscillatory behavior, localized deformation, or deployment anomalies throughout the extension process.

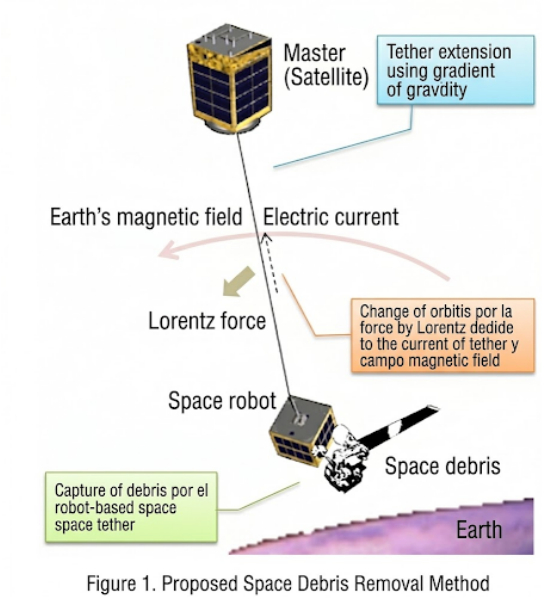


Figure 2.1: Conceptual configuration of the STARS-II mission, illustrating the Mother Satellite (MS) and Daughter Satellite (DS) connected by an electrodynamic tether (EDT) [1].

Similar observational limitations were present in larger tether missions such as NASA's TSS-1 and TSS-1R programs. Although these missions incorporated onboard imaging systems for operational monitoring, they did not include a dedicated stereo-vision architecture designed for geometric tether reconstruction.

When the tether failed during the TSS-1R mission, the available visual data were insufficient to directly characterize the full structural dynamics of the rupture event. As a result, post-flight analysis relied heavily on telemetry interpretation and indirect estimation methods derived from dynamic models and flight data.

These precedents highlight a common technological limitation in previous tethered missions: the lack of dedicated visual systems capable of continuously reconstructing the tether's three-dimensional geometry under orbital conditions. This limitation becomes particularly critical when characterizing transient oscillations, local deformations, deployment anomalies, or out-of-plane motion.

These observational limitations motivate the need to evaluate reconstruction methods specifically adapted to the optical and geometric conditions expected during orbital tether observation.

2.2 Theoretical basis and reconstruction of slender structures

The optical reconstruction of thin structures presents a fundamentally different problem compared to the conventional three-dimensional reconstruction of volumetric objects. Standard photogrammetry techniques were originally developed for textured surfaces observed under diffuse illumination, where stable visual features can be extracted and matched across multiple images.

However, tether reconstruction in orbit introduces a combination of constraints that significantly complicates the use of traditional feature-based methods:

- Extremely thin geometry
- Limited or nonexistent surface texture

- Strong directional illumination.
- High radiometric contrast
- Predominantly featureless backgrounds

Under these conditions, the tether may occupy only a fraction of a pixel at long observation distances, while simultaneously lacking distinguishable surface patterns required for conventional feature matching.

For this reason, the reconstruction problem addressed in the LINKU mission must be treated primarily as a geometric and radiometric detection problem rather than as a conventional texture-matching problem.

2.2.1 Standard photogrammetry and its limitations

Traditional photogrammetry is based on the extraction and correspondence of visual features between multiple images. Methods such as Structure-from-Motion (SfM) and Semi-Global Matching (SGM) estimate three-dimensional geometry by matching textured image regions observed from different viewpoints.

These methods operate effectively when the observed scene contains:

- Stable texture patterns
- Diffuse illumination
- Sufficient spatial contrast
- Volumetric structures occupying multiple pixels

Under terrestrial conditions, surrounding environmental texture and ambient illumination help stabilize the feature-matching process. However, the orbital environment expected for tether observation differs substantially from these assumptions.

In the LINKU observation scenario, the tether is expected to appear as a thin, nearly textureless structure against a low-radiance background. Under these conditions, conventional feature descriptors cannot reliably identify stable correspondences between stereo images.

Additionally, highly directional solar illumination may produce localized specular reflections and saturation effects that violate the brightness-consistency assumptions typically required by standard photogrammetry algorithms [2, 3].

As observation distance increases, the tether projection may also enter the sub-pixel regime, where the apparent width of the structure becomes smaller than a single image pixel. In this regime, traditional patch-based stereo correspondence methods become fundamentally unreliable because the target no longer occupies a sufficiently large image region for stable feature extraction.

These limitations become particularly critical under representative high-contrast orbital-like imaging conditions, where conventional photogrammetry struggles to maintain reliable feature correspondence for thin tether structures.

Figure 2.2 reflects the fundamental limitation identified by Kniaz et al. [2], conventional image-based reconstruction methods rely heavily on stable visual features and texture-rich observations. As the observed structure becomes thinner and less textured, feature correspondence becomes increasingly unreliable, motivating the development of alternative reconstruction approaches based on geometric continuity, radiometric analysis, and line-based representations rather than traditional texture-based feature matching.

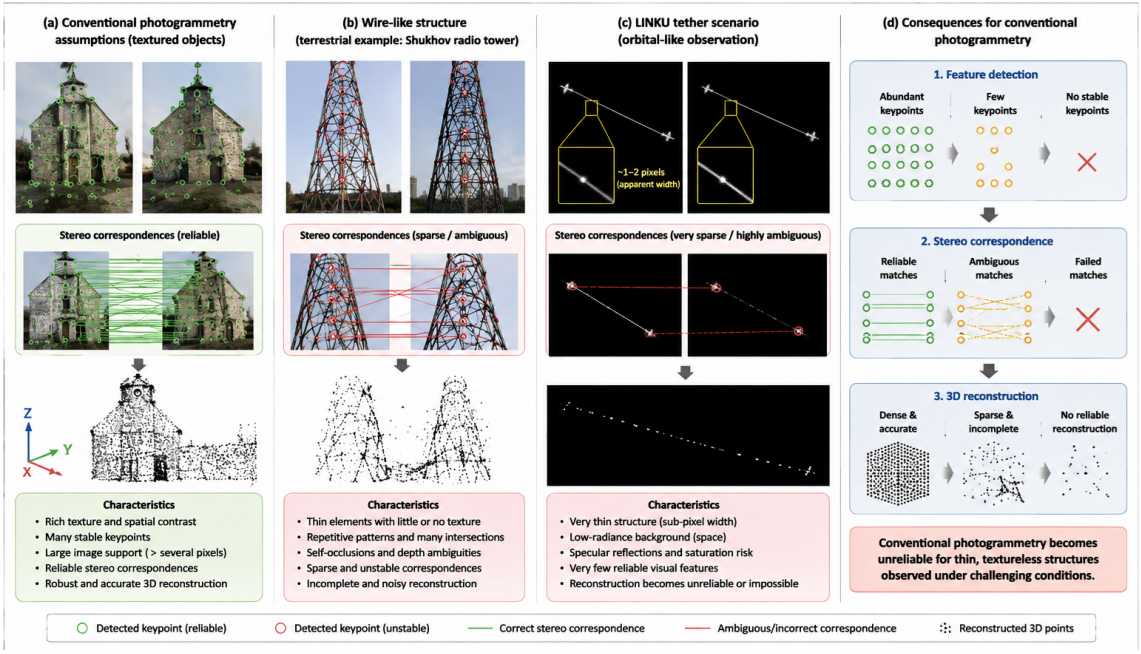


Figure 2.2: Conceptual illustration of the challenges associated with conventional photogrammetry when applied to thin wire-like structures and orbital tether observation, inspired by the reconstruction limitations discussed by Kniaz et al. [2]. (a) presents a conventional photogrammetry scenario, where textured surfaces provide a large number of stable visual features that can be reliably detected and matched between images. Under these conditions, Structure-from-Motion (SfM) and stereo reconstruction algorithms can establish dense correspondences and generate accurate three-dimensional reconstructions. (b) illustrates a wire-like structure, where the available visual information is significantly reduced due to the thin geometry and repetitive patterns. As a result, the number of distinctive features decreases and correspondence ambiguity increases. (c) represents the orbital tether observation scenario considered in LINKU, where the tether appears as a thin, low-texture structure observed against a dark background under directional illumination. In this regime, the tether may occupy only a few pixels or even operate under sub-pixel conditions, further reducing the availability of reliable feature descriptors. (d) summarizes the consequences for conventional photogrammetry, showing the progressive degradation of feature detection, stereo correspondence, and three-dimensional reconstruction performance as the scene transitions from textured objects to thin wire-like and tether-like structures.

2.2.2 Edge-based extraction: Alternative for textureless geometry

Because texture-based correspondence becomes unreliable for thin structures, alternative reconstruction strategies must rely primarily on geometric contrast rather than on surface texture.

In terrestrial industrial inspection applications, edge-based and centerline-based extraction methods are commonly used to detect slender structures such as metallic wires, filaments, or bonding traces [3]. Instead of identifying textured image patches, these methods estimate the geometric location of the structure boundary or centerline relative to the surrounding background.

This approach is particularly relevant for the LINKU mission because the tether is expected to appear primarily as a radiometric line-like feature rather than as a textured object.

Sub-pixel line extraction methods, such as those derived from Steger's framework [4], provide a mathematical basis for estimating the centerline position of thin structures, even when the apparent width approaches the pixel-resolution limit. These methods exploit intensity gradients and scale-space analysis to estimate geometric location with sub-pixel precision.

For the LINKU stereo vision system, these principles provide the theoretical foundation for a reconstruction approach centered on:

- Tether segmentation
- Centerline estimation
- Stereo geometric correspondence
- Curve-based triangulation

By focusing on geometric continuity and radiometric contrast rather than texture matching, the proposed approach is more compatible with the expected optical conditions for orbital tether observation.

As shown in Figure 2.3, the presented workflow follows the principles established by Steger [4], in which line-like structures are not characterized by texture-based features but by their radiometric profiles, gradient behavior, and sub-pixel localization properties. This interpretation is particularly relevant for orbital tether observation, where the target may occupy only a few pixels or even operate within the sub-pixel regime, making conventional feature-based photogrammetry unreliable.

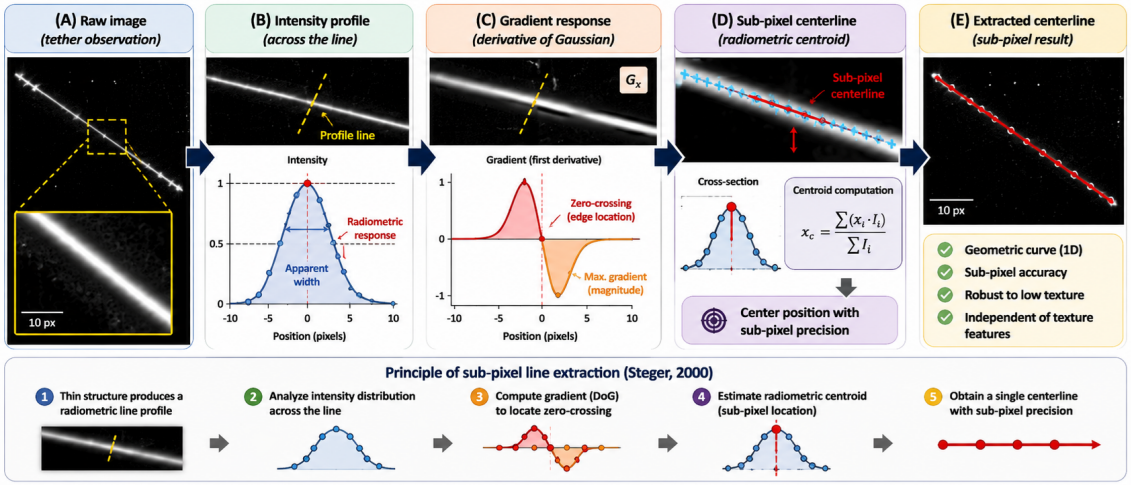


Figure 2.3: Conceptual illustration of the sub-pixel line extraction process for thin structures based on the line-profile interpretation framework proposed by Steger [4]. (A) shows the raw image of a thin tether-like structure, which appears as a finite-width radiometric response due to the combined effects of the optical system, sensor sampling, and image blur. (B) illustrates the intensity profile extracted across the line, where the structure is represented by a localized radiometric distribution rather than a perfectly resolved geometric object. (C) presents the corresponding gradient response, obtained via derivative-based filtering, which reveals the intensity transitions associated with the line boundaries and provides information on line localization. (D) shows the estimation of the line center using radiometric centroiding and sub-pixel localization, enabling inference of the centerline position with precision greater than the sensor pixel spacing. Finally, (E) presents the extracted centerline representation, where the thin structure is reduced to a single geometric trajectory suitable for subsequent curve tracking, stereo correspondence, and three-dimensional reconstruction.

2.2.3 The limitations of emerging AI methods for space

Recent advances in Artificial Intelligence (AI), including convolutional neural networks (CNNs), neural rendering methods, and Neural Radiance Fields (NeRFs), have demonstrated promising capabilities for image segmentation and three-dimensional reconstruction in terrestrial environments [2, 5, 6].

These approaches can outperform traditional reconstruction methods when large representative datasets and stable imaging conditions are available. However, their direct application to orbital tether observation remains highly challenging.

Most currently available datasets used for AI-based reconstruction contain textured objects, ambient illumination, structured environments, and dense visual context. In contrast, the LINKU observation scenario involves highly directional illumination, extreme background contrast, sparse visual reference, and thin sub-pixel structures.

This mismatch between terrestrial training datasets and the expected orbital environment limits the direct applicability of current AI-based approaches.

Additionally, many neural reconstruction methods require substantial computational resources that are not compatible with lightweight onboard processing constraints typically associated with nanosatellite platforms.

For these reasons, AI-assisted reconstruction is treated within the LINKU project as an exploratory future-oriented capability intended to complement, rather than replace, the primary geometry-driven reconstruction approach.

Figure 2.4 illustrates the contrast between contemporary AI-assisted reconstruction scenarios and the orbital tether observation conditions considered in LINKU. This comparison highlights the domain gap that currently limits the direct application of AI-based reconstruction methods to the proposed mission.

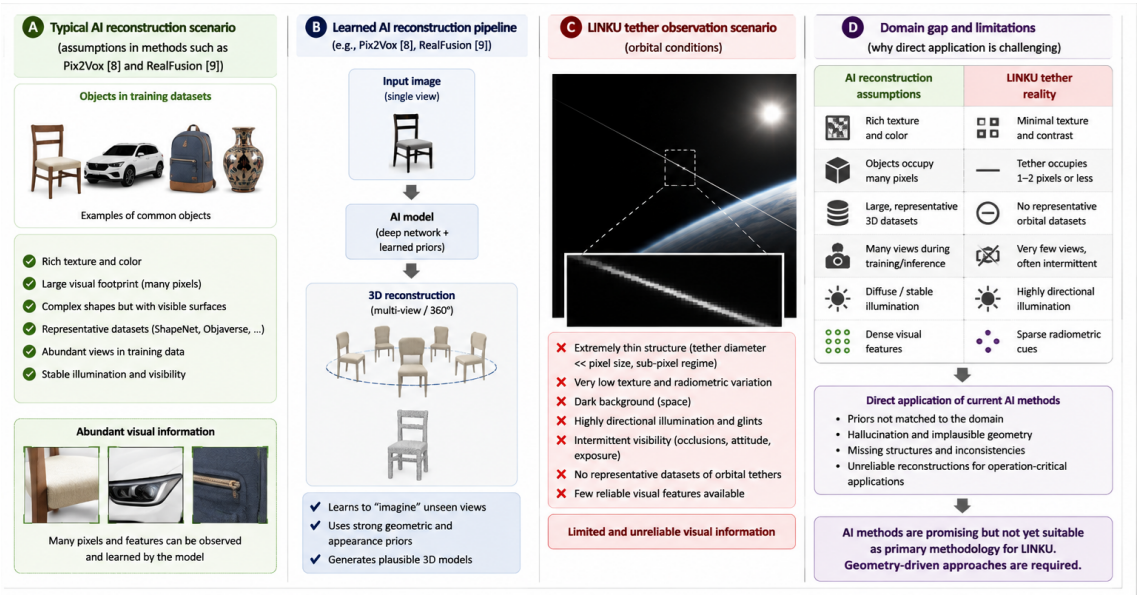


Figure 2.4: Conceptual comparison between the assumptions of modern AI-assisted reconstruction methods and the visual conditions expected for orbital tether observation. (A) illustrates the rich visual information typically available in AI reconstruction datasets, including textured objects, abundant image features, and representative training samples. (B) summarizes the reconstruction process employed by methods such as Pix2Vox [5] and RealFusion [6], where learned geometric priors are used to infer complete three-dimensional object representations from one or multiple images. (C) presents the LINKU observation scenario, characterized by an extremely thin tether observed under sub-pixel conditions, directional illumination, low texture, and limited visual information. (D) highlights the resulting domain gap, showing that the assumptions underlying current AI reconstruction methods differ significantly from the orbital tether observation problem. Consequently, LINKU adopts a deterministic geometry-driven reconstruction framework as its primary operational methodology, while AI-assisted approaches are considered future augmentation mechanisms.

2.3 Motivation for the LINKU approach

The analysis presented throughout this chapter reveals a consistent technological gap in the observation of tethered systems in orbit.

Previous missions demonstrated the operational feasibility and scientific relevance of tethered satellite architectures, but they lacked dedicated systems capable of directly reconstructing tether geometry during flight operations. At the same time, conventional photogrammetry techniques remain strongly dependent on texture-based correspondence assumptions that are incompatible with the expected optical conditions of orbital tether observation.

Although emerging AI-based methods offer promising future capabilities, their dependence on representative datasets and high computational requirements currently limits their direct applicability to this problem.

In response to these limitations, the LINKU project proposes a specialized stereo reconstruction strategy focused specifically on the geometric characteristics of thin tether structures under high-contrast orbital-like illumination conditions.

Rather than relying primarily on textured feature matching, the proposed approach prioritizes:

- Geometric continuity
- Radiometric contrast
- Centerline estimation
- Stereo geometric constraints

This methodology forms the foundation of the LINKU reconstruction pipeline and motivates the experimental validation strategy presented in the following chapters.

Chapter 3

Theoretical and optical analyses

This chapter establishes the theoretical and optical basis used to define the main design constraints of the LINKU stereo vision system. After identifying the need for a dedicated geometric reconstruction approach, the next step is to determine whether the tether can be visually observed and reconstructed under the expected orbital conditions.

The analysis focuses on three principal design drivers, the dynamic behavior of the tethered formation, the optical field of view required to maintain visual coverage, and the geometric resolution needed to detect a thin tether over the expected separation range. These constraints define the trade-off between situational awareness during deployment and geometric accuracy during post-deployment characterization.

3.1 Dynamic constraints and field of view estimation

Understanding the expected motion of the tethered formation is a prerequisite for defining the camera field of view and preserving line-of-sight during the mission. The stereo vision system must be able to observe the tether during the deployment process, when the configuration may exhibit transient attitude motion, and during the post-deployment regime, when the tethered system is expected to evolve toward gravity-gradient stabilization.

In this context, gravity-gradient stabilization refers to the tendency of the tether axis and the connected satellite masses to align approximately along the local orbital radial direction. In the Local Vertical Local Horizontal reference frame, this corresponds to the Nadir–Zenith axis, with Nadir pointing toward the Earth's center and Zenith pointing away from it. Therefore, the tethered configuration is expected to oscillate around this radial equilibrium direction rather than remain perfectly static.

The physical origin of this behavior is the differential gravitational acceleration experienced along the length of the tethered system. The gravitational acceleration at a distance r from the Earth's center can be expressed as:

$$g(r) = \frac{\mu}{r^2} \tag{3.1}$$

where $\mu \approx 3.986 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$ is Earth's standard gravitational parameter.

Because the two tethered masses are located at slightly different radial distances from the Earth's center, they experience slightly different gravitational accelerations. This differential effect produces a restoring tendency that aligns the tethered system with the local radial direction and generates librational motion around the Nadir-Zenith axis.

For the purpose of field-of-view estimation, the optical design must account for both deployment transients and the expected post-deployment oscillatory behavior. Dynamic trends identified in

Deliverable D4 indicate that the tethered system response is sensitive to initial attitude deviations and to the relationship between body-angular and orbital-frame angular velocities. These simulations also suggest that out-of-plane deviations may induce cone-like motion around the Nadir direction, requiring additional angular margin in the optical design. A representative stable case considered in D4 includes initial pitch and roll deviations of approximately 10°.

Since the final long-term stabilization envelope is still under definition, the field of view must be sized conservatively. The optical design should therefore accommodate expected librational motion, possible out-of-plane deviations, deployment transients, and residual pointing uncertainties associated with body-fixed cameras.

3.1.1 Design trade-off: Field of View vs. Resolution

The definition of the camera field of view involves an inherent trade-off between visual coverage and tether geometric reconstruction.

On one hand, the system requires a sufficiently wide field of view to maintain visibility of the tether and companion satellite during deployment and early post-deployment motion. Since the cameras are fixed to the satellite body, relative attitude variations, pendular motion, and cone-like oscillations can shift the tether's projection across the image plane. A wider field of view increases the probability of preserving visual contact with the tethered configuration under transient motion and residual ADCS pointing errors.

On the other hand, the tether is expected to have a millimeter-scale diameter. Increasing the field of view reduces the number of pixels sampling the tether, which directly affects the ability to detect its edges, estimate its centerline, and reconstruct its geometry. For this reason, a narrower field of view is preferable for geometric reconstruction, because it increases the projected pixel occupancy of the tether and improves spatial measurement capability.

These two requirements are antagonistic:

- **Situational awareness requires a wide field of view**. This is particularly important during deployment and early post-deployment operations, when the tethered configuration may move across the image plane due to transient attitude dynamics.
- **Geometric reconstruction requires higher angular resolution**. This favors a narrower field of view, allowing the tether projection to occupy more pixels and improving the reliability of centerline extraction and stereo correspondence.

Based on the dynamic trends identified in D4, a wide-angle configuration with a diagonal field of view greater than approximately 60ř is considered a conservative choice for maintaining visual coverage under uncertain attitude and librational conditions. Conversely, a narrower-angle configuration in the range of approximately 38ř to 44ř diagonal is preferable for supporting more accurate tether geometric reconstruction.

A single optical configuration cannot fully optimize both requirements simultaneously. A narrow lens improves tether sampling but increases the risk of losing the tether or companion satellite during transient motion. A wide lens improves coverage but reduces geometric resolution. Therefore, the LINKU concept considers a hybrid optical approach based on complementary camera configurations: one optimized for wide-field situational awareness and another optimized for higher-resolution geometric reconstruction.

3.2 Comparative optical analysis for COTS sensors

To establish the geometric and radiometric detection limits of the LINKU stereo vision system, two representative Commercial Off-The-Shelf (COTS) camera configurations were analyzed. This analysis evaluates the relationship among field of view, pixel density, depth of field, and tether detectability under expected operational conditions.

The optical analysis follows three principal steps:

1. Estimation of the Depth of Field (DoF) and hyperfocal distance required to maintain tether visibility over the expected operational separation range of up to approximately 100 m.
2. Determination of the geometric resolution limit, defined as the maximum distance at which the tether subtends at least one pixel.
3. Evaluation of the radiometric detectability of the tether in the sub-pixel regime.

A custom analytical tool was developed in Python to simultaneously calculate and visualize these quantities. The resulting analysis establishes the operational optical limits that drive the stereo vision system's reconstruction strategy.

3.2.1 Reference camera configurations and Depth of Field

The Circle of Confusion (CoC) is defined by the one-pixel criterion, which corresponds to the minimum resolvable blur diameter on the sensor. The CoC is therefore approximated as:

$$c = \frac{\text{sensor width}}{\text{horizontal resolution}} \tag{3.2}$$

Using the thin-lens relation, the image distance ν for an object located at distance μ is:

$$\nu = \frac{f \cdot \mu}{\mu - f} \tag{3.3}$$

where f is the focal length.

The Depth of Field is estimated through the hyperfocal distance H, defined as:

$$H = \frac{f^2}{N \cdot c} + f \tag{3.4}$$

where:

- f is the focal length
- N is the aperture F-number
- c is the Circle of Confusion

Given a focus distance s, the near and far limits of the Depth of Field are:

$$D_n = \frac{H \cdot s}{H + s} \tag{3.5a}$$

$$D_f = \frac{H \cdot s}{H - s} \tag{3.5b}$$

valid for $s < H$. When $s \geq H$, the far limit extends toward infinity.

The analysis compares two representative 12 MP camera configurations differing primarily in focal length, optical format, and pixel pitch. The two reference optical configurations analyzed are summarized below.

The reference camera configurations were selected according to four practical and optical criteria: COTS availability, high pixel resolution, compatibility with the intended embedded processing architecture, and the possibility of evaluating different field-of-view conditions. Configuration 1 was selected because the Raspberry Pi HQ Camera provides a 12.3 MP sensor with interchangeable C-mount optics, allowing the same sensor to be evaluated under wide-angle and narrower-angle lens settings. In particular, the 2.8 mm setting was considered to assess a wide-field observation

condition. Configuration 2 was considered as a higher-resolution compact alternative with a fixed integrated lens, useful for evaluating the effect of increased pixel density when lens interchangeability is not available.

Table 3.1: Comparative optical performance of the two reference COTS configurations.

Parameter	**Configuration 1: IMX477 Varifocal Lens**	**Configuration 2: OV64A40 Integrated Lens**
Camera module	Raspberry Pi HQ Camera	Raspberry Pi Camera Module 3
Sensor	Sony IMX477	Omnivision OV64A40
Resolution	4056×3040 px	9152×6944 px
Sensor width	6.287 mm	7.4 mm
Focal length analyzed	2.8 mm	5.1 mm
Lens type	Varifocal C-mount lens, 2.8–12 mm	Integrated fixed lens
Aperture	F/1.6	F/1.8
Pixel pitch / CoC	1.55 μm	0.81 μm
FOV, wide condition	$\approx 120^\circ$ diagonal	$\approx 84^\circ$ diagonal
FOV, narrow/tele condition	$\approx 38^\circ$ diagonal at 12 mm	Not applicable; fixed $\approx 84^\circ$
Hyperfocal distance, H	≈ 3.16 m	≈ 17.88 m
Focus locked at	≈ 3.07 m	≈ 15.2 m
DoF range	1.558–103.3 m	8.215–101.5 m
Total DoF	≈ 101.78 m	≈ 93.31 m
Geometric limit, Z_{geo}, for 0.8 mm tether	≈ 1.44 m	≈ 5.04 m
Pixel occupancy at 100 m, 1 mm tether	≈ 0.018 px	≈ 0.063 px
Near-field coverage	Better close-range coverage, ≥ 1.5 m	More restrictive close range, ≥ 8.2 m
Far-field pixel density	Lower	Higher
Main optical role	Wide-field situational awareness	Improved pixel sampling at longer range

The comparison highlights the fundamental optical trade-off between wide-field situational awareness and long-range detectability of geometric tethers. While Configuration 1 provides broader angular coverage and improved near-field visibility, Configuration 2 increases far-field tether sampling capability by providing a higher effective pixel density.

These two configurations illustrate the principal optical trade-off of the LINKU stereo vision system. Wide-angle optics improve deployment visibility and tolerance to attitude motion, while narrower configurations improve geometric sampling and reconstruction.

The comparison should therefore be interpreted as an analysis of complementary optical behaviors rather than as a direct stereo pairing between different optical systems. In a final stereo implementation, both cameras used for geometric reconstruction must share calibrated intrinsic and extrinsic parameters.

3.2.2 Geometric resolution limit and Near-Field analysis

The ability to geometrically resolve the tether depends on its projected width on the image sensor. A critical parameter is therefore the maximum observation distance at which the tether subtends at least one pixel.

Using the thin-lens magnification relation, the projected tether width in pixels can be approximated as:

$$w_{\text{proj}} = \left(\frac{f}{d-f}\right)\left(\frac{N_h}{W_{\text{sensor}}}\right) D_t$$

where:

- f is the focal length
- d is the observation distance
- D is the tether diameter
- N_h is the horizontal sensor resolution

Setting:

$$w_{proj} = 1pixel$$

allows estimation of the geometric resolution limit Z_{geo}, corresponding to the maximum distance at which the tether occupies at least one full pixel.

For the geometric-resolution analysis, a conservative tether diameter of 0.8 mm was adopted to represent a worst-case optical detectability condition. This value is used only for the analytical estimation of the geometric resolution limit. In contrast, the 1 mm diameter corresponds to the physical tether proxy used during the experimental validation campaign.

Assuming a worst-case tether diameter of:

$$D_t = 0.8mm$$

the resulting geometric limits are:

Configuration 1

$$Z_{\text{geo},1} \approx 1.44 \text{ m}$$

Configuration 2

$$Z_{\text{geo},2} \approx 5.04 \text{ m}$$

This result shows that Configuration 2 extends the geometric detection regime significantly farther than Configuration 1 due to its higher effective pixel density.

Within this near-field regime, the tether occupies at least one pixel and conventional edge-based geometric extraction remains viable for estimating tether boundaries and centerlines.

However, focusing the optical system for improved far-distance detectability also affects near-field sharpness. For example, when Configuration 2 is focused near the geometric threshold distance, the recalculated near Depth-of-Field limit becomes approximately:

$$D_n \approx 3.92m$$

This implies that tether segments located closer than approximately 3.9 m may appear increasingly blurred when the optical system is optimized for larger-distance geometric detection.

Consequently, the stereo vision system must tolerate a certain degree of near-field blur in exchange for improved geometric sampling at larger distances.

3.2.3 Radiometric detection limit in the Far-Field regime

Beyond the geometric resolution limit, the tether enters the sub-pixel observation regime. Under these conditions, the tether no longer occupies a complete pixel, making direct geometric edge estimation increasingly unreliable.

To evaluate this regime, the projected tether width was estimated at the maximum expected separation distance of approximately 100 m, assuming a tether diameter of 1 mm.

The resulting projected widths are:

Configuration 1

$$w_{proj}(100m) \approx 0.018 pixels$$

Configuration 2

$$w_{proj}(100m) \approx 0.063 pixels$$

These results confirm that the tether remains firmly within the sub-pixel regime at long observation distances, regardless of the selected COTS configuration.

Consequently, increasing sensor resolution alone does not fundamentally solve the detection problem. Even extremely high-resolution sensors would still observe the tether as a sub-pixel structure at long range while simultaneously introducing additional challenges associated with processing load, storage requirements, and reduced radiation tolerance caused by smaller pixel sizes.

Under these conditions, tether detectability becomes primarily a radiometric problem rather than a purely geometric one.

Despite occupying only a fraction of a pixel, the tether may still produce a detectable radiometric response due to the extreme contrast conditions expected in orbit. Under highly directional solar illumination, the tether may generate localized specular reflections against the low-radiance space background, producing detectable intensity peaks on the image sensor.

Therefore, in the far-field regime, the reconstruction strategy must transition from direct geometric edge fitting toward radiometric centroid estimation and intensity-based detection methods capable of tracking sub-pixel radiometric signatures.

3.2.4 Custom analytical tool

To support the optical analysis and evaluate the operational limits of the stereo vision system, a custom analytical tool was developed in Python. This framework estimates the relationship between optical configuration, tether detectability, geometric resolution, and observation distance under representative orbital-like conditions.

The analytical workflow begins with the definition of the sensor and optical input parameters, including:

- Image resolution
- Focal length
- Aperture
- Focus distance
- Tether diameter assumptions

Using these parameters, the framework computes the corresponding optical quantities required for the analysis, including:

- Circle of Confusion (CoC)
- Hyperfocal distance
- Near and far Depth-of-Field limits
- Projected tether width in pixels
- Geometric resolution thresholds

The tool then evaluates the evolution of tether detectability as a function of observation distance, allowing identification of the transition between the geometric detection regime and the sub-pixel

radiometric regime. Figure 3.1 illustrates the graphical interface of the analytical framework used for these calculations.

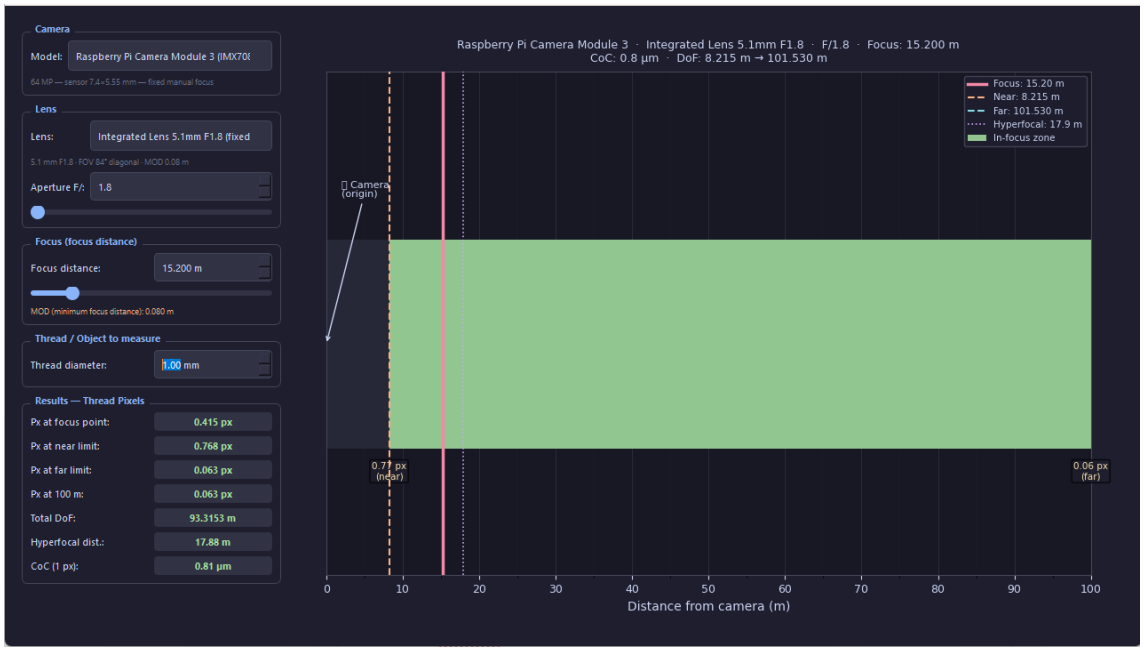


Figure 3.1: Graphical interface of the custom analytical software developed to evaluate the optical performance of the LINKU stereo vision system. The visualization corresponds to the Hawkeye-based configuration and illustrates the relationship between observation distance, Depth of Field limits, hyperfocal distance, and tether pixel occupancy across the expected operational range of up to 100 m. The interface includes an interactive legend to identify the in-focus region and the transition between geometric and sub-pixel detection regimes.

The framework also generates comparative visualizations between different optical configurations, enabling rapid evaluation of the trade-off between:

- Wide-field situational awareness
- Tether geometric localization capability
- Long-range radiometric detectability

Although the analytical model is based on simplified thin-lens assumptions, it provides a practical first-order estimation of the optical feasibility limits associated with tether observation under the expected mission conditions. The resulting analysis serves as a preliminary design tool for evaluating COTS optical configurations prior to experimental validation.

3.3 Orbital illumination physics

The optical detectability of the tether is strongly influenced by the illumination conditions expected in orbit. Unlike terrestrial environments, where diffuse ambient illumination and textured backgrounds are commonly available, orbital observation is dominated by highly directional solar radiation and extreme radiometric contrast between illuminated objects and deep space.

Under these conditions, tether visibility depends not only on geometric resolution, but also on the interaction between the tether surface, solar illumination geometry, and sensor viewing direction. Consequently, understanding the radiometric behavior of thin structures in orbit is essential for evaluating the feasibility of long-range tether detection.

3.3.1 Illumination geometry in LEO

In LEO, the Sun behaves approximately as a collimated illumination source due to its large distance relative to spacecraft dimensions. As a result, incident solar radiation can be modeled as nearly parallel rays across the tethered system.

The average solar irradiance near Earth orbit is approximately $E \approx 1360.8 W/m^2$ under standard solar conditions [7].

This illumination regime differs substantially from terrestrial imaging conditions, where atmospheric scattering and environmental reflections contribute significantly to scene brightness. In orbit, the background is predominantly low-radiance deep space, and radiometric visibility is therefore dominated by direct solar reflection.

Consequently, the apparent brightness of the tether depends strongly on:

- Solar incidence angle
- Observation angle
- Tether surface material properties
- Local tether orientation relative to the camera line-of-sight

Figure 3.2 shows the simplified illumination geometry considered for tether observation under orbital-like conditions.

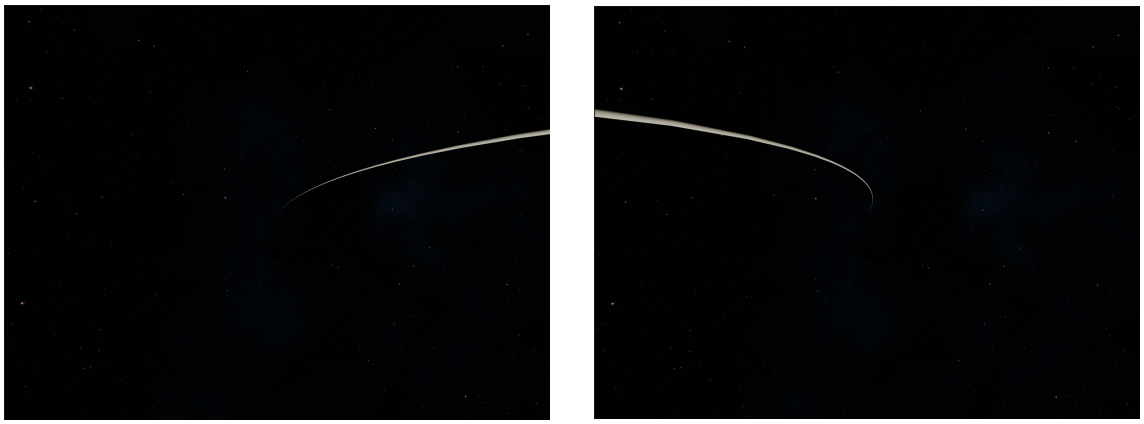


Figure 3.2: Simulated stereo image pair of the tether under high-contrast orbital-like illumination conditions. The left and right images represent a calibrated stereo pair with identical optical parameters and a fixed baseline of 100 mm. The visualization illustrates the expected appearance of a thin tether against a low-radiance background, highlighting the radiometric contrast and localized brightness variations associated with directional illumination. The figure is intended to represent the expected optical observation conditions rather than a final flight-camera configuration.

Because the tether diameter is extremely small relative to the observation distance, even minor orientation changes can significantly modify the observed radiometric response. In particular, localized specular reflections may temporarily increase the apparent intensity of the tether against the dark background, enabling detection even when the tether occupies only a fraction of a pixel.

3.3.2 Specular reflection and sub-pixel visibility

At large observation distances, the tether enters the sub-pixel regime described in Section 3.2.3, where direct geometric edge reconstruction becomes increasingly unreliable. Under these conditions, tether visibility depends primarily on radiometric contrast rather than on explicit geometric resolution.

If the tether surface exhibits partially specular behavior, incident solar radiation may generate localized reflection peaks observable by the image sensor. These reflections may produce detectable

intensity variations even when the projected tether width is substantially smaller than one pixel.

This effect becomes particularly relevant under high-contrast orbital conditions because the surrounding background radiance is extremely low. As a result, relatively small reflected signals may become detectable if the sensor dynamic range and exposure conditions are appropriately configured. Figure 3.3 illustrates a conceptual example of stereo tether observation under directional illumination conditions.

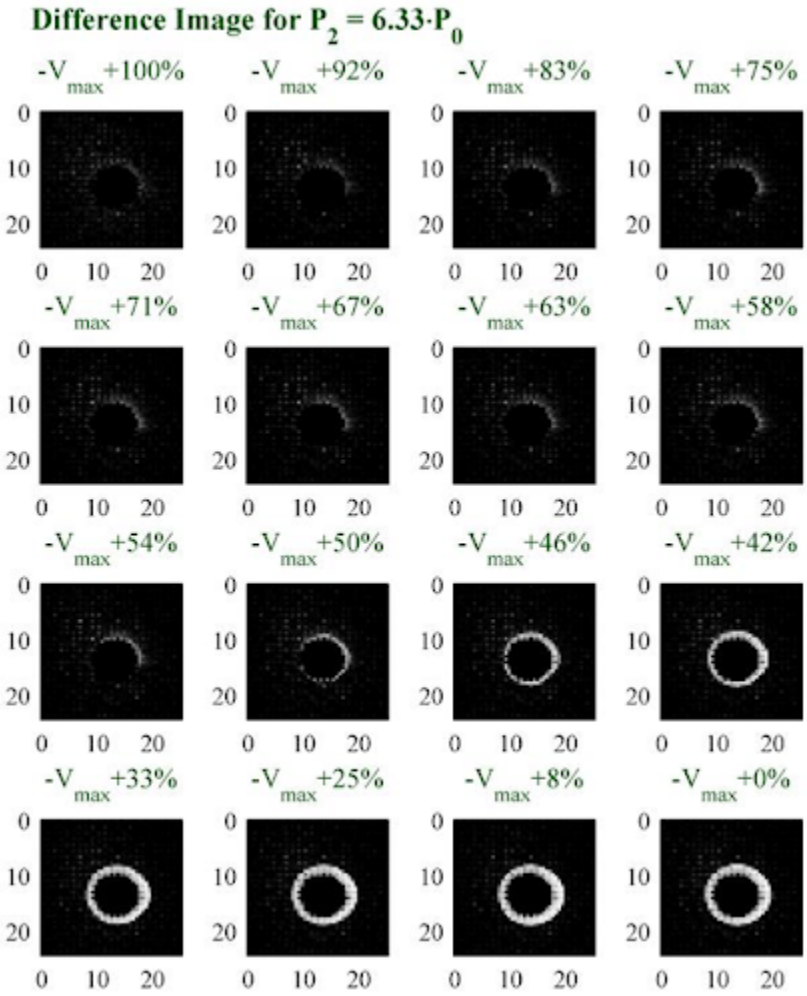


Figure 3.3: Empirical demonstration of blooming across a high-contrast boundary in a CMOS sensor, adapted from Ailuri et al. [8]. The difference image highlights the spatial spreading of charge from the saturated region into adjacent dark pixels, illustrating how radiometric overflow may artificially enlarge the apparent width of thin bright structures under high-contrast illumination conditions.

However, this visibility mechanism also introduces several challenges:

- Reflections may appear intermittently depending on tether orientation
- Brightness may vary rapidly along the tether length
- Saturation or blooming effects may occur under favorable reflection geometry [8, 9]

Blooming effects in CMOS sensors may spread saturated charge into neighboring pixels, artificially enlarging the apparent width of thin structures and distorting radiometric measurements [8]. Consequently, tether detection cannot rely exclusively on stable brightness assumptions. Instead, the reconstruction strategy must tolerate non-uniform radiometric behavior and intermittent visibility conditions.

For the present mission concept, blooming is considered only as a possible aid for initial detectability rather than as a reliable geometric measurement mechanism. Saturation-induced charge spreading may increase the apparent footprint of an otherwise sub-pixel tether, facilitating intensity-based detection under high-contrast conditions. However, this apparent width does not represent the true physical diameter of the tether and should not be interpreted as a direct geometric measurement.

Under these conditions, accurate reconstruction must rely primarily on radiometric centroiding and stereo geometric constraints rather than on the apparent width of saturated image regions [9, 8].

These effects further reinforce the limitations of conventional feature-based photogrammetry methods discussed in Chapter 2, since standard correspondence algorithms typically assume relatively stable brightness consistency between stereo views [2, 3]. Figure 3.4 illustrates how bright line-like structures may produce radiometric spreading and blooming artifacts on CMOS image sensors.

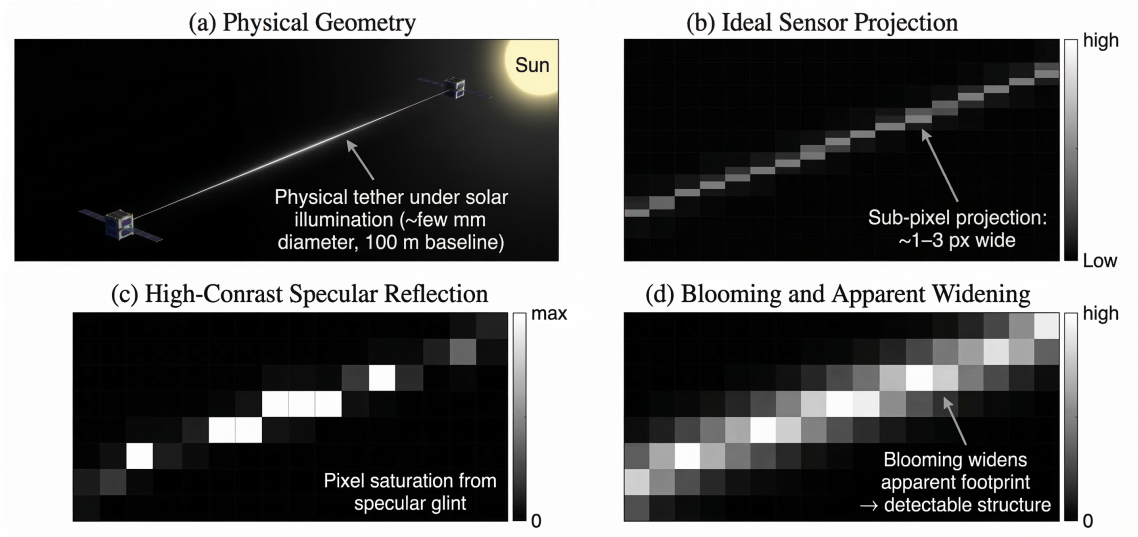


Figure 3.4: Author-generated conceptual illustration of a possible imaging response of a thin tether under high-contrast orbital-like illumination conditions. The figure is intended as an idealized visual aid and does not represent a measured or quantitatively validated blooming width. (a) Simplified physical geometry. (b) Ideal sub-pixel sensor projection. (c) Localized saturation associated with a bright tether response. (d) Qualitative apparent footprint enlargement that may occur due to saturation-induced blooming.

3.3.3 Implications for optical detection strategy

The orbital illumination conditions analyzed in the previous sections directly influence the reconstruction strategy adopted for the LINKU stereo vision system.

As a consequence of the sub-pixel visibility, specular reflection, saturation, and blooming effects discussed above, the optical problem cannot be treated purely as a conventional geometric imaging problem throughout a significant portion of the operational range.

Under sub-pixel observation conditions, the tether progressively ceases to be conventionally observable as a fully resolvable geometric object and instead becomes primarily detectable through its radiometric response within the image plane.

Due to these conditions, tether detectability depends on a combination of the following:

Geometric sampling, which determines the spatial projection of the tether onto the sensor plane.

Radiometric contrast, which governs the distinguishability of the tether signal relative to the

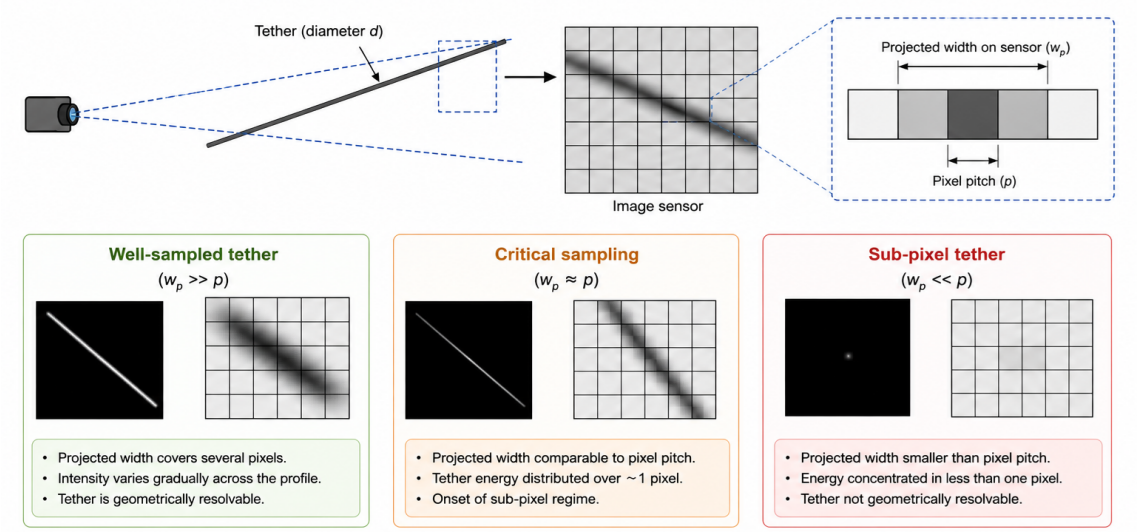


Figure 3.5: Conceptual illustration of the influence of geometric sampling on tether detectability under orbital observation conditions. The figure shows how the tether is projected onto the image sensor and how its apparent width w_p relates to the sensor pixel pitch p. Three sampling regimes are presented: well-sampled, critical sampling, and sub-pixel observation. As the projected tether width decreases below the pixel size, the tether progressively loses its directly resolvable geometric structure and becomes primarily observable through its radiometric response. This transition highlights the increasing importance of radiometric-based detection and localization techniques under sub-pixel observation conditions.

background.

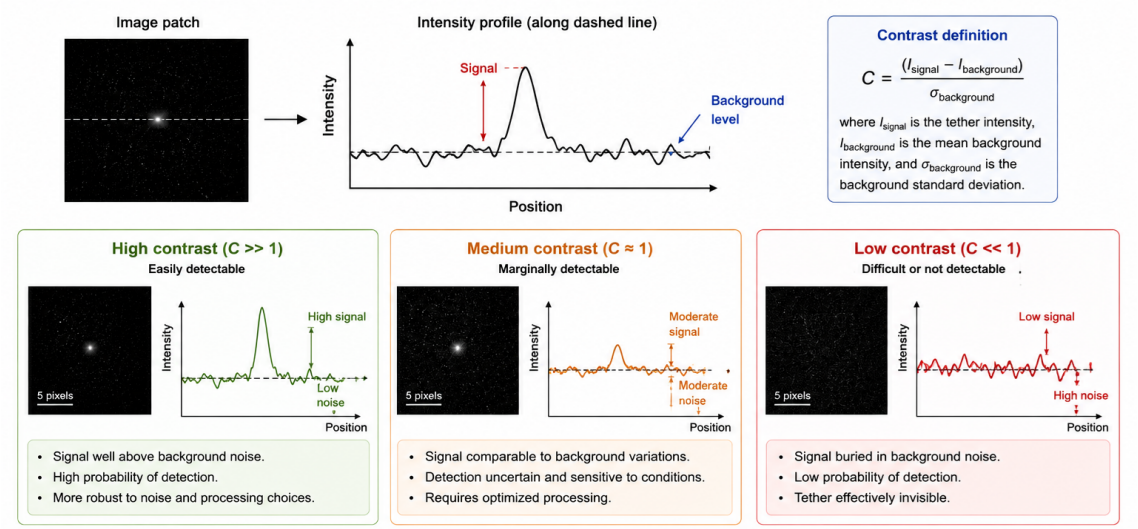


Figure 3.6: Conceptual illustration of the influence of radiometric contrast on tether detectability under sub-pixel orbital observation conditions. The upper portion of the figure shows the relationship between the tether signal and the surrounding background intensity, highlighting how detectability depends on the signal level relative to background variations and noise. The lower portion presents three representative contrast regimes: high contrast, medium contrast, and low contrast. As radiometric contrast decreases, the tether signal becomes progressively more difficult to distinguish from the background, reducing detection reliability. The figure emphasizes that even when the tether is not geometrically resolvable, it may remain detectable if sufficient radiometric contrast exists between the tether response and the surrounding background. This behavior reinforces the importance of radiometric-based detection and localization techniques under sub-pixel observation conditions.

Directional illumination, which controls brightness variations through observation geometry and specular reflection effects.

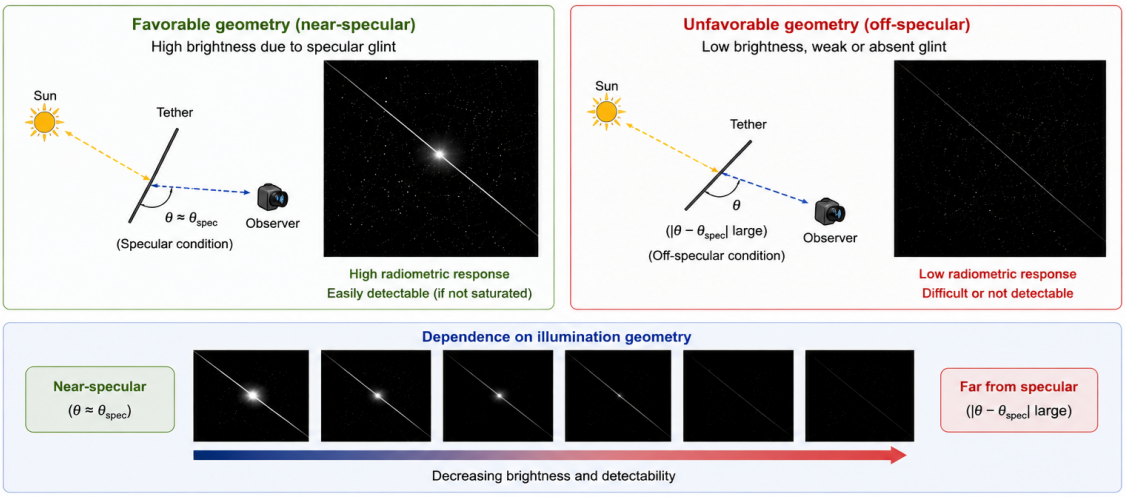


Figure 3.7: Conceptual illustration of the influence of directional illumination on tether detectability under orbital observation conditions. The figure shows how the apparent brightness of the tether depends on the relative geometry between the Sun, the tether, and the observer. Under favorable near-specular conditions, reflected sunlight produces a strong radiometric response, increasing tether visibility and detectability. In contrast, under off-specular conditions, the reflected energy reaching the sensor is significantly reduced, resulting in weak radiometric signals and decreased detectability. The lower portion of the figure illustrates the progressive reduction in tether brightness as the observation geometry moves away from the specular reflection condition. This behavior highlights the strong dependence of tether visibility on illumination geometry and explains why a tether may alternate between highly detectable and nearly invisible states despite no change in its physical characteristics.

Temporal visibility conditions, which introduce intermittent detectability due to orbital motion and changing illumination geometry.

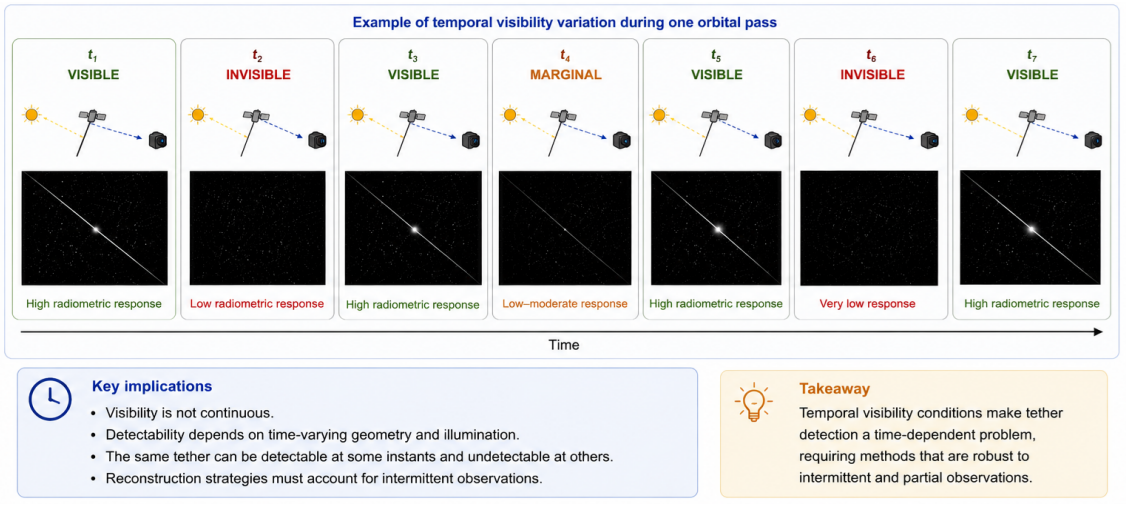


Figure 3.8: Conceptual illustration of temporal visibility effects on tether detectability under orbital observation conditions. The figure shows how orbital motion, spacecraft attitude variations, and changing Sun–tether–observer geometry can cause the same tether to alternate between visible, marginally visible, and invisible states without any change in its physical properties. This time-dependent radiometric behavior motivates reconstruction strategies capable of operating with partial and intermittently visible observations.

This motivates the use of geometry-driven reconstruction methods capable of exploiting radiometric

signatures, intensity-based extraction, geometric continuity, and stereo correspondence under sub-pixel observation conditions. These methods include:

- Centerline estimation
- Intensity-based extraction
- Geometric curve tracking
- Radiometric centroiding
- Sub-pixel localization techniques

Under these conditions, the apparent tether width produced by saturation-induced blooming should not be interpreted as a direct geometric measurement, reinforcing the need for reconstruction approaches based on radiometric localization and constrained stereo geometry rather than on raw saturated image regions.

Additionally, the strong dependency between tether visibility and illumination geometry suggests that experimental validation should incorporate representative high-contrast lighting conditions capable of approximating the directional illumination behavior expected in orbit.

For this reason, the experimental design incorporates controlled illumination configurations intended to reproduce the radiometric challenges associated with orbital tether observation.

Chapter 4

Experimental and processing approaches

This chapter defines the experimental and methodological framework used to evaluate the feasibility of optical tether observation and geometric reconstruction under representative orbital-like conditions. Based on the optical and radiometric constraints established previously, the methodology was designed to progressively evaluate the limitations of conventional reconstruction methods and validate alternative approaches specifically adapted to thin, textureless tether structures.

The proposed validation strategy follows a progressive transition from conventional feature-based photogrammetry toward geometry-driven reconstruction and exploratory AI-assisted processing. Rather than treating all reconstruction approaches equivalently, the methodology intentionally prioritizes physically interpretable geometric extraction methods capable of operating under sparse visual conditions and extreme radiometric contrast.

The complete framework is organized into three global development phases:

- Phase I: Evaluation of conventional photogrammetry methods under terrestrial and space-emulated visual conditions to establish the operational limitations of standard feature-matching approaches.
- Phase II: Development and validation of a custom geometry-driven reconstruction pipeline based on edge extraction, radiometric centroiding, and stereo geometric constraints.
- Phase III: Exploratory evaluation of AI-assisted reconstruction techniques intended to support future improvements in segmentation robustness and complex topology reconstruction.

Together, these phases establish a progressive methodological roadmap for the LINKU stereo vision system.

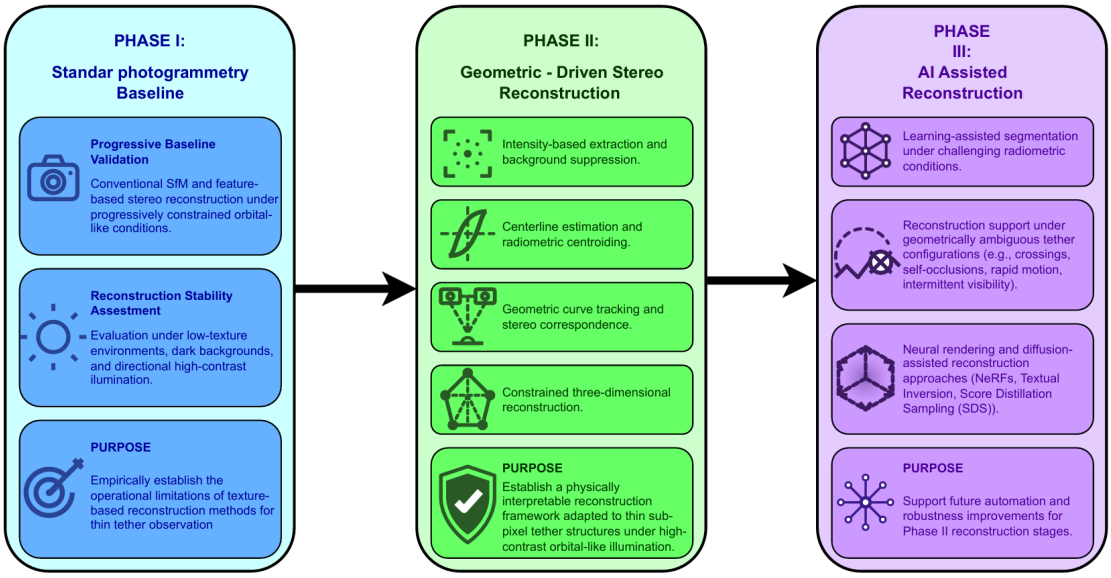


Figure 4.1: Methodological roadmap of the LINKU stereo vision system validation strategy. The framework progressively transitions from conventional texture-based photogrammetry toward a geometry-driven stereo reconstruction methodology and exploratory AI-assisted enhancements under increasingly constrained orbital-like optical conditions. The roadmap highlights the progressive evolution from feature-based reconstruction failure analysis to physically interpretable tether reconstruction adapted to thin sub-pixel structures.

Rather than directly attempting end-to-end reconstruction under fully space-emulated conditions, the proposed methodology progressively isolates the principal failure modes associated with thin tether observation, including low texture, sub-pixel geometry, directional illumination, and radiometric instability.

This progressive validation strategy enables analysis of the reconstruction problem with increasing optical and computational complexity while maintaining traceability between the optical constraints identified in Chapter 3 and the observed reconstruction behavior under representative orbital-like conditions.

As a result, the framework establishes a transition from conventional texture-based photogrammetry to a dedicated geometry-driven stereo reconstruction methodology adapted to thin, sub-pixel tether structures under high-contrast, orbital-like illumination.

4.1 Phase I: Standard photogrammetry baseline

Phase I establishes the experimental baseline used to evaluate conventional photogrammetry methods under progressively constrained visual conditions representative of orbital tether observation. Its purpose is to empirically establish the operational limitations of conventional texture-based photogrammetry for thin tether observation under representative orbital-like optical conditions.

This phase assesses the applicability of feature-based stereo reconstruction approaches to geometries and illumination conditions relevant to the LINKU mission. In particular, the methodology evaluates the influence of:

- Environmental texture
- Geometric scale
- Directional illumination
- High-contrast backgrounds

On the operation of conventional Structure-from-Motion (SfM) and stereo matching algorithms.

The experiments were conducted using Agisoft Metashape as the reference photogrammetry platform. The evaluation considered:

- Camera alignment behavior
- Disparity-map generation
- Stereo correspondence consistency
- Point-cloud reconstruction capability

As illustrated in the methodological roadmap presented in Figure 4.1, the Phase I methodology is organized into two principal processing stages:

Progressive baseline validation

Phase I was structured as a progressive validation sequence intended to incrementally approximate the optical conditions expected during orbital tether observation.

The methodology begins with the reconstruction of a volumetric object containing easily distinguishable geometric features under controlled environmental conditions. This initial stage establishes a reference baseline for evaluating conventional feature-based reconstruction behavior under textured terrestrial environments.

Subsequently, the validation sequence transitions toward thin tether observation using a tether proxy operating under progressively constrained visual conditions. This progression allows evaluation of conventional stereo reconstruction methods as the observed geometry evolves from feature-rich volumetric structures toward thin sub-pixel line-like objects representative of the LINKU operational scenario.

Reconstruction stability assessment

This second stage evaluates reconstruction stability under progressively constrained visual environments representative of orbital-like observation conditions.

The analysis considers the influence of:

- Reduced environmental texture
- Dark backgrounds
- Directional high-contrast illumination
- Geometric scale reduction

On the stability of conventional photogrammetry algorithms.

Particular attention is placed on evaluating:

- Camera alignment stability
- Disparity-map consistency
- Stereo correspondence reliability
- Point-cloud reconstruction capability

Under conditions where conventional texture-based feature matching becomes increasingly limited.

This validation framework establishes a controlled methodological progression for evaluating the operational applicability of conventional photogrammetry, as shown in Figure 4.2, prior to implementing the geometry-driven reconstruction methodology introduced in Phase II.

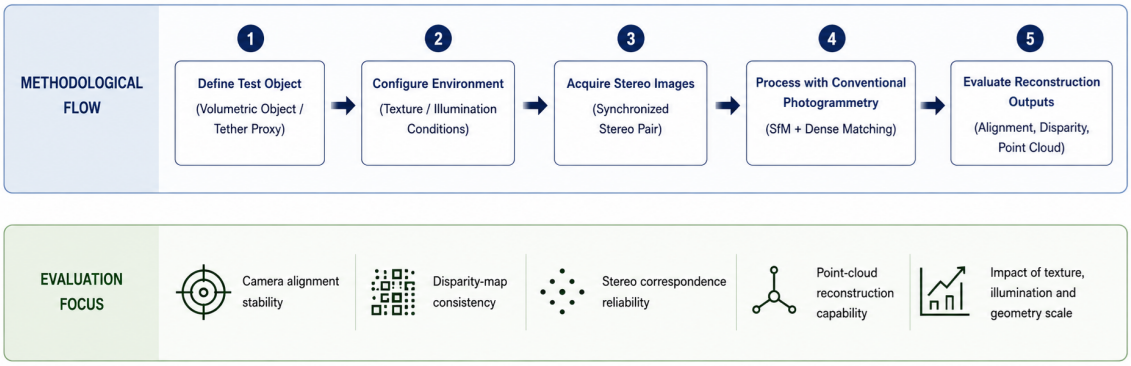


Figure 4.2: Methodological flow and evaluation focus adopted during Phase I. The validation sequence progressively transitions from controlled volumetric-object reconstruction toward thin tether observation under increasingly constrained orbital-like optical conditions.

Traditional dense stereo pathway retained in the current software

Alongside the external Agisoft Metashape experiments described above, the LINKU software itself still retains a conventional, cable-agnostic dense stereo processing path. It is not a separate tool: it is the same `process_stereo_pair` routine used throughout this work, running with no cable mask supplied, which is exactly what happens if the operator skips the segmentation-tuning step described in Phase II. Figure 4.3 shows this pathway for completeness, since it remains available in the current implementation as the software's own conventional/baseline processing mode, distinct from the dedicated geometry-driven pipeline introduced in Phase II.

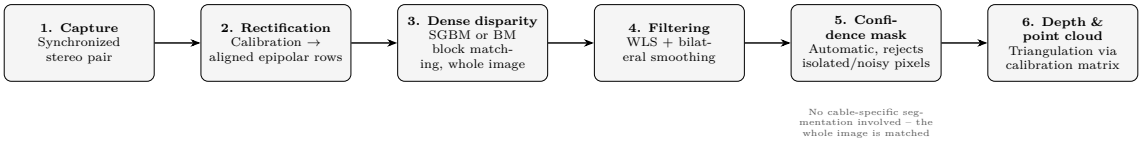


Figure 4.3: Traditional dense stereo pipeline still present in the current software (`processing/stereo_processor.py`, `process_stereo_pair`). This is the pathway the software falls back to automatically whenever no cable-specific mask has been configured through the segmentation tuner: it treats the whole image as a conventional dense stereo scene, with no thread-specific extraction step. The confidence mask (stage 5) is generated automatically from local disparity consistency and pixel density, not from a manually or AI-selected filter, and is used to discard unreliable points before triangulation. This pathway corresponds to the dense-stereo experiment discussed in Section 6.2.1, where the absence of a cable-specific mask allowed background disparity artifacts to propagate into the reconstruction.

4.2 Phase II: Geometry-driven stereo reconstruction

Phase II is the primary reconstruction approach of the LINKU camera system. After explaining the methodological limitations identified during Phase I, Phase II introduces a dedicated reconstruction framework tailored to thin tether structures operating under low-texture, high-contrast, orbital-like optical conditions.

Unlike conventional photogrammetry approaches, the proposed methodology does not rely primarily on texture-based feature correspondence. Instead, the reconstruction strategy treats the tether as a continuous geometric structure whose detectability is governed mainly by radiometric contrast, geometric continuity, and constrained stereo geometry, as shown in Figure 4.4.

This phase establishes a physically interpretable reconstruction methodology that operates under the sub-pixel observation conditions derived in Chapter 3.

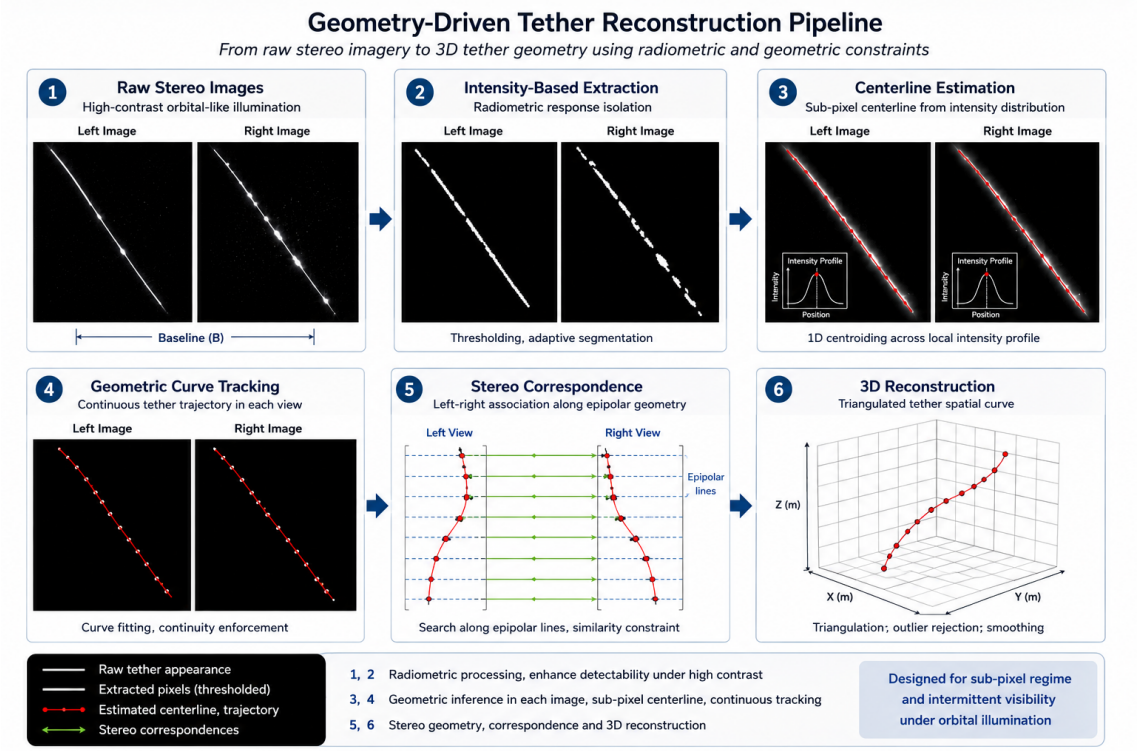


Figure 4.4: Conceptual overview of the geometry-driven reconstruction pipeline proposed for thin tether observation in the sub-pixel regime. The workflow illustrates the progressive transition from raw stereo image acquisition under high-contrast orbital-like illumination toward constrained three-dimensional tether reconstruction. The methodology includes: (1) acquisition of synchronized stereo images, (2) intensity-based tether extraction and background suppression, (3) sub-pixel centerline estimation through radiometric centroiding, (4) geometric curve tracking for continuous tether representation, (5) stereo correspondence establishment using epipolar geometric constraints, and (6) three-dimensional reconstruction through triangulation of the stereo-associated tether geometry. The figure highlights the transition from conventional texture-dependent processing toward radiometric and geometry-driven reconstruction methods adapted to thin low-texture structures under orbital-like optical conditions.

Figure 4.4 presents the conceptual view of this pipeline. Figure 4.5 complements it by naming the specific algorithms used in the current software implementation, including the classical and AI-assisted variants evaluated for each stage.

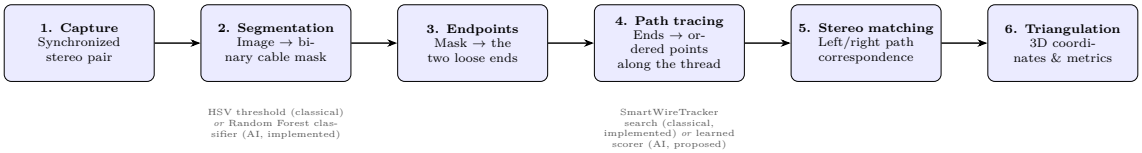


Figure 4.5: Implementation-level view of the Phase II (geometry-driven) reconstruction pipeline. The six boxes are the processing stages, each labeled with the transformation it performs. Where a stage currently offers more than one implementation – segmentation and path tracing – the classical method already implemented and the AI-assisted alternative are noted underneath; all other stages have a single, classical implementation today. Algorithm-level detail for each stage is given in the surrounding text rather than in the diagram.

As illustrated in the methodological roadmap presented in Figure 4.1, the Phase II methodology is organized into four principal processing stages:

Intensity-based extraction and background suppression

Once stereo images have been acquired from both cameras, the reconstruction process begins with intensity-based extraction and background suppression under controlled illumination conditions designed to maximize radiometric contrast between the tether and the surrounding background. Since the orbital environment is expected to exhibit minimal texture, the methodology prioritizes isolating radiometric response and localized contrast enhancement over conventional feature extraction.

This initial stage focuses on isolating the tether's radiometric signature while minimizing the effects of background noise, localized reflections, and non-uniform illumination.

Under sub-pixel observation regimes, the tether may no longer remain conventionally observable as a fully resolvable geometric structure. Instead, its detectability becomes increasingly dependent on the isolation of localized intensity variations generated within the image plane.

The resulting segmented representation establishes the initial geometric reference used for the subsequent centerline estimation and stereo reconstruction stages.

Centerline estimation and radiometric centroiding

Following intensity-based extraction and background suppression, the tether is represented in both stereo images as a thin line-like radiometric structure whose apparent position must be estimated with sub-pixel precision.

At this stage, the methodology estimates the geometric centerline of the tether and its radiometric centroid from localized intensity distributions. Instead of relying on conventional textured keypoints, the reconstruction framework uses edge-based localization and radiometric analysis to estimate the apparent tether trajectory in the image plane.

This approach is conceptually aligned with line-profile analysis methods such as Steger's framework [4], in which line-like structures are characterized by local intensity gradients and scale-space behavior. In this context, centerline estimation allows the tether to be represented as a continuous geometric trajectory inferred from its radiometric response rather than from explicitly resolvable geometric boundaries.

Under sub-pixel observation conditions, radiometric centroiding becomes particularly important because the tether may no longer remain fully observable as a conventionally resolvable structure. Consequently, the apparent tether position is estimated from the radiometric distribution associated with the observed line response, enabling sub-pixel localization of the tether trajectory prior to stereo correspondence and three-dimensional reconstruction.

Geometric curve tracking and stereo correspondence

Once the tether centerline has been estimated in both stereo views, the methodology performs geometric curve tracking and stereo correspondence in order to establish consistent associations between the tether trajectories observed in each image plane.

Rather than treating the tether as a collection of independent feature points, the reconstruction framework models it as a continuous geometric structure whose trajectory can be constrained spatially and radiometrically across both stereo observations. This approach prioritizes geometric continuity and trajectory coherence instead of conventional dense texture matching.

At this stage, epipolar geometry is used to constrain the stereo correspondence search along the extracted tether trajectories, reducing ambiguity in low-texture regions and improving geometric consistency between both image planes.

Because the tether behaves as a continuous line-like structure under sub-pixel observation conditions, geometric curve tracking becomes particularly important for preserving trajectory continuity during stereo association. Consequently, the methodology prioritizes constrained stereo correspon-

dence and continuous geometric tracking rather than independent point-feature matching.

Constrained three-dimensional reconstruction

The final stage of the methodology consists of constrained three-dimensional reconstruction based on the stereo correspondence relationships established between both image planes.

At this stage, the stereo-associated tether trajectories are processed using constrained stereo geometry to infer the three-dimensional spatial configuration of the tether under sub-pixel observation conditions. Unlike conventional dense photogrammetry approaches, the methodology does not seek to reconstruct volumetric point clouds or texture-rich surfaces. Instead, the reconstruction framework focuses on recovering the continuous geometric behavior of the tether throughout the observed operational range.

The primary objective of this stage is to reconstruct the spatial trajectory of the tether, curvature evolution, oscillatory behavior, and temporal geometric dynamics.

Because the tether behaves as a continuous, line-like structure with limited geometric observability, the reconstruction process prioritizes trajectory continuity and constrained geometric consistency across stereo observations over dense feature correspondence.

The resulting framework establishes the primary operational reconstruction methodology of the LINKU stereo vision system and serves as the basis for the experimental hardware implementation and validation environments described in Chapter 5.

4.3 Phase III: AI-Assisted reconstruction (Exploratory)

Phase III investigates the future integration of Artificial Intelligence (AI) techniques intended to complement the geometry-driven reconstruction framework established in Phase II.

This phase will not replace the deterministic reconstruction methodology based on constrained stereo geometry, geometric curve tracking, and radiometric localization, but rather evaluate the potential of AI-assisted approaches to improve the robustness of intensity-based extraction, geometric trajectory continuity, and operation under visually ambiguous conditions.

As illustrated in the methodological roadmap in Figure 4.1, the exploratory AI-assisted framework primarily supports the reconstruction stages already established during Phase II.

Current implementation status

Phase III groups together three AI-related directions with markedly different levels of maturity. Rather than presenting all three as equally exploratory, this report distinguishes explicitly between the direction that has already produced a working implementation, the direction for which only data-collection infrastructure exists, and the direction that remains a documented idea without any corresponding code. Figure 4.6 summarizes this distinction, which is elaborated in the three subsections below.

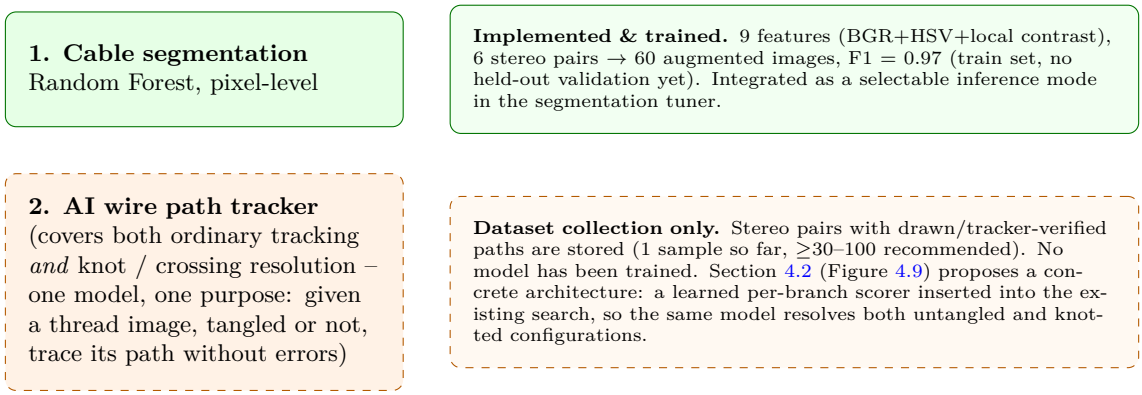


Figure 4.6: Current maturity of the two AI-related components considered within the LINKU camera system software, as of the software state accompanying this report. Only the segmentation model (1) is operational today. Wire path tracking and topological knot/crossing resolution (2) are treated as a single AI component – not two separate systems – because both solve the same underlying problem: given a segmented thread image, determine the correct continuous path from one end to the other, regardless of how entangled the thread is.

1. Learning-assisted intensity-based extraction

The first exploratory direction considers using learning-based methods to improve intensity-based extraction and background suppression under challenging radiometric conditions.

Although the deterministic extraction approach developed in Phase II prioritizes physically interpretable processing, AI-assisted methods may provide additional robustness under conditions involving non-uniform illumination, intermittent tether visibility, partial saturation, and localized blooming artifacts, as shown in Figure 4.7.

Unlike the other two exploratory directions discussed in this section, this direction has already produced a first working implementation, integrated into the same laboratory software used for the Phase II experiments. A pixel-level Random Forest classifier was trained to segment the tether from the background, using nine features per pixel (RGB, HSV, and a local-contrast measure derived from a difference of Gaussian blurs). The training set consisted of 6 manually annotated stereo pairs, expanded to 60 images through horizontal/vertical flipping and brightness augmentation (±35 levels), and evaluated with an F1-score of 0.97 (precision 0.94, recall 0.99). This evaluation was performed on the training set itself; a held-out validation split has not yet been implemented, so the reported score should be read as an upper bound on current performance rather than a generalization estimate. The trained model is selectable as an alternative to manual HSV thresholding directly within the segmentation tuning interface, and every new mask produced through either method can be added to the training set, allowing the model to be iteratively retrained as more laboratory sessions are collected.

Potential further approaches include:

- Convolutional neural networks (CNNs), which would replace the current hand-crafted per-pixel features with learned spatial features
- Full semantic segmentation architectures (e.g. U-Net-style encoder-decoders)

- Hybrid image-enhancement pipelines

This stage assists the intensity-based extraction and background suppression processes while preserving compatibility with the geometry-driven reconstruction framework established in Phase II.

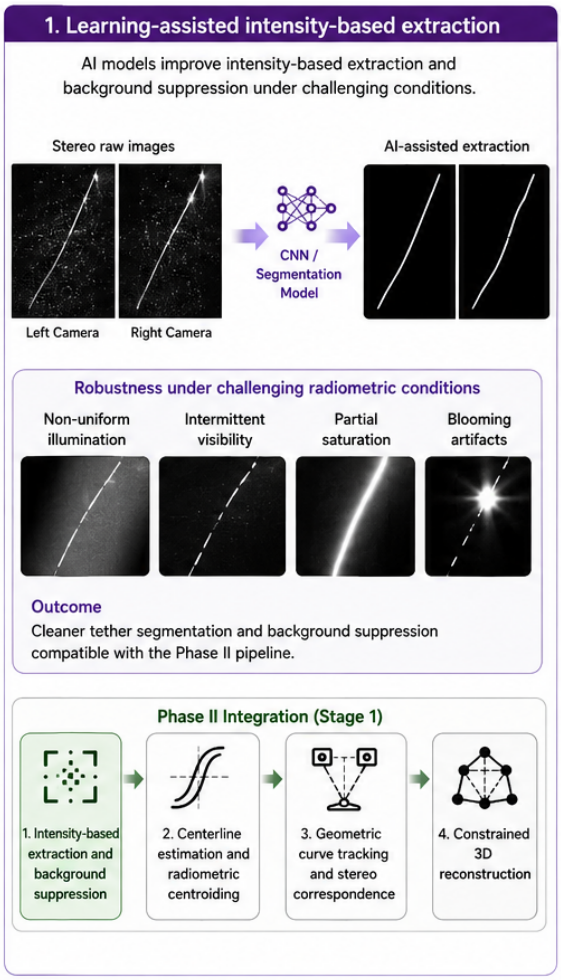


Figure 4.7: Conceptual overview of learning-assisted intensity-based extraction under challenging radiometric conditions. AI-based segmentation approaches are investigated to improve tether extraction and background suppression in scenarios involving non-uniform illumination, intermittent visibility, partial saturation, and localized blooming artifacts. This stage enhances the robustness of the initial radiometric extraction process while preserving compatibility with the deterministic geometry-driven framework.

2. AI-assisted wire path tracking (including knot / crossing resolution)

The second exploratory direction investigates the use of AI-assisted methods to support geometric curve tracking and stereo correspondence under geometrically ambiguous tether configurations.

This includes scenarios involving tether crossings, partial self-occlusion, rapid oscillatory motion, and discontinuous radiometric visibility, as seen in Figure 4.8. Earlier drafts of this report discussed "learning-assisted path tracking" and "topological knot solving" as two separate future capabilities. They are treated here as a single AI component instead, because they solve the same underlying problem: given a segmented thread image, determine the correct continuous path from one end to the other. An untangled thread simply presents fewer ambiguous decision points along that path than a self-crossing or knotted one.

This direction is presently at an earlier stage than the segmentation model described above. The current software already collects the training data it would require: every path traced through the deterministic tracker, or drawn manually by an operator, is stored together with its stereo image pair and segmentation mask in a dedicated dataset (1 stereo sample so far; 30–100 are recommended before training is attempted). However, no learning-based tracker has been trained or implemented to date.

Rather than leaving the target architecture undefined, Figure 4.9 proposes a concrete design that builds directly on the classical tracker already implemented (Section 4.2, Figure 4.5): the same step-by-step search over the segmented mask is kept unchanged, and only the local decision made at each branch point is replaced – from a fixed, hand-tuned scoring formula to a lightweight classifier trained on the paths already being collected. Because that classifier is invoked identically at every branch point, whether there are none (a straight, untangled thread) or many (a knotted one), a single trained model would cover both cases without requiring a separate topology-reasoning module.

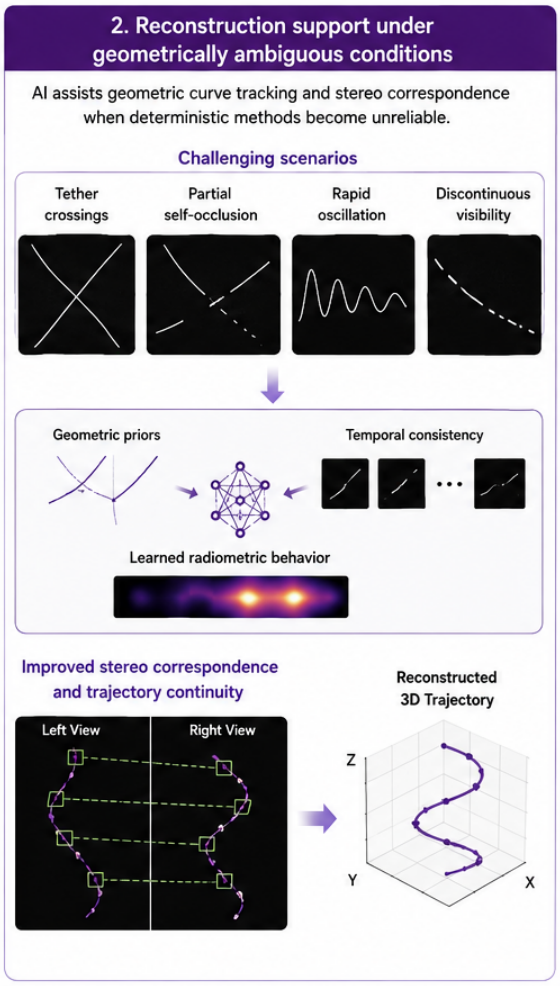


Figure 4.8: Conceptual overview that illustrates reconstruction support under geometrically ambiguous conditions. AI-assisted methods are explored to improve geometric curve tracking and stereo correspondence in scenarios involving tether crossings, partial self-occlusion, rapid oscillatory motion, and discontinuous radiometric visibility. The figure highlights the incorporation of temporal consistency, geometric priors, and learned radiometric behavior to support continuity in reconstruction when deterministic correspondence becomes unreliable.

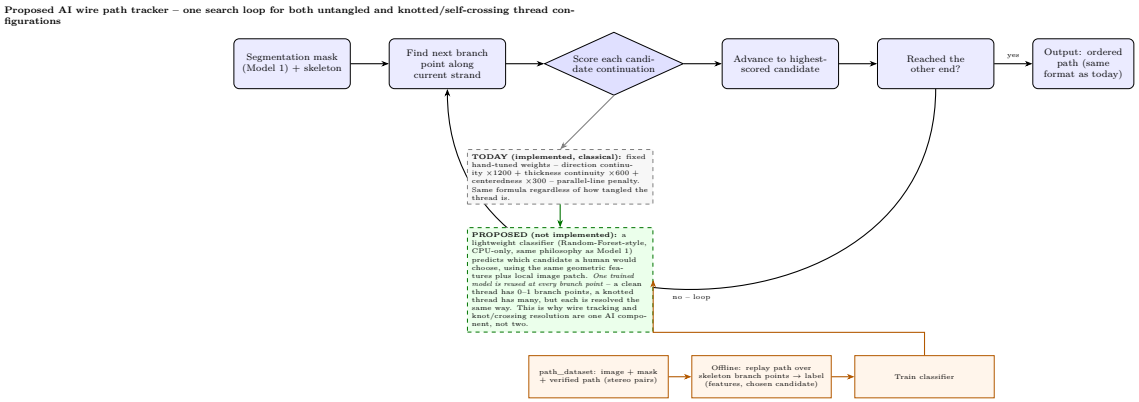


Figure 4.9: Proposed architecture for the AI-assisted wire path tracker. The overall search loop (top row) is unchanged from the classical SmartWireTracker already implemented today: it walks the segmented thread step by step and, at each branch point, scores the candidate continuations to pick the most likely one. The only proposed change is replacing the fixed, hand-tuned scoring formula with a learned scorer trained on the same `path_dataset` already being collected (bottom row). Because the same scorer is invoked at every branch point regardless of how many exist, this single component naturally handles both simple, untangled threads (few or no branch points) and heavily tangled or self-crossing configurations (many branch points) – it is one AI model solving one problem, not two separate systems for "tracking" and "knot solving".

3. Neural rendering and future reconstruction approaches

The final exploratory direction considers the potential future integration of neural rendering approaches, including Neural Radiance Fields (NeRFs) and related implicit scene representation techniques.

These methods may offer future capabilities for multi-view geometric interpretation, radiometric interpolation, and reconstruction of sparse visual structures under limited observation conditions, as illustrated in Figure 4.10.

Potential future implementations may additionally incorporate techniques such as *Textual Inversion for tether-specific radiometric adaptation* and *Score Distillation Sampling (SDS) for geometry-consistent neural rendering optimization*

However, the direct application of these approaches to orbital tether reconstruction remains constrained by several practical limitations, including a lack of representative datasets, computational complexity, sparse visual texture, and highly dynamic illumination conditions. For this reason, AI integration is currently treated as a future-oriented augmentation layer rather than as the primary operational reconstruction methodology of the LINKU stereo vision system.

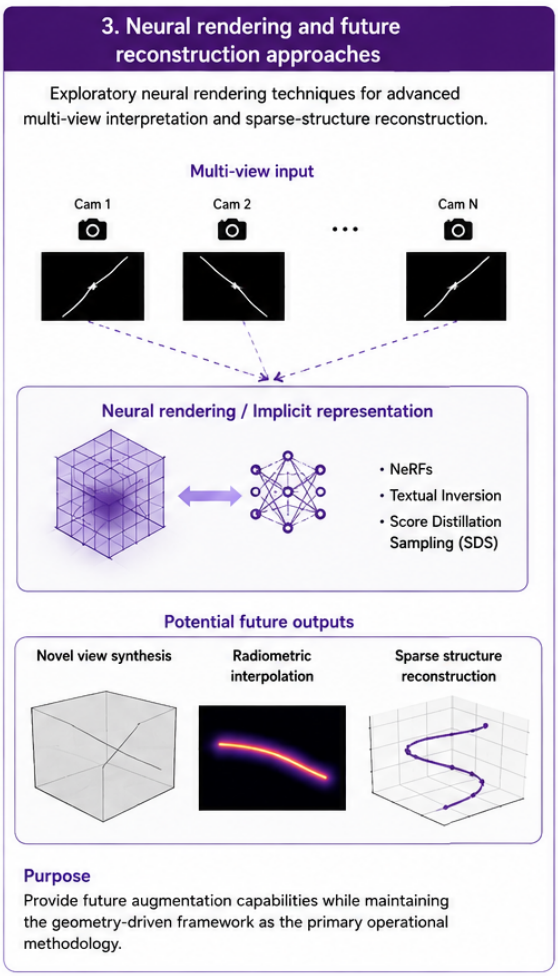


Figure 4.10: Conceptual overview that presents exploratory neural rendering and future reconstruction approaches based on implicit scene-representation techniques such as Neural Radiance Fields (NeRFs), Textual Inversion, and Score Distillation Sampling (SDS). These methods are investigated for future applications involving multi-view geometric interpretation, radiometric interpolation, sparse-structure reconstruction, and novel-view synthesis under limited observation conditions.

The resulting Phase III framework establishes a preliminary research direction for future system maturation while preserving the geometry-driven reconstruction methodology of Phase II as the principal operational strategy for reconstruction.

Chapter 5

Experimental setup and hardware configuration

This chapter describes the experimental setup and hardware configuration developed to validate the LINKU stereo vision system under controlled laboratory conditions.

The experimental framework was designed to progressively reproduce the principal optical conditions associated with orbital tether observation while maintaining controlled and repeatable image acquisition. Particular emphasis was placed on reproducing low-texture environments, dark backgrounds, directional high-contrast illumination, and sub-pixel tether visibility conditions.

Unlike conventional terrestrial stereo vision validation platforms, the proposed setup was specifically adapted for the observation of thin tether-like structures operating under conditions where environmental texture becomes limited and radiometric effects increasingly dominate the apparent image response.

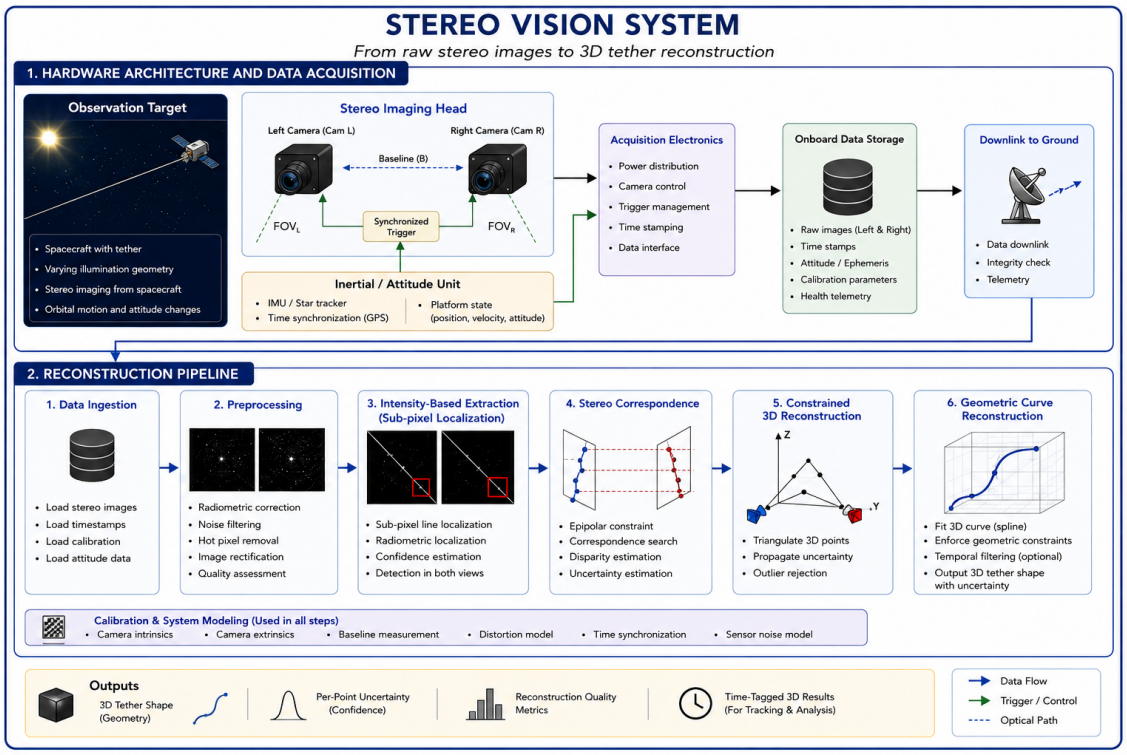


Figure 5.1: Stereo vision system architecture and reconstruction pipeline.

The experimental configuration was organized to support two principal validation objectives. First, it provides a controlled baseline for evaluating conventional feature-based photogrammetry under progressively constrained visual conditions. Second, it enables controlled evaluation of the geometry-driven stereo reconstruction stages introduced in Phase II under orbital-like imaging conditions. Figure 5.1 provides an overview of the LINKU stereo vision system considered throughout this chapter. The diagram summarizes the hardware architecture responsible for stereo image acquisition, the geometry-driven reconstruction pipeline, and the principal outputs generated during tether reconstruction. This system-level representation serves as a reference framework for the experimental validation activities described in the following sections.

Accordingly, the chapter is organized into two principal blocks:

5.1 Hardware architecture and data acquisition

The imaging system consists of a stereo pair of synchronized image sensors interfaced with a centralized processing and acquisition platform. The architecture was designed to emulate the operational principles of the proposed flight stereo vision system while maintaining sufficient flexibility for laboratory characterization and optical testing.

Particular attention was placed on optical repeatability, geometric stability, synchronization consistency, and configurable imaging conditions.

The hardware platform enables controlled evaluation of stereo correspondence behavior, disparity generation, sub-pixel tether visibility, and geometric reconstruction stability under progressively constrained orbital-like optical conditions.

5.1.1 Camera sensors and optics

The stereo imaging system was implemented using two Raspberry Pi High Quality Cameras based on the Sony IMX477 image sensor. The sensor provides a native resolution of 4056×3040 pixels (12.3 MP), a pixel pitch of 1.55 μm, and a sensor width of 6.287 mm. These characteristics provide sufficient spatial sampling capability for the tether observation scenarios analyzed in Chapter 3 while remaining representative of the proposed flight architecture.

The prototype configuration was intentionally selected to remain consistent with the final system design. Although early conceptual studies considered substantially higher-resolution sensors, the flight baseline converged toward a 12 MP-class architecture to balance image resolution, data volume, processing requirements, and robustness against potential radiation-induced degradation effects. Consequently, the IMX477 provides a representative platform for validating the near-field and far-field observation strategies proposed for the LINKU stereo vision system.

The use of interchangeable C-mount optics enabled the evaluation of multiple focal-length configurations during the optical trade-off analyses presented in Chapter 3. Based on these analyses, an $8mm$ focal-length configuration was selected for the experimental validation campaign. This configuration was found to provide an appropriate compromise between field-of-view coverage and spatial sampling performance for tether observation.

Each sensor was equipped with an Arducam C1508ZM04 C-mount lens with a focal length of $8mm$ and a manually adjustable aperture ranging from f/1.6 to f/16. The resulting field of view on the IMX477 sensor is approximately 42.9° in the horizontal direction and 32.8° in the vertical direction, providing sufficient angular coverage throughout the expected tether deployment region.

These values are obtained from the sensor geometry, $\mathrm{FOV} = 2\arctan\left(W/2f\right)$, using the IMX477 active area of 6.287×4.712 mm and the 8 mm focal length, and they agree with the 44° horizontal figure the lens manufacturer publishes for this lens on a $1/2.3''$ IMX477. The lens specification also quotes $58.4° \times 44.6°$; those figures apply to the lens's own $2/3''$ optical format, which is larger than the IMX477, and must not be used for this camera. An earlier revision of this report quoted them, overstating the horizontal field of view by about 36%.

5.1.2 Stereo rig and mounting configuration

The stereo imaging system was implemented using a dual-camera configuration mounted on a custom support structure, as shown in Figure 5.2. The cameras were separated by a fixed baseline of 100 mm, corresponding to the geometric constraints imposed by the available mounting envelope of the proposed satellite platform. This baseline provides a compromise between stereo depth sensitivity, triangulation accuracy, and mechanical integration feasibility.

The camera support was fabricated using fused deposition modeling (FDM) additive manufacturing and designed to maintain parallel optical alignment while minimizing relative sensor motion and vibration during image acquisition. This configuration ensures consistent stereo geometry and stable epipolar constraints throughout the experimental campaigns.

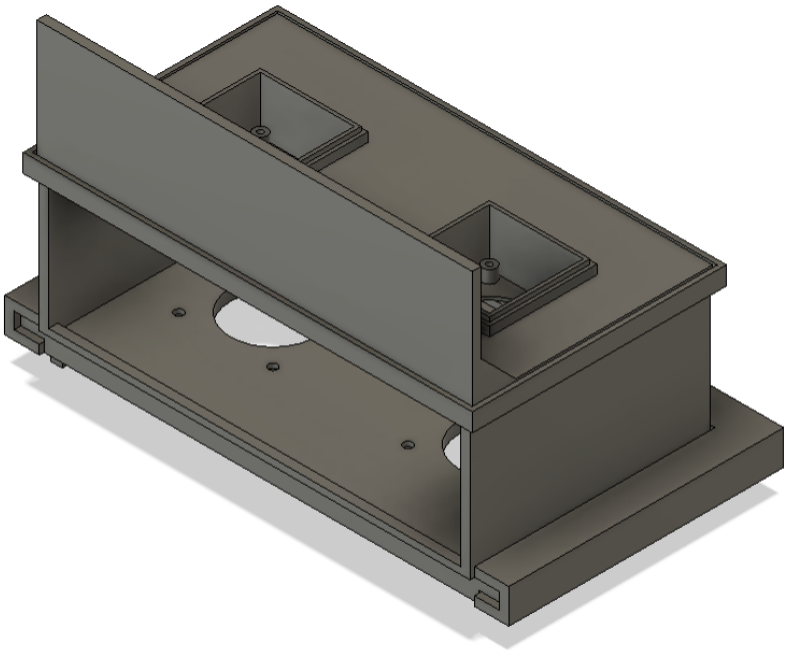


Figure 5.2: Experimental stereo rig configuration with fixed 100 mm baseline and rigid dual-camera mounting structure used for stereo image acquisition during the laboratory validation campaign.

5.1.3 Processing unit and synchronization

Stereo image acquisition, synchronization, and data management were performed using a Raspberry Pi Compute Module 5 (CM5). The platform was selected due to its support for dual MIPI CSI-2 camera interfaces, high data throughput, compact architecture, and compatibility with the stereo imaging hardware described in Section 5.1.1.

The CM5 is based on a 64-bit quad-core Arm Cortex-A76 processor operating at 2.4 GHz and provides up to 16 GB of LPDDR4X memory, enabling simultaneous acquisition and management of high-resolution stereo image streams generated by the dual IMX477 camera system as illustrated in Figure 5.3 The platform also provides an architecture conceptually consistent with the distributed processing constraints anticipated for a future flight implementation.

Because dedicated hardware triggering interfaces are not directly exposed through the standard development configuration, image synchronization was implemented through software-based acquisition control using the libcamera framework. The adopted synchronization strategy relies on low-latency sequential triggering between both cameras.

For the quasi-static tether dynamics considered during laboratory testing, with deployment velocities below approximately 5 cm/s, the resulting inter-frame delay remains sufficiently small to

preserve stereo correspondence and maintain geometric consistency during reconstruction.

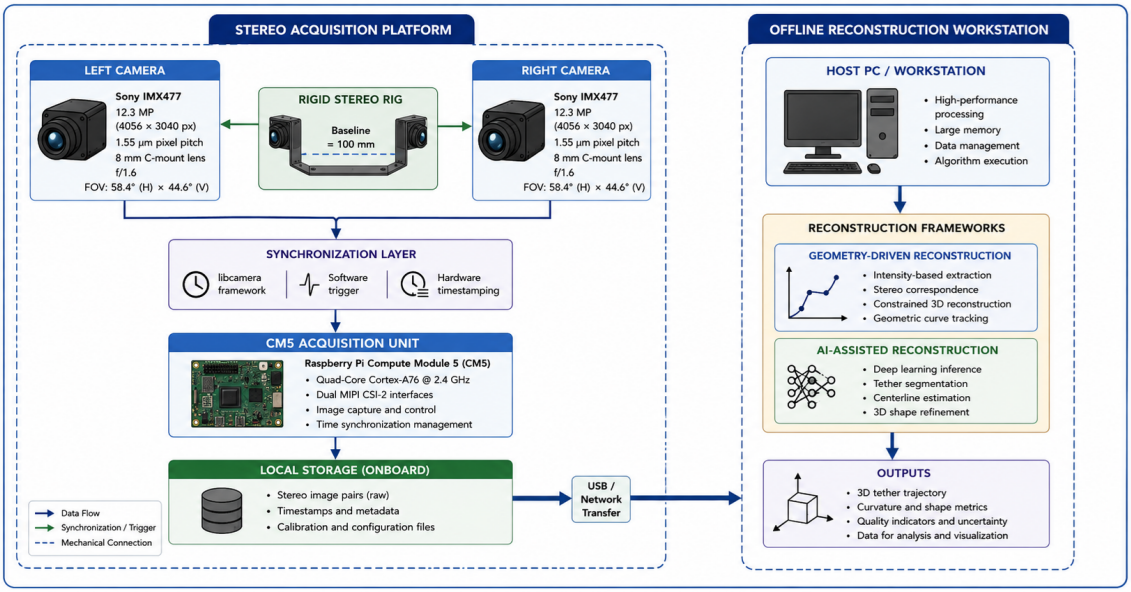


Figure 5.3: Hardware architecture and data acquisition flow of the LINKU stereo vision system. Dual IMX477 cameras acquire synchronized stereo imagery through the CM5 acquisition platform, which manages image capture, timestamping, and data transfer to the ground processing workstation.

5.2 Experimental validation environment

The experimental validation environment was developed to reproduce, under controlled laboratory conditions, the principal optical characteristics associated with orbital tether observation. The objective was not to replicate the complete space environment, but rather to emulate the visual conditions that directly influence tether detectability and stereo reconstruction performance.

The experimental setup was therefore designed around the optical constraints identified in Chapter 3, including low-texture backgrounds, high radiometric contrast, directional illumination, and sub-pixel tether visibility. Particular attention was given to controlling illumination conditions and background radiance in order to isolate the effects of geometric sampling and radiometric response on the reconstruction process.

The resulting validation environment provides a repeatable platform for evaluating both conventional feature-based photogrammetry and the geometry-driven reconstruction methodology introduced in Chapter 4. By progressively reducing the availability of environmental texture while maintaining controlled imaging conditions, the setup enables systematic assessment of the transition from conventional image-based reconstruction toward radiometric and geometry-driven tether observation strategies.

5.2.1 Validation philosophy and test scenarios

The experimental validation strategy was organized around two complementary observation scenarios designed to evaluate reconstruction performance under progressively constrained visual conditions.

The first scenario represents a terrestrial baseline environment in which conventional photogrammetry algorithms operate under favorable conditions. A volumetric target with clearly observable geometric features is imaged in the presence of environmental texture, enabling reliable feature extraction, stereo correspondence, and three-dimensional reconstruction. This scenario establishes a reference performance level for conventional image-based reconstruction techniques.

The second scenario represents an orbital-like observation environment specifically designed for

tether detection and reconstruction. In this configuration, environmental texture is intentionally minimized, background radiance is suppressed, and directional illumination is introduced to reproduce the high-contrast imaging conditions expected during orbital tether observation. Under these conditions, the target progressively transitions from a conventionally resolvable geometric object toward a radiometrically detectable structure, requiring reconstruction methods based on intensity localization, geometric continuity, and constrained stereo correspondence.

Together, these two scenarios provide a controlled framework for evaluating the limitations of conventional feature-based photogrammetry and the applicability of geometry-driven reconstruction methods under conditions representative of the LINKU mission concept.

5.2.2 Baseline terrestrial validation scenario

The baseline terrestrial validation scenario was established to provide a reference environment in which conventional feature-based photogrammetry methods could operate under favorable visual conditions. This configuration verifies the nominal performance of stereo reconstruction algorithms before introducing the optical constraints associated with orbital tether observation.

A volumetric object with well-defined geometric features was selected as the observation target. Unlike the tether considered in the LINKU mission, the object occupied a significant number of image pixels and contained sufficient visual texture to support reliable feature detection and matching. This configuration enabled the generation of stable stereo correspondences and three-dimensional reconstructions using standard photogrammetric techniques as presented in Figure 5.4.

To evaluate the influence of environmental texture on reconstruction performance, two imaging conditions were considered. In the first condition, the target was placed within a conventional laboratory environment containing visible background features. This configuration provided abundant visual references for feature extraction and served as the nominal reconstruction baseline.

In the second condition, the same target was imaged against a uniform dark background with minimal environmental texture. Although the geometric characteristics of the object remained unchanged, the reduction of surrounding visual references allowed the sensitivity of conventional reconstruction methods to background texture availability to be assessed.

These experiments establish a controlled reference for interpreting the results obtained in the orbital-like observation environment described in the following sections. By maintaining the same imaging hardware while progressively reducing scene texture, the setup enables direct comparison between conventional feature-based reconstruction and the geometry-driven methodologies developed for tether observation.

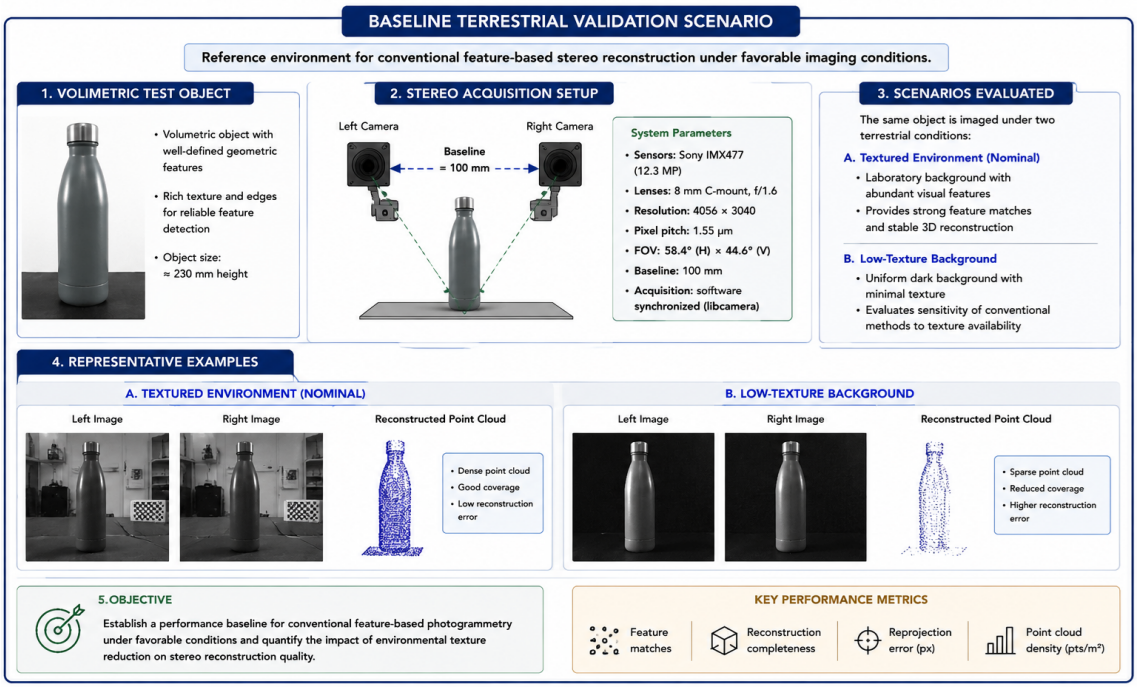


Figure 5.4: Baseline terrestrial validation scenario. A volumetric target with well-defined geometric features is observed under two controlled terrestrial conditions: a textured environment and a low-texture background. The comparison enables the evaluation of conventional feature-based stereo reconstruction performance and establishes a reference baseline against which the orbital-like tether observation scenarios can be assessed.

5.2.3 Orbital-like observation environment

The orbital-like observation environment was developed to reproduce, under controlled laboratory conditions, the principal optical characteristics associated with tether observation in orbit. Unlike the baseline terrestrial scenario described previously, this configuration intentionally minimizes environmental texture and background radiance while introducing highly directional illumination conditions.

This environment will not replicate all aspects of the space environment, but rather to reproduce the visual constraints that directly affect tether detectability and stereo reconstruction performance. Particular emphasis was placed on generating conditions representative of low-texture scenes, high radiometric contrast, and sub-pixel target observation.

The experimental configuration was designed to support the validation of the geometry-driven reconstruction methodology presented in Chapter 4. By progressively reducing the availability of conventional visual features, the setup enables evaluation of reconstruction strategies based on radiometric localization, geometric continuity, and constrained stereo correspondence.

The orbital-like environment consists of three principal elements: a light-isolated enclosure, a directional illumination system, and a tether proxy representative of the expected mission target. Together, these elements provide a controlled platform for evaluating tether observation under conditions that approximate the optical challenges associated with orbital operation.

5.2.3.1 Dark enclosure

A dedicated dark enclosure was constructed to isolate the experimental volume from ambient illumination and external visual references. The enclosure provides a controlled low-radiance environment intended to approximate the absence of background texture and stray light typically encountered during orbital observation.

The enclosure was fabricated from laser-cut medium-density fiberboard (MDF) panels and has internal dimensions of approximately $600mm \times 800mm \times 500mm$. All interior surfaces were coated with matte black paint to reduce diffuse reflections and suppress unwanted radiometric contributions from the surrounding environment.

By minimizing background brightness and environmental texture, the enclosure increases the radiometric contrast between the observed target and the surrounding scene. This condition is particularly important for the evaluation of thin tether-like structures, whose detectability depends primarily on their radiometric response rather than on conventional texture-based image features.

Figure 5.5 presents the dark enclosure used throughout the experimental validation campaign, including both the computer-aided design model and the fabricated implementation. The resulting environment provides a repeatable and controllable observation volume for the orbital-like experiments described in the following sections.

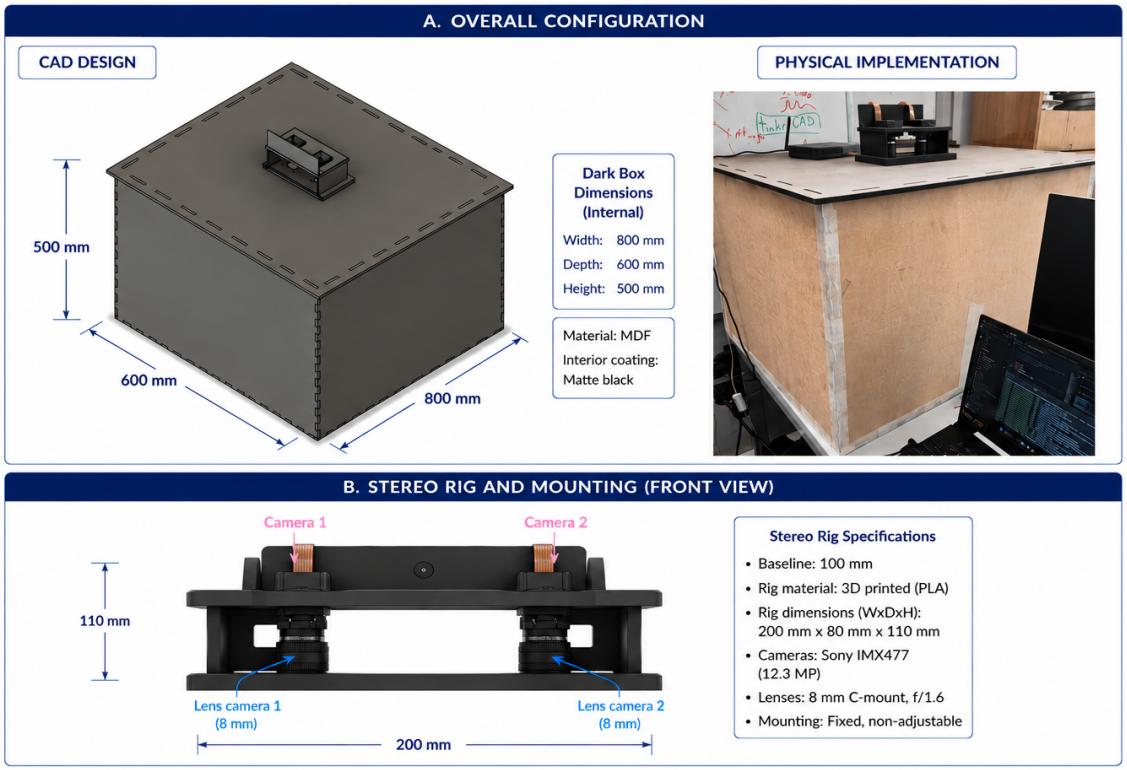


Figure 5.5: Dark enclosure and stereo vision system configuration. (A) CAD model and physical implementation of the light-tight MDF enclosure used to reproduce low-texture, high-contrast orbital-like observation conditions. The enclosure provides an internal volume of approximately $800mm \times 600mm \times 500mm$ and incorporates a matte-black interior coating to minimize reflections and background radiance. (B) Front view of the stereo rig used for image acquisition. The system consists of two Sony IMX477 cameras equipped with 8 mm C-mount lenses and mounted on a rigid structure with a fixed 100 mm stereo baseline, ensuring stable and repeatable stereo geometry during the experimental campaigns.

5.2.3.2 Directional illumination system

Reproducing the geometric characteristics of solar illumination within a laboratory-scale environment remains challenging. Under orbital conditions, the Sun behaves as an effectively collimated source with an apparent angular diameter of approximately 0.5°, producing nearly parallel rays, sharp shadow boundaries, and strong specular reflections.

For the experimental campaign described in this work, a high-intensity directional LED source was employed to generate controlled illumination inside the dark enclosure. Although the source does

not reproduce the full geometric characteristics of solar radiation, it provides sufficient radiometric contrast to evaluate tether detectability, intensity-based extraction, and centerline localization under high-contrast observation conditions.

The adopted illumination strategy was intentionally designed to emphasize radiometric visibility effects rather than absolute solar irradiance levels. This approach is consistent with the primary objective of the validation campaign, namely the assessment of geometry-driven reconstruction performance under conditions where tether detectability becomes strongly dependent on illumination geometry.

A similar challenge has been discussed by Sun et al. [10], who demonstrated that realistic solar simulation requires not only spectral matching but also controlled beam collimation, volumetric illumination uniformity, and directional consistency. Their LED-based solar simulator achieved approximately one solar constant irradiance with a beam divergence close to $\pm 3^{\circ}$, highlighting the importance of geometric illumination properties when reproducing solar observation conditions in laboratory environments.

5.2.3.3 Tether proxy and observation target

A tether proxy was employed to provide a controllable and repeatable representation of the tether structure considered in the LINKU mission concept. The proxy consisted of a thin cable with an external diameter of approximately 1 mm, selected to reproduce the small apparent dimensions and limited image occupancy characteristic of long-range tether observation.

The target was positioned within the dark enclosure and observed under the directional illumination conditions described previously. By combining a low-texture background with a thin elongated structure, the experimental configuration creates imaging conditions in which target detectability becomes increasingly dependent on radiometric response rather than on conventional texture-based image features.

Unlike the volumetric object used in the baseline terrestrial scenario, the tether proxy exhibits a highly anisotropic geometry with limited visual texture and a significantly reduced projected width within the image plane. These characteristics make the target representative of the reconstruction challenges addressed by the geometry-driven methodology developed in Chapter 4.

The tether proxy therefore serves as the primary observation target for evaluating intensity-based extraction, centerline estimation, geometric curve tracking, stereo correspondence, and constrained three-dimensional reconstruction under orbital-like observation conditions.

5.2.4 Representative experimental observations

Representative observations acquired within the orbital-like validation environment are presented to illustrate the imaging conditions obtained using the experimental setup described in the previous sections. These observations provide qualitative verification that the combination of the dark enclosure, directional illumination system, and tether proxy successfully reproduces the principal optical characteristics required for geometry-driven tether reconstruction.

Figure 5.6 shows a representative image of the 1 mm tether proxy acquired under directional illumination together with the corresponding extraction result. The observed image exhibits high radiometric contrast between the tether and the surrounding background, while maintaining minimal environmental texture within the field of view. This configuration is consistent with the orbital-like observation conditions discussed in Chapter 3.

The extracted tether response demonstrates that the illumination and background suppression strategy provide sufficient target visibility for subsequent processing stages, including intensity-based extraction, centerline estimation, geometric curve tracking, and stereo correspondence. Although the tether occupies only a small fraction of the image area, its radiometric signature remains sufficiently distinguishable to support reliable geometric localization.

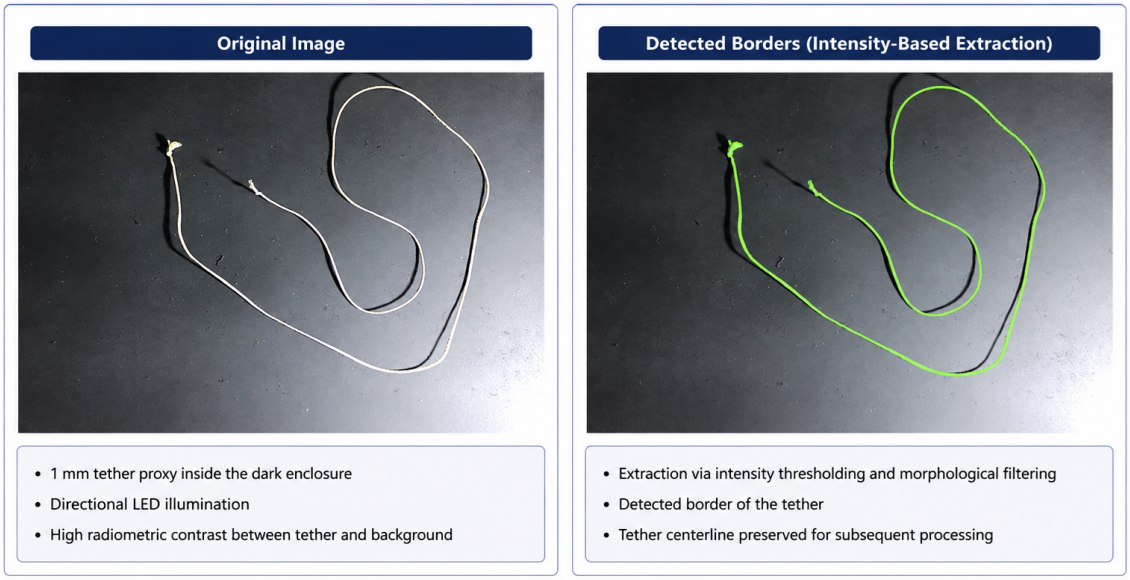


Figure 5.6: Representative tether observation and intensity-based extraction.

These observations confirm that the experimental environment provides a suitable validation platform for evaluating the geometry-driven reconstruction methodology under controlled high-contrast conditions representative of orbital tether observation.

5.3 Computational resources for offline reconstruction

The experimental validation campaign generated stereo image datasets that were processed offline using dedicated computing resources. Separating image acquisition from reconstruction processing enabled systematic evaluation of the proposed methodologies while avoiding the computational and power constraints associated with embedded flight hardware.

This approach provided full access to intermediate processing results, facilitating algorithm verification, parameter tuning, and quantitative performance assessment under the different observation scenarios described in the previous sections. Two complementary reconstruction approaches were evaluated throughout the study: a geometry-driven methodology based on radiometric and geometric constraints, and an exploratory AI-assisted reconstruction framework based on recent neural three-dimensional reconstruction techniques.

5.3.1 Offline execution of the Geometry-driven reconstruction pipeline

The geometry-driven reconstruction methodology developed in Section 4.2 was executed offline using the stereo image datasets acquired during the experimental validation campaign. This processing strategy enabled systematic evaluation of the proposed reconstruction framework while maintaining complete access to intermediate results for verification and analysis.

The complete reconstruction workflow, summarized in Figure 4.4, was applied consistently to all datasets acquired under both the baseline terrestrial scenario and the orbital-like observation environment. Using a common processing configuration ensured that reconstruction performance could be evaluated under different imaging conditions without introducing variations associated with algorithm implementation or parameter selection.

All reconstruction experiments were executed on a workstation equipped with an Intel Core i9 processor and 64 GB of DDR4 memory. The computational requirements of the geometry-driven methodology remained relatively modest, allowing complete offline execution using conventional workstation hardware without dependence on dedicated high-end GPU acceleration.

The computational environment was intentionally separated from the image acquisition platform described in Section 5.1. This modular architecture allowed the experimental setup to focus on controlled data collection while providing sufficient flexibility for algorithm development, validation, and quantitative performance assessment under the different observation scenarios investigated in this work.

5.3.2 AI-assisted reconstruction resources

Exploratory AI-assisted reconstruction experiments were conducted to evaluate the potential applicability of recent neural three-dimensional reconstruction techniques to tether observation problems. Unlike the geometry-driven methodology, these approaches rely on learned scene representations and generative models that require greater computational resources.

The AI-assisted experiments were executed using the same workstation platform employed for the geometry-driven reconstruction tasks, incorporating an Intel Core i9 processor, 64 GB of DDR4 memory, and an NVIDIA RTX 2070 GPU. For reconstruction models whose memory requirements exceeded the available local GPU resources, cloud-based execution was performed using Google Colab Pro environments equipped with NVIDIA V100 and A100 accelerators.

The resulting computational requirements were higher than those associated with the geometry-driven reconstruction pipeline. While the geometry-driven methodology could be executed efficiently on conventional workstation hardware and remains compatible with future embedded implementations, the AI-assisted approaches required dedicated GPU acceleration and, in some cases, access to high-performance cloud computing resources.

These observations reinforce the role of geometry-driven reconstruction as the primary operational strategy for the LINKU stereo vision system, while AI-assisted methods remain exploratory tools intended to support future research into reconstruction under visually ambiguous observation conditions.

Chapter 6

Experimental results and discussion

This chapter presents the experimental results from the validation campaign for the LINKU stereo vision system. The analysis is organized to progressively evaluate the limitations of conventional photogrammetric approaches, the performance of the proposed geometry-driven reconstruction methodology, and the potential role of AI-assisted techniques for future system evolution.

The experimental results are analyzed with respect to the optical constraints identified in Chapter 3, the geometry-driven and AI-assisted reconstruction methodologies introduced in Chapter 4, and the validation platform and orbital-like observation environment described in Chapter 5.

Particular emphasis is placed on assessing the system's ability to detect, localize, and reconstruct thin tether-like structures under low-texture and high-contrast observation conditions. These conditions are representative of the visual challenges expected during orbital tether deployment and post-deployment characterization phases.

The chapter first examines the performance of conventional photogrammetry under space-emulated conditions to establish a reference baseline. The results are then compared with those obtained using the proposed geometry-driven reconstruction framework, followed by an assessment of exploratory AI-assisted reconstruction approaches. Together, these analyses provide a comprehensive evaluation of the strengths, limitations, and operational suitability of the proposed stereo vision system.

6.1 Validation objectives and experimental datasets

The experimental campaign was designed to evaluate the feasibility of reconstructing tether geometry under conditions representative of orbital observation.

The validation process focused on three principal objectives:

1. Assessing the limitations of conventional feature-based photogrammetry.
2. Validating the geometry-driven reconstruction methodology proposed in Chapter 4.
3. Investigating the potential applicability of AI-assisted reconstruction techniques.

Two experimental datasets were employed throughout the validation campaign. The first consisted of a volumetric reference object (Figure 5.3), while the second utilized a 1 mm tether proxy representative of the geometry considered in the LINKU mission concept (Figure 5.5). These datasets were used to evaluate both conventional photogrammetric techniques and the geometry-driven reconstruction methodology under controlled laboratory conditions.

The tether reconstruction experiments were evaluated under two representative geometric configurations.

- Scenario A corresponds to a predominantly planar configuration in which the tether remained approximately parallel to the image plane, resulting in limited depth variation.
- Scenario B introduces significant out-of-plane deformation and depth variation, creating a more challenging reconstruction problem that better represents realistic deployment dynamics.

All datasets were acquired using the calibrated stereo imaging platform described in Chapter 5 and processed using identical calibration parameters to ensure consistency throughout the experimental campaign.

6.2 Evaluation of conventional photogrammetry

Before developing the geometry-driven reconstruction methodology, a series of experiments were conducted to evaluate the suitability of conventional photogrammetric techniques for tether observation under orbital-like conditions. These tests were intended to establish a reference baseline and identify the principal failure modes associated with feature-based stereo reconstruction in low-texture environments.

The evaluation focused on the experimental targets introduced in Chapter 5. The volumetric reference object (Figure 5.3) was used to assess the behavior of conventional dense stereo reconstruction under reduced-texture conditions, while the tether proxy (Figure 5.5) was employed to evaluate the applicability of commercial Structure-from-Motion (SfM) techniques to thin tether-like geometries.

The results demonstrate that although conventional photogrammetry performs adequately when sufficient texture is available, its performance rapidly degrades as environmental features disappear. These observations provide direct experimental evidence supporting the optical limitations discussed in Chapter 3 and motivate the development of the geometry-driven reconstruction strategy presented in Chapter 4.

6.2.1 Volumetric object reconstruction under low-texture conditions

A preliminary validation experiment was performed using the volumetric reference object introduced in Figure 5.3. The objective was to evaluate the behavior of conventional dense stereo reconstruction when operating inside the low-texture environment provided by the dark enclosure described in Section 5.2.3.1.

The reconstruction was generated using a standard dense stereo workflow based on disparity estimation and three-dimensional point triangulation. Although the object remained clearly visible in the stereo imagery, the absence of surrounding visual features significantly degraded the reliability of the matching process.

Figure 6.1 illustrates the origin of the reconstruction failure. The disparity map in Figure 6.1(a) correctly identifies the volumetric target through its high disparity values, which correspond to the shortest camera-to-object distances. However, non-physical disparity responses also appear throughout regions that should correspond to empty background space. These erroneous matches propagate into the reconstructed coordinate maps shown in Figures 6.1(b) and 6.1(c), where spatially inconsistent points emerge outside the physical boundaries of the object.

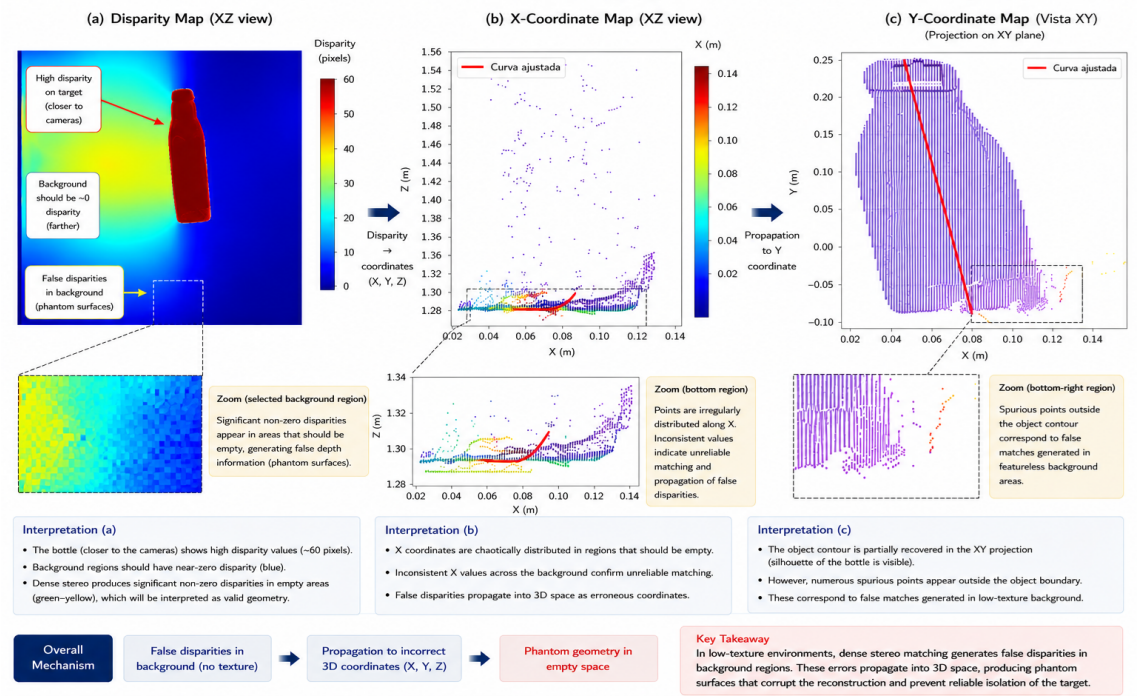


Figure 6.1: Failure mechanism of conventional dense stereo reconstruction under Low-texture conditions.

The resulting disparity artifacts generate phantom geometric structures that do not correspond to any real surface within the scene. Although portions of the target geometry remain identifiable, the presence of false correspondences introduces substantial ambiguity into the three-dimensional reconstruction process.

The consequences of these errors are presented in Figure 6.2. The raw point cloud contains numerous outliers and phantom surfaces due to incorrect disparity estimates in low-texture regions.

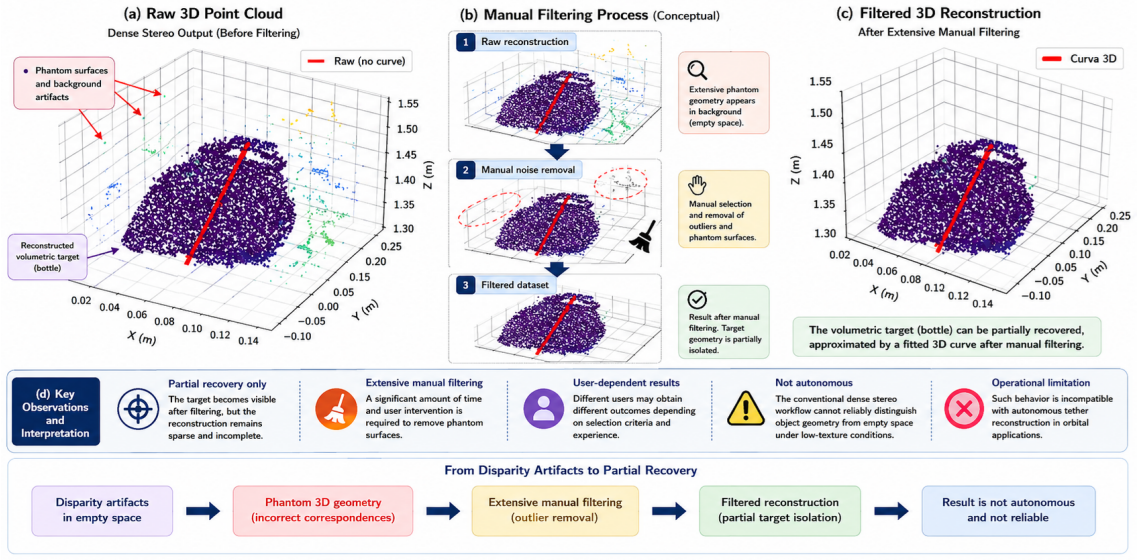


Figure 6.2: Filtered three-dimensional reconstruction after manual removal of disparity artifacts. The figure shows consequences of disparity estimation error on the final 3D reconstruction, which is the manual filtering for target isolation.

These results demonstrate that conventional dense stereo reconstruction remains strongly dependent on environmental texture and visual context. While the target object can still be partially

recovered, the reconstruction process requires significant user intervention and is therefore not autonomous. Such behavior would be incompatible with operational tether-monitoring scenarios, where robust, automatic reconstruction must be achieved under similarly low-texture orbital observation conditions.

6.2.2 Tether reconstruction against textured and texturedless background conditions

To further investigate the limitations of conventional photogrammetric reconstruction, the tether proxy introduced in Figure 5.5 was processed using a commercial Structure-from-Motion (SfM) workflow. Unlike the volumetric reconstruction experiment presented in Section 6.2.1, the objective of this test was to evaluate whether a thin tether-like structure could be reconstructed under conditions representative of the LINKU mission scenario.

Figure 6.3 presents the reconstruction obtained when the tether was observed within a textured environment containing multiple visual features, including floor patterns, object edges, and surrounding structures. Under these conditions, the SfM pipeline successfully detected image features, established reliable correspondences, aligned the camera poses, and generated a valid three-dimensional reconstruction. However, inspection of the reconstructed scene reveals that the majority of the recovered geometry originates from the surrounding environment rather than from the tether itself. The reconstruction process therefore remains strongly dependent on environmental texture.

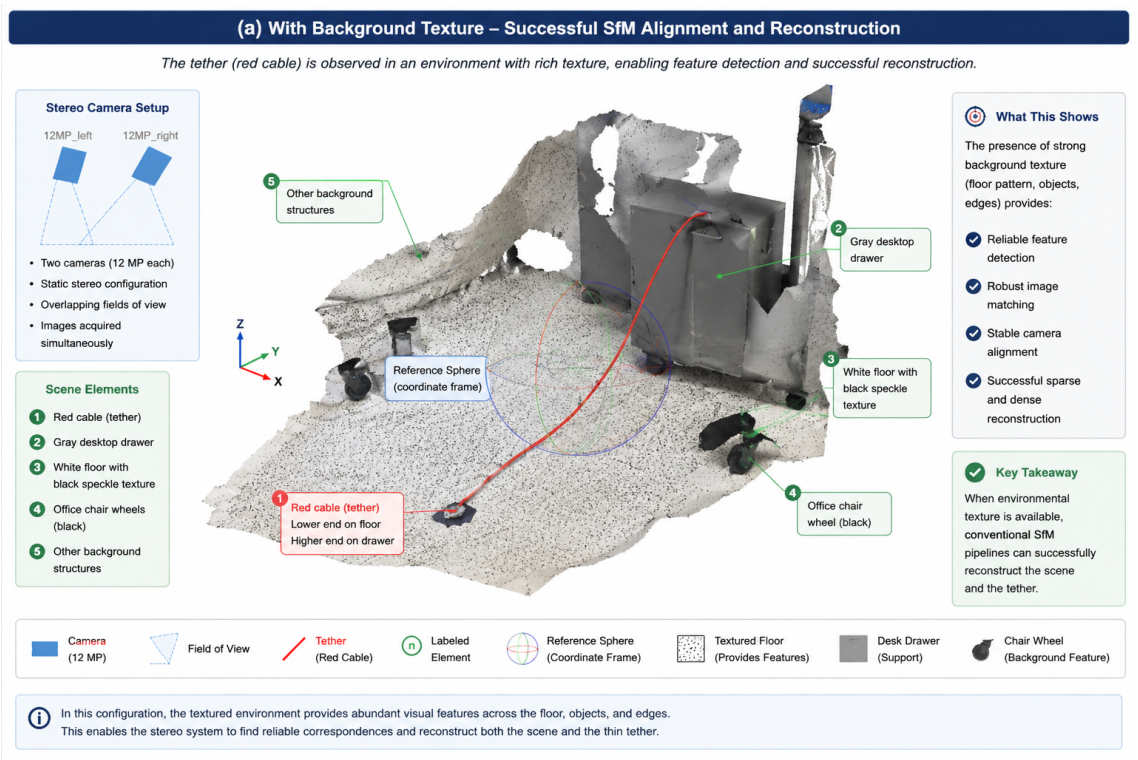


Figure 6.3: Structure-from-Motion reconstruction supported by environmental texture features.

To evaluate the intrinsic reconstructability of the tether, a second experiment was performed in which the background was intentionally removed through image segmentation prior to processing. The resulting stereo pair retained only the isolated tether masks, eliminating the influence of environmental texture and ensuring that the reconstruction process relied exclusively on the visual information contained within the tether itself.

The outcome of this experiment is shown in Figure 6.4. Despite the successful isolation of the tether, the SfM pipeline failed to generate a valid reconstruction and returned a "Zero Resolution" error. No reliable camera alignment, sparse point cloud, or three-dimensional model could be ob-

tained. This failure occurred even though the background had already been removed, eliminating the possibility that environmental structures or disparity artifacts could influence the reconstruction process.

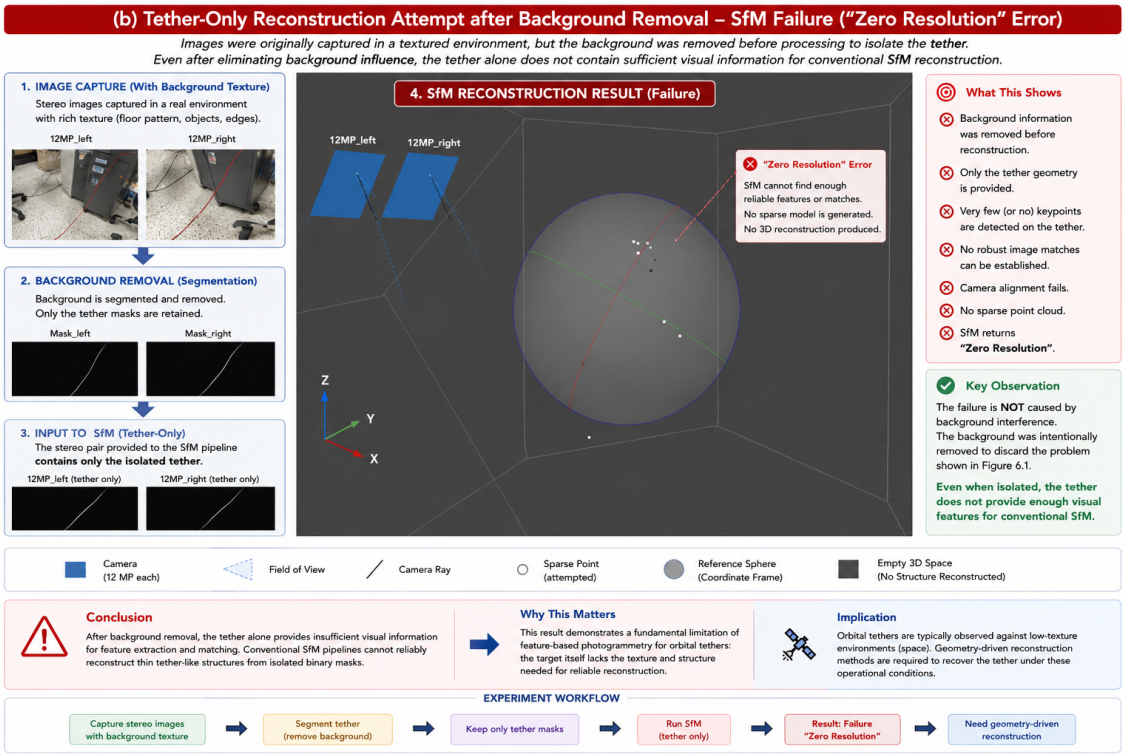


Figure 6.4: Tether-only reconstruction attempt resulting in a "Zero resolution" failure.

The results demonstrate that the primary limitation is not background interference but insufficient visual information within the tether itself. As a thin, elongated structure occupying only a small number of pixels in each image, the tether provides very few distinctive keypoints for conventional feature-extraction and matching algorithms. Consequently, the correspondence stage becomes underconstrained, preventing robust image alignment and subsequent three-dimensional reconstruction.

The comparison between Figures 6.3 and 6.4 provides direct experimental evidence of a fundamental limitation of conventional feature-based photogrammetry. While SfM can successfully reconstruct scenes with abundant environmental texture, it cannot reliably reconstruct an isolated tether even after perfect segmentation. Therefore, the reconstruction challenge is not merely a segmentation problem but a structural limitation of feature-based approaches when applied to thin, low-texture objects.

These observations directly motivate the geometry-driven reconstruction methodology introduced in Chapter 3. Rather than relying on sparse image features, the proposed approach exploits radiometric localization, centerline estimation, geometric continuity constraints, and stereo correspondence strategies specifically designed for thin tether-like structures operating under low-texture orbital observation conditions.

6.2.3 Implications for orbital tether observation

The results presented in Sections 6.2.1 and 6.2.2 reveal two complementary limitations of conventional photogrammetric reconstruction when applied to orbital tether observation.

The dense stereo reconstruction experiment demonstrated that disparity-based methods remain highly sensitive to the absence of environmental texture. Although portions of the target geometry could be recovered, substantial phantom structures and false correspondences were generated

throughout regions that physically correspond to empty space. The resulting reconstructions required extensive manual filtering and could not be considered operationally autonomous.

The Structure-from-Motion evaluation exposed an even more fundamental limitation. In the case of the tether proxy, the reconstruction process failed because the target itself did not provide sufficient visual features to support reliable feature extraction and image matching. As environmental texture was progressively removed, the reconstruction quality degraded until complete failure occurred.

These observations are consistent with the optical analysis presented in Chapter 3. Under orbital observation conditions, tether visibility is governed not only by geometric sampling but also by radiometric contrast, directional illumination, and sub-pixel detectability effects. Consequently, the tether cannot be treated as a conventional textured object suitable for feature-based photogrammetry across a significant portion of the operational range.

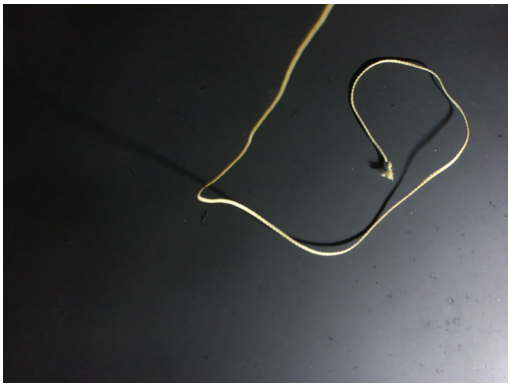
The experimental evidence therefore motivates the transition toward the geometry-driven reconstruction methodology introduced in Chapter 4. Rather than relying on sparse image features, the proposed approach exploits radiometric signatures, geometric continuity, centerline estimation, and stereo correspondence constraints specifically adapted to thin tether-like structures. This strategy directly addresses the limitations observed in conventional photogrammetric reconstruction and provides the foundation for the results presented in the following sections.

6.3 Geometry-driven reconstruction results

The geometry-driven reconstruction methodology introduced in Chapter 4 was evaluated using the tether datasets described in Section 6.1, after establishing the limitations of conventional photogrammetric methods in Section 6.2. The objective was to determine whether geometric and radiometric constraints could provide reliable tether reconstruction under low-texture conditions where conventional feature-based approaches failed.

Two representative tether configurations were analyzed, as shown in Figures 6.5a and 6.5b. Scenario A corresponds to a predominantly planar configuration, in which the tether is arranged approximately parallel to the image plane and exhibits limited depth variation. Scenario B introduces stronger out-of-plane deformation, looped geometry, and depth variation, thereby representing a more demanding reconstruction case.

(a) Scenario A - Tether proxy positioned on the XY plane inside the dark enclosure.



(b) Scenario B - Tether proxy with one end elevated relative to the XY plane inside the dark enclosure.

Figure 6.5: Experimental Scenarios A and B for validating the geometry-driven tether reconstruction process.

For clarity, the results are presented by experimental scenario. Each scenario is analyzed through the same reconstruction sequence: tether extraction, stereo correspondence establishment, and three-dimensional reconstruction. This organization allows the complete reconstruction process to be evaluated first for the simpler planar case and then for the more complex volumetric case.

6.3.1 Reconstruction Workflow for Scenario A

Scenario A was used to evaluate the performance of the geometry-driven reconstruction framework under a predominantly planar tether configuration. In this case, the expected depth variation is limited, and the stereo displacement between corresponding tether points should remain relatively uniform.

The first processing stage consisted of extracting the tether from the stereo image pair. As shown in Figure 6.6, the tether was successfully isolated as a continuous binary trace in both the left and right images. The extracted curves preserve the principal topology of the tether and provide a clean geometric representation for subsequent correspondence analysis.

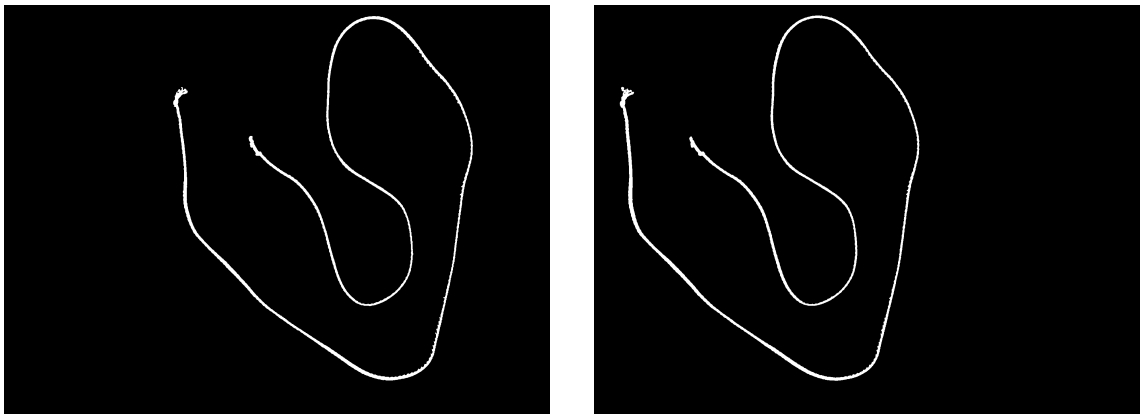


Figure 6.6: (Left) left camera image, (Right) right camera image. Tether extraction results for Scenario A (planar geometry). The binary traces successfully isolate the thin structure against the black background with a relatively uniform horizontal shift.

After extraction, stereo correspondence was established between the left and right tether traces. Figure 6.7 shows that the matching lines are approximately parallel and relatively uniform, which is consistent with the limited depth variation of the planar configuration.

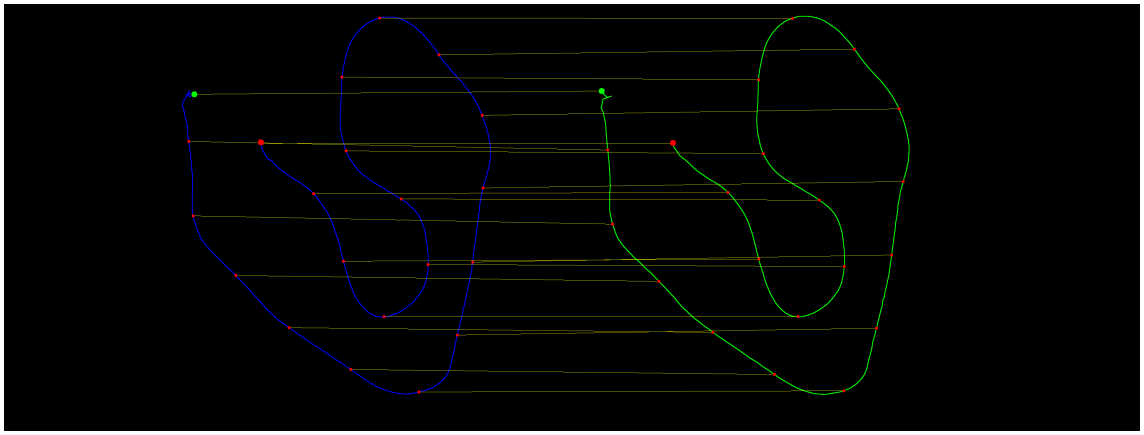


Figure 6.7: Stereo correspondence results for Scenario A. The parallel and uniform matching lines indicate consistent depth across the planar geometry.

The matched tether points were then triangulated to recover the three-dimensional tether trajectory. Figure 6.8 presents the final 3D reconstruction obtained for Scenario A. The reconstructed geometry preserves the overall shape of the tether and confirms that the proposed method can recover spatially coherent tether trajectories without relying on environmental texture or conventional feature matching.

The complete reconstruction sequence for Scenario A is summarized in Figure 6.9, which integrates the stereo image pair, extracted traces, correspondence results, and final 3D representation.

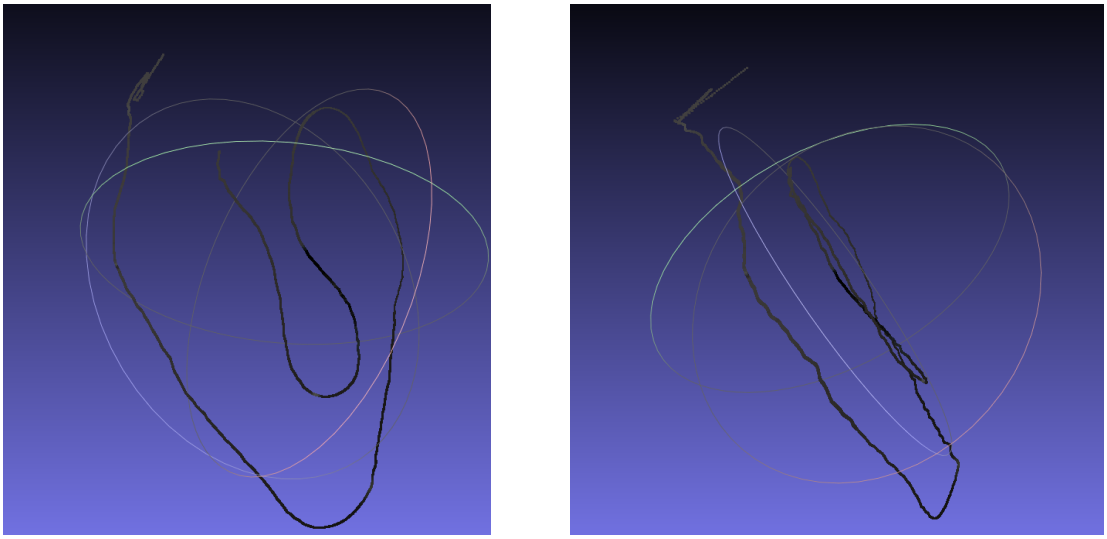


Figure 6.8: Final three-dimensional tether reconstruction for Scenario A. The spatial coherence confirms successful triangulation of a relatively flat geometry.

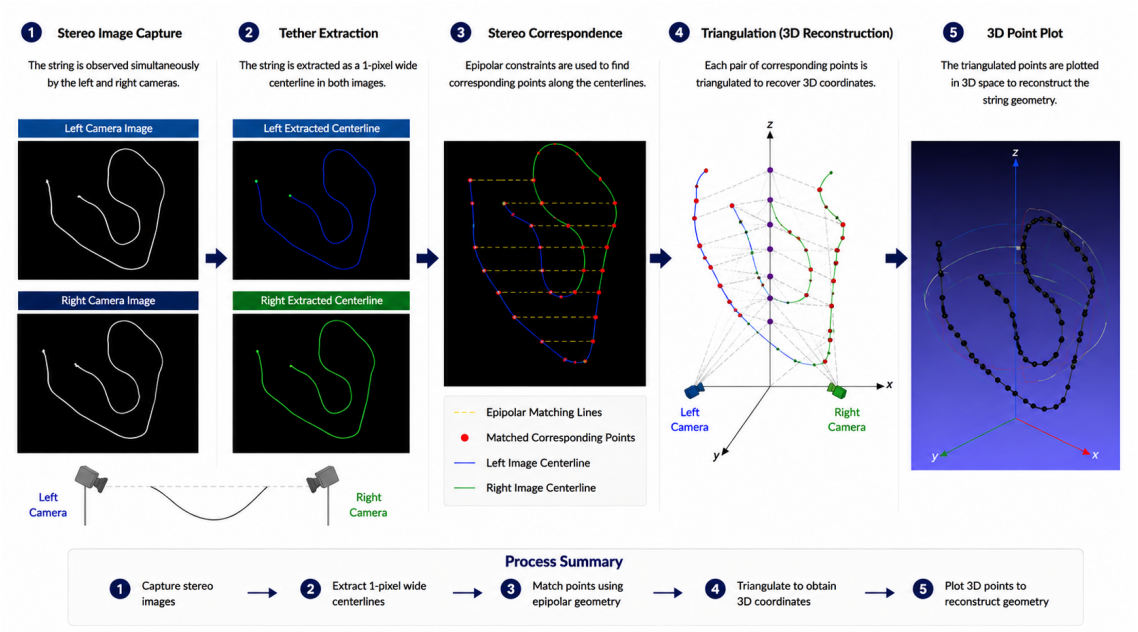


Figure 6.9: Geometry-driven Stereo reconstruction workflow from image capture to three-dimensional tether representation in scenario A.

6.3.2 Reconstruction Workflow for Scenario B

Scenario B was used to evaluate the robustness of the geometry-driven reconstruction framework under a more complex tether configuration. Unlike Scenario A, this case includes out-of-plane deformation, perspective foreshortening, and stronger depth variation along the tether.

The first processing stage again consisted of tether extraction from the stereo image pair. As shown in Figure 6.10, the method was able to isolate continuous tether traces in both camera views despite the increased geometric complexity. The extracted curves preserve the main shape of the tether, including looped and deformed regions.

Stereo correspondence was then established between the left and right extracted traces. Figure 6.11 shows that the matching lines exhibit more variable lengths than in Scenario A. This behavior is consistent with the stronger depth variation present in the volumetric configuration.

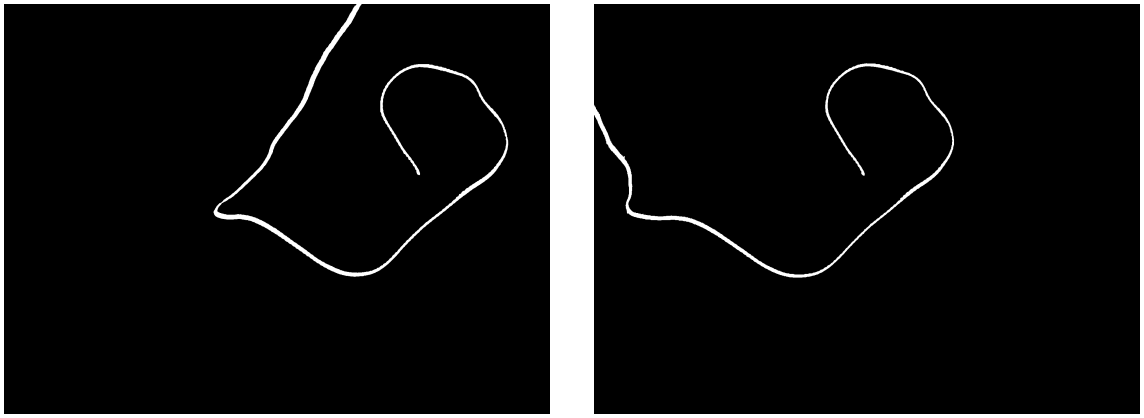


Figure 6.10: (Left) left camera image, (Right) right camera image. Tether extraction results for Scenario B (volumetric geometry). The algorithm maintains continuous trace extraction despite the complex loops and perspective foreshortening caused by the tether extending toward the camera lenses.

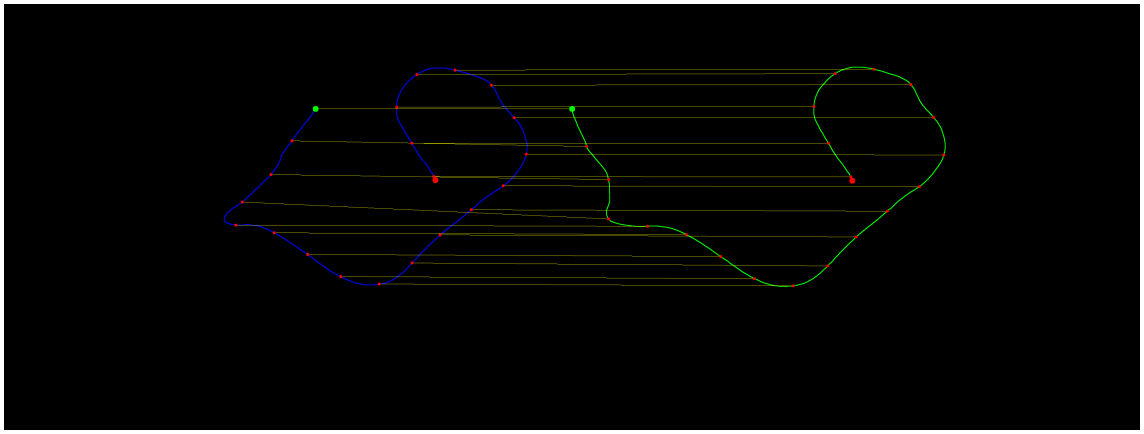


Figure 6.11: Stereo correspondence for Scenario B. The noticeably varying lengths of the horizontal matching lines explicitly illustrate the variation in depth along the continuous curve.

The matched points were triangulated to recover the three-dimensional tether geometry. Figure 6.12 presents the final 3D reconstruction obtained for Scenario B. The result demonstrates that the proposed methodology can recover out-of-plane tether deformation and depth variation under low-texture observation conditions.

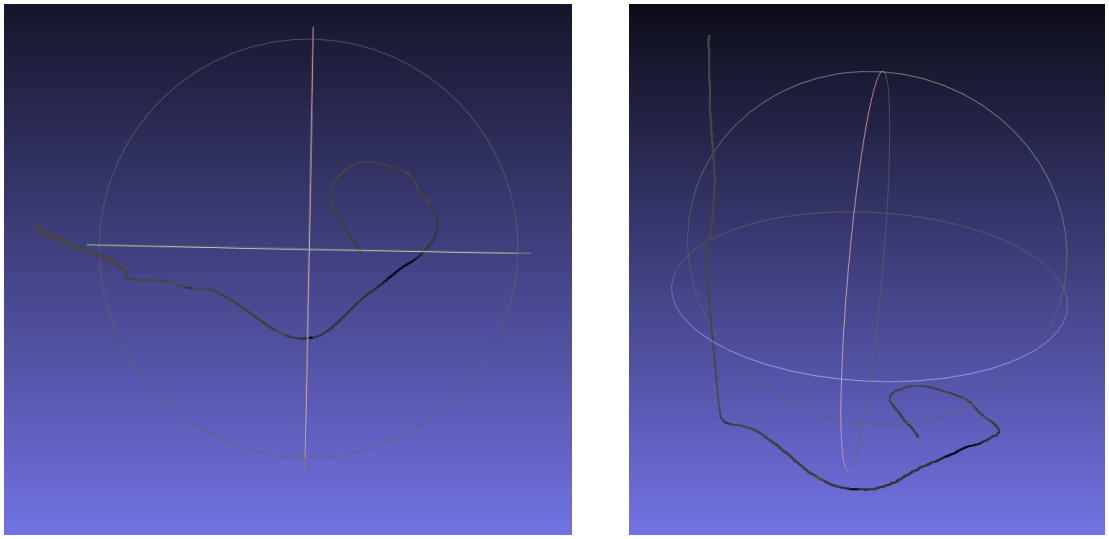


Figure 6.12: Final three-dimensional tether reconstruction for scenario B. The rotated perspective on the right side explicitly demonstrates the successful recovery of Z-axis depth, confirming the system's capability to monitor complex tether deformations.

The complete reconstruction sequence for Scenario B is summarized in Figure 6.13, which shows the progression from stereo image acquisition to the final three-dimensional tether representation.

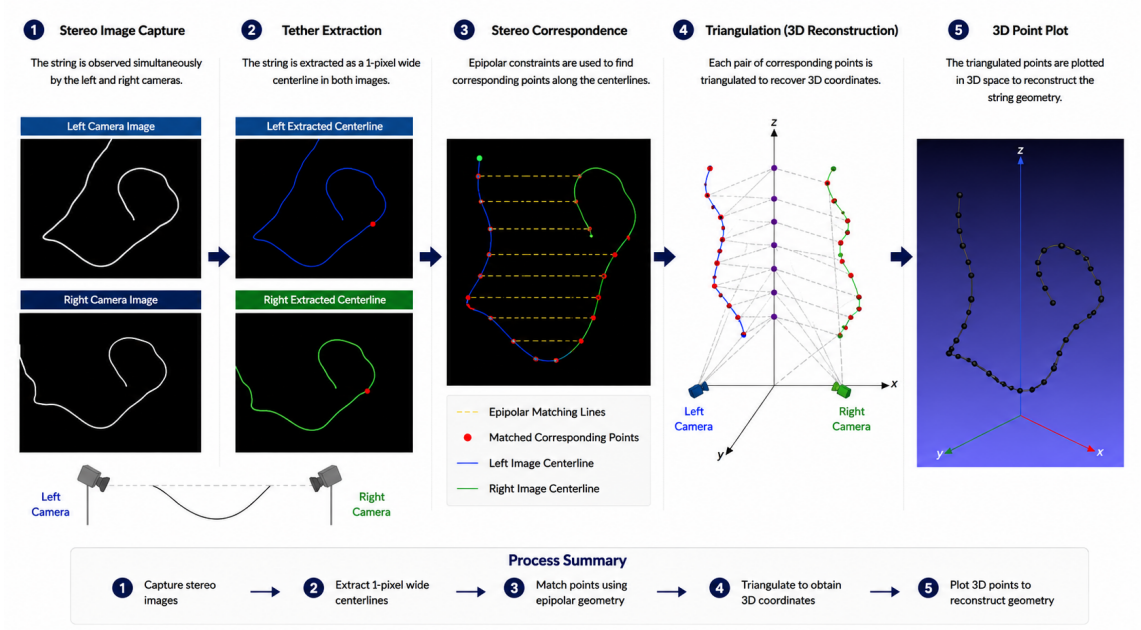


Figure 6.13: Geometry-driven Stereo reconstruction workflow from image capture to three-dimensional tether representation in scenario B.

6.3.3 Comparative assessment of scenarios A and B

The results obtained for both scenarios demonstrate that the geometry-driven reconstruction framework can operate under different tether configurations. Scenario A validated the method under a simpler planar condition, where the stereo disparity remains relatively uniform and the reconstructed tether geometry exhibits limited depth variation.

Scenario B provided a more demanding test case, introducing stronger depth variation, perspective effects, and out-of-plane deformation. The successful reconstruction of this configuration demonstrates that the method is not limited to planar tether geometries and can recover more

representative three-dimensional shapes.

The main difference between both cases is observed in the stereo correspondence stage. Scenario A produced more uniform matching patterns, while Scenario B produced variable correspondence lengths associated with depth changes along the tether. This behavior confirms that the reconstruction pipeline is sensitive to geometric depth variation and can translate stereo disparity differences into three-dimensional tether structure.

6.3.4 Topological crossing analysis

Although the geometry-driven framework successfully reconstructed both tether configurations, complex topological conditions remain an important challenge. In particular, crossings, overlaps, and self-intersections may introduce ambiguity when multiple tether segments project onto nearby or overlapping image regions.

The results indicate that these ambiguities do not prevent the reconstruction of the overall tether trajectory, but they can affect local continuity interpretation in highly entangled configurations. This limitation motivates future improvements based on temporal tracking, disparity consistency, and topology-aware processing.

These results contrast directly with the failures reported in Section 6.2. While conventional photogrammetric approaches were unable to reconstruct the tether reliably under low-texture conditions, the geometry-driven methodology produced stable and physically plausible reconstructions for both planar and volumetric tether configurations.

6.4 Assessment of AI-assisted reconstruction

This section does not report an operational reconstruction result; it summarizes an exploratory assessment that was conducted to investigate the potential of AI-based techniques for tether observation. The objective was not to develop an operational reconstruction system but to evaluate whether recent advances in neural reconstruction could inform future versions of the LINKU architecture. This assessment is distinct from the operational segmentation model reported in Section 4.2 and summarized in Figure 4.6: this section concerns end-to-end neural *reconstruction* of three-dimensional geometry, which remains exploratory, whereas AI-based *segmentation* has already been implemented and evaluated as a supporting preprocessing step.

The assessment focused on the capabilities, limitations, and potential integration pathways of neural methods when applied to thin tether-like structures observed under constrained visual conditions.

6.4.1 Evaluation of neural reconstruction methods

The exploratory investigation considered neural reconstruction approaches capable of inferring three-dimensional geometry from a limited number of images. In particular, Neural Radiance Field (NeRF) methodologies and RealFusion-based reconstruction frameworks were reviewed due to their ability to generate volumetric representations from sparse observations.

The assessment confirmed that these methods can infer plausible three-dimensional structures by leveraging geometric priors learned during training. This capability demonstrates the potential of neural approaches to recover information that may be difficult to extract using purely geometric methods.

However, the computational requirements of these techniques remain substantially higher than those of the geometry-driven framework developed in this work. Current implementations typically require dedicated GPU acceleration and are not compatible with real-time execution on embedded platforms representative of the proposed flight architecture.

6.4.2 Observed limitations

The exploratory evaluation identified several limitations that currently restrict the use of generative AI models as primary reconstruction tools for scientific tether observation:

- Preference for visual plausibility over physical fidelity. Reconstructed outputs often appeared visually coherent but were not always consistent with the actual geometry present in the input observations.
- Generation of non-existent features. The models occasionally introduced artificial textures, surface details, or geometric structures that were not present in the original images.
- Limited geometric reliability. The reconstructed representations could not guarantee that the inferred shape accurately corresponded to the true three-dimensional tether configuration.
- Lack of deterministic behavior. Similar inputs may produce different reconstructions, complicating repeatability and quantitative validation.
- Insufficient suitability for scientific measurements. Because reconstruction accuracy is more critical than visual realism for tether monitoring, AI-generated hallucinations represent a significant source of uncertainty.

These observations indicate that fully generative reconstruction approaches are not currently suitable as primary sources of scientific tether geometry without additional physical constraints, geometric validation mechanisms, and mission-specific training datasets.

6.4.3 Potential roles of AI within LINKU

Although AI-based reconstruction was not adopted as the primary reconstruction strategy, the assessment identified several areas where neural methods could strengthen the existing geometry-driven framework.

One promising application is AI-assisted wire path tracking, which subsumes both ordinary path tracing and topological knot / crossing disambiguation as a single capability rather than two separate ones. As discussed in Section 6.3.4, self-intersections and overlapping tether segments may generate multiple valid interpretations of the observed geometry; a tracker that decides which branch to follow at each ambiguous point is solving exactly the same problem whether the thread is mostly straight (few or no ambiguous points) or heavily entangled (many of them). Section 4.2 (Figure 4.9) proposes a concrete design for this component, built by replacing the fixed, hand-tuned decision rule inside the already-implemented tracker with a learned one.

A second opportunity is tether segmentation, where a first implementation already exists: a pixel-level Random Forest classifier, described in Section 4.2, is currently used as a selectable alternative to manual threshold tuning. Lightweight neural architectures, such as U-Net-based models, represent a natural next iteration of this same component and could further improve mask generation under challenging illumination conditions, including specular reflections and high-contrast observations.

These results suggest that the most effective role of AI within LINKU is as a supporting technology for perception and ambiguity resolution, while geometric reconstruction remains the responsibility of the physics-based stereo pipeline.

6.4.4 Controlled comparison of two path-tracing strategies

The claim that crossing disambiguation is the limiting step of geometric curve tracking was tested directly, by implementing a second tracer on a different principle and measuring both against known ground truth.

The tracer used throughout the Phase II experiments walks the segmentation mask one pixel at a time and selects, at each ambiguous point, the continuation favoured by a fixed weighted rule. The alternative converts the skeleton of the same mask into a graph, in which nodes are the points where strands meet and edges are the stretches where the cable runs alone, and then answers a different question at every node: which two of the branches meeting there belong to the same

strand passing through. Once every node has been answered, the cable is read off the graph by a bounded search for the route that joins the two endpoints while covering the greatest length of distinct cable. The two formulations differ in the unit of decision, a single step against a pair of whole branches, and therefore in the evidence available to each.

Evaluation protocol

Both tracers were run on sixty synthetic scenes drawn from a generator whose parameters were previously matched to the statistics of the laboratory masks, using random seeds disjoint from any used during development. Both receive the same mask and the same endpoints, produced by the operational detector, so the comparison isolates stage three of the pipeline. The score is the F1 agreement between the traced path and the known centreline, with a tolerance of two and a half cable radii.

Two properties of the measurement are worth stating because the first version of it was wrong in a way that flattered one method. The agreement score is computed after filling in the pixels between consecutive returned vertices, since a tracer that returns two points a hundred pixels apart has drawn a straight line across whatever lies between them, and scoring only the returned vertices never looks there. Separately, every returned path is audited for the fraction of its pixels that fall outside the mask, which is the direct measure of cable that was invented rather than observed. That quantity matters beyond the score: stage six triangulates the traced path and stage seven converts it into arc length in metres, so an invented segment becomes a measurement that is wrong without looking wrong.

Results

Table 6.1 reports the mean agreement by geometric family. The graph formulation reaches 98.9 against 89.0 over the sixty scenes, winning on twenty, losing on two and tying on thirty-eight, with 0.09 per cent of its pixels falling off the mask against 0.57 per cent.

Table 6.1: Agreement between the traced path and the known centreline, mean F1 per cent over sixty synthetic scenes. Both tracers receive identical masks and identical endpoints.

Geometry	Scenes	Pixel walk	Skeleton graph
Catenary	3	100.0	100.0
Smooth curve	2	100.0	100.0
Straight	1	100.0	100.0
Coil	6	100.0	100.0
Loop	25	86.5	99.3
Knot	14	81.0	99.8
Tangle	9	93.9	95.2
All	60	89.0	98.9

The separation is entirely in the self-intersecting families. On the catenary, smooth and straight scenes, which are the geometries a tensioned tether between two spacecraft actually presents, the two tracers are indistinguishable at 100.0. The advantage of the graph formulation appears only where the cable crosses itself, rising to 18.8 points on knots and 12.7 on loops. This is a quantitative confirmation of the qualitative argument of Section 6.3.4: crossing disambiguation, not curve following, is what limits geometric tracking, and it is the only place where a different formulation changes the outcome.

Figure 6.14 shows three representative scenes. The thick line is the known centreline, so cable left exposed is cable the tracer did not reach.

What the comparison required correcting

Three findings emerged from the comparison that apply to the pipeline independently of which tracer is adopted, and are recorded here because each was a silent failure rather than an error.

The first concerns endpoint detection rather than tracing. On one coil scene the pixel walk scored

37.9, which initially read as a tracing failure. The skeleton in fact contained both correct tips, within one pixel of the true ends, and they were lost by the clustering step that collapses the two or three pixel fork left at a real tip by thinning. That step used a fixed radius of eighty pixels, and the two ends of that particular coil lie fifty-six pixels apart, so they merged into a single candidate; with one candidate there is no pair to evaluate, and the detector fell through to a fallback that returned two arbitrary points. Supplied with correct endpoints, the same tracer scores 100.0 on the same scene. The clustering radius is now derived from the measured cable radius, and clustering is discarded whenever it would leave fewer than two candidates.

The second concerns the model of the traced route. Treating the route as a simple path, in which each skeleton edge is used at most once, makes the correct answer unreachable as soon as the cable crosses itself, because merging a crossing into a single graph node means both strands share the short edges that reach it. Measured against the ground truth, one knot scene traverses three of its edges twice and one tangle scene traverses a seven hundred pixel edge twice. The quantity consumed once is therefore not an edge but a half-edge, the act of leaving a given node along a given edge, which permits an edge to be walked once from each end and never twice the same way.

The third concerns the objective. An intermediate version permitted repeated traversal only of dead-end branches, which recovered hairpins and simultaneously reduced the catenary and smooth families from 100.0 to 89.0, because an objective that maximises distance walked rewards a detour as soon as a repeated edge is counted again. Counting each edge once however often it is walked removes that incentive, so a second traversal is taken only when it reaches cable that is otherwise unreachable.

Scope of these results

These measurements are made on synthetic masks, where the centreline is known by construction, and they should not be read as a reconstruction accuracy figure. Two limits in particular bound their transfer. The generator returns a connected mask in fifty-nine of sixty scenes, whereas the segmentation models produce masks in several disconnected pieces on real frames, so the benchmark cannot exercise behaviour across breaks. And the agreement score measures the path in the image plane; metric accuracy in three dimensions additionally depends on the stereo calibration, which the system's own quality flag currently rejects.

6.5 Comparative discussion

The experimental investigation evaluated three distinct approaches for tether reconstruction under space-emulated observation conditions: conventional photogrammetry, geometry-driven stereo reconstruction, and AI-assisted reconstruction. Each approach exhibited different capabilities, limitations, and operational requirements.

To facilitate comparison, this section walks through both pipelines stage by stage, starting from the same stereo capture, and states exactly where each one succeeds or fails under the experimental conditions investigated in this work, rather than summarizing them as qualitative criterion-by-criterion ratings.

6.5.1 Conventional photogrammetry vs geometry-driven reconstruction

Both pipelines start from the same input: a synchronized stereo image pair. From that point, they diverge immediately.

Conventional photogrammetry proceeds as feature detection (keypoints) → feature matching → triangulation. This pipeline breaks at the very first stage: a tether that occupies only 1–3 pixels does not provide enough distinctive keypoints for reliable detection, so feature matching and triangulation never receive usable input. This limitation persisted even after the tether was perfectly isolated from all background texture, producing the "Zero Resolution" failure reported in Section 6.2.2 (Figure 6.4). Acceptable performance can still be achieved for volumetric objects with abundant texture under favorable imaging conditions, but the approach fails as soon as it is applied to a thin structure against a featureless background.

The geometry-driven pipeline proceeds instead as intensity extraction and segmentation → skeleton and endpoint detection → curve tracking (Figure 4.5) → stereo correspondence via normalized cross-correlation along the epipolar line → triangulation. None of these stages require keypoints: radiometric contrast and geometric continuity are sufficient on their own. This is precisely why the pipeline succeeds where the conventional one fails, and it produced stable reconstructions for both the planar and volumetric tether configurations (Figures 6.8 and 6.12).

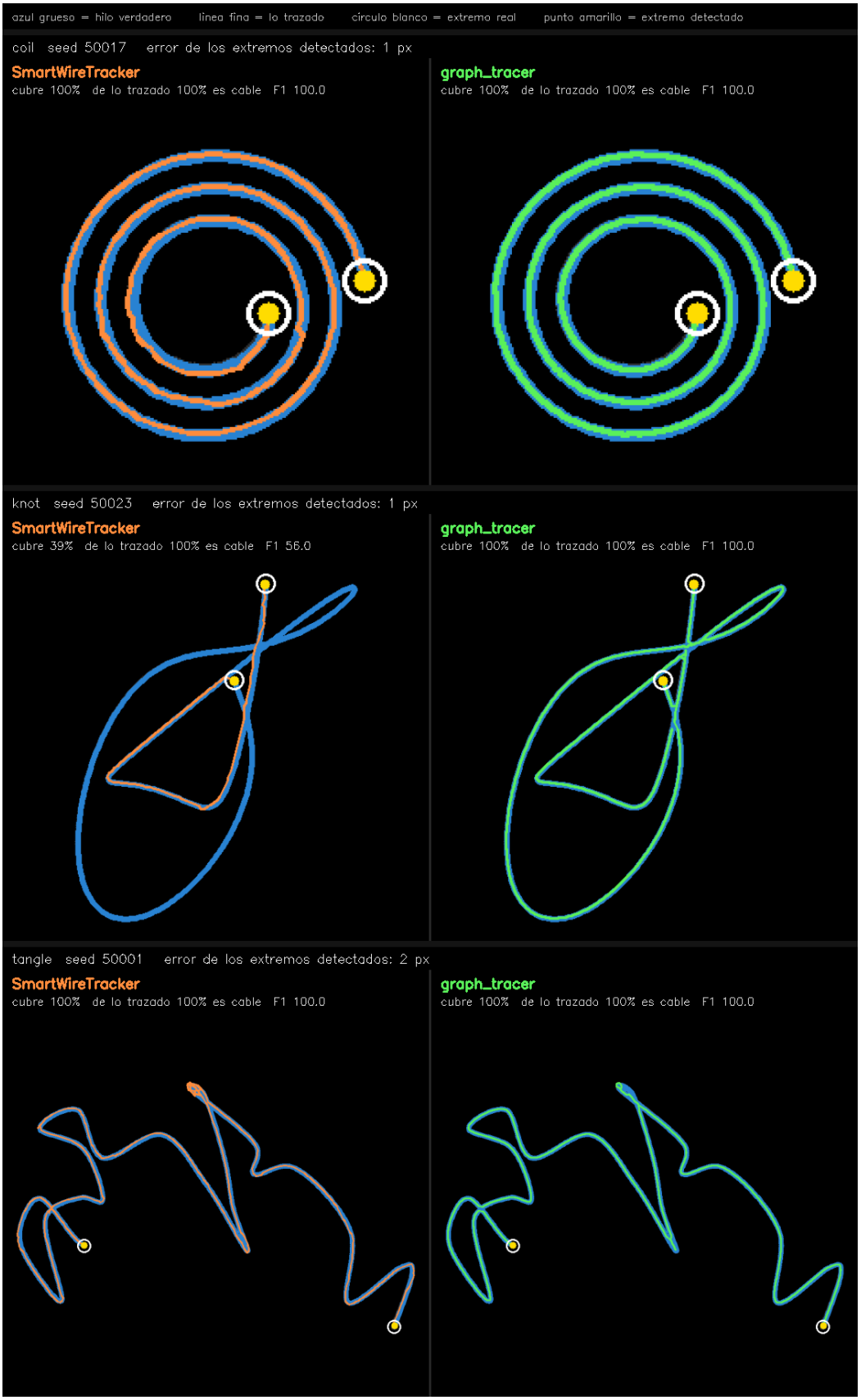


Figure 6.14: The same three masks traced by the pixel walk, left, and by the skeleton graph, right. The thick underlying line is the known centreline, the thin line is the returned path, hollow circles mark the true endpoints and filled circles the detected ones. Both tracers receive identical input. On the coil and the tangle the two agree at 100.0; on the knot the pixel walk reaches 39 per cent of the cable against 100 per cent, and the exposed centreline shows precisely which strands it did not follow.

6.5.2 Geometry-driven vs AI-assisted reconstruction

Unlike the comparison above, geometry-driven reconstruction and AI-assisted wire path tracking are not two competing pipelines evaluated side by side experimentally – only the former has been implemented and tested. The comparison here is therefore architectural: Figure 4.9 (Section 4.2) shows that the proposed AI component does not replace the geometry-driven search; it reuses the exact same step-by-step tracker already implemented and validated in this chapter, and only swaps the fixed, hand-tuned branch-scoring rule for a learned one trained on operator-verified paths.

This has two direct consequences for how the two approaches compare. First, every property that makes the geometry-driven framework attractive for a scientific mission – a deterministic search, physically grounded features (direction, thickness, centeredness), and traceable intermediate results – is preserved in the proposed AI variant, because the search itself is not learned, only the local scoring weights are. Second, the risk categories usually associated with generative AI reconstruction (hallucinated geometry, non-deterministic outputs, lack of physical grounding) apply to the end-to-end neural methods assessed in Section 6.2 (NeRF, RealFusion), not to this proposed component: a mis-scored branch produces a locally wrong path segment that remains subject to the same backtracking and downstream consistency checks as the classical tracker, rather than an unconstrained hallucinated shape.

The geometry-driven framework provides physically interpretable and repeatable reconstructions directly linked to the observed image measurements. This characteristic is particularly important for scientific missions, where reconstruction fidelity and traceability are more critical than visual realism. The exploratory AI assessment revealed promising capabilities in perception-oriented tasks, particularly for segmentation (already implemented, Figure 4.6) and, prospectively, for path tracking through topologically ambiguous configurations. However, the tendency of fully generative models to introduce non-physical structures limits their suitability as primary reconstruction engines. Consequently, AI is currently better positioned as a complementary technology that enhances specific, well-bounded stages of the reconstruction pipeline rather than replacing the underlying geometric framework.

6.5.3 Implications for future tether observation missions

The results obtained throughout this chapter indicate that future tether observation systems should prioritize physics-based geometric reconstruction as the primary source of scientific measurements. Conventional photogrammetric methods were shown to be inadequate for thin, low-texture structures, while the proposed geometry-driven framework successfully recovered tether geometry under both planar and volumetric configurations.

The exploratory assessment of AI-based techniques further suggests that neural methods are most valuable when applied to perception and ambiguity-resolution tasks rather than direct geometric reconstruction. In particular, future developments may benefit from AI-assisted wire path tracking – covering both ordinary tracing and topological knot/crossing resolution as one capability – and from further iterations of the already-implemented segmentation model to improve robustness under challenging illumination conditions.

These findings support a hybrid architecture in which geometric reconstruction provides physically interpretable measurements, while AI modules enhance robustness in specific operational scenarios. Such an approach offers a practical path toward reliable tether monitoring systems for future orbital deployment and inspection missions.

Chapter 7

Future work and system maturation

The results presented in Chapter 6 demonstrated that geometry-driven stereo reconstruction provides a viable solution for three-dimensional tether observation under low-texture conditions where conventional photogrammetric methods fail. The experimental validation established the feasibility of recovering tether geometry through geometric continuity, radiometric localization, and constrained stereo correspondence.

Despite these encouraging results, several opportunities remain for improving reconstruction robustness, increasing operational autonomy, and advancing the system toward a flight-ready implementation. The observations obtained during testing identified specific challenges related to topological ambiguities, temporal consistency, assessment of reconstruction confidence, and onboard deployment constraints.

This chapter outlines a roadmap for the proposed framework's maturation, as shown in Figure 7.1. The discussion focuses on algorithmic enhancements for challenging observation scenarios, the integration of targeted AI-assisted perception tools, the hardware transition toward flight-compatible architectures, and the long-term evolution of the system into an operational tether-monitoring instrument for future space missions.

7.1 Reconstruction robustness enhancements

The geometry-driven reconstruction framework successfully recovered tether geometries under the experimental conditions evaluated in this work. However, several opportunities for improvement were identified during testing, particularly in scenarios involving complex tether configurations, partial occlusions, and ambiguous geometric interpretations, as illustrated in Figure 7.2. The following developments are proposed to increase reconstruction robustness while preserving the physical interpretability of the current methodology.

7.1.1 Topological crossing resolution

One of the principal challenges identified during the experimental evaluation was the presence of topological crossings, where multiple tether segments overlap within the image plane. Although such events did not prevent reconstruction of the overall tether geometry, they introduced localized ambiguities regarding the true connectivity of the observed structure.

Future work should focus on incorporating dedicated crossing-resolution strategies that can distinguish foreground and background tether branches. One possible approach is to integrate disparity-consistency checks, in which local depth information from stereo correspondence is used to determine the most physically plausible continuation path through an intersection. This would allow the reconstruction process to maintain geometric continuity even in highly entangled tether configurations.

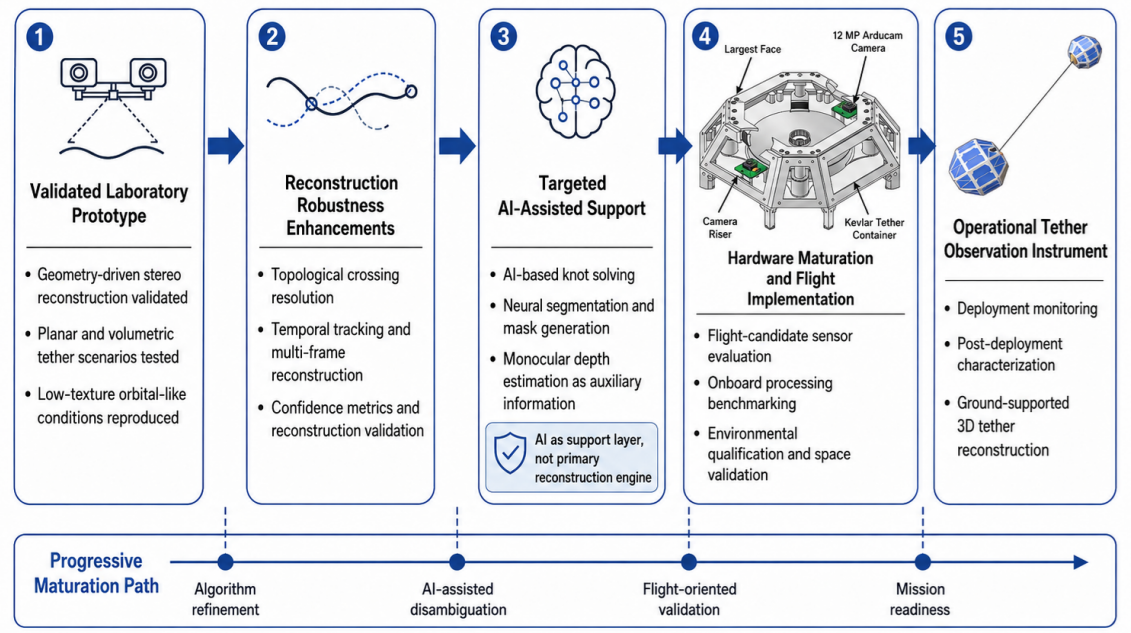


Figure 7.1: Technology maturation roadmap for the LINKU stereo vision system, showing the progression from laboratory validation toward a flight-compatible tether observation instrument.

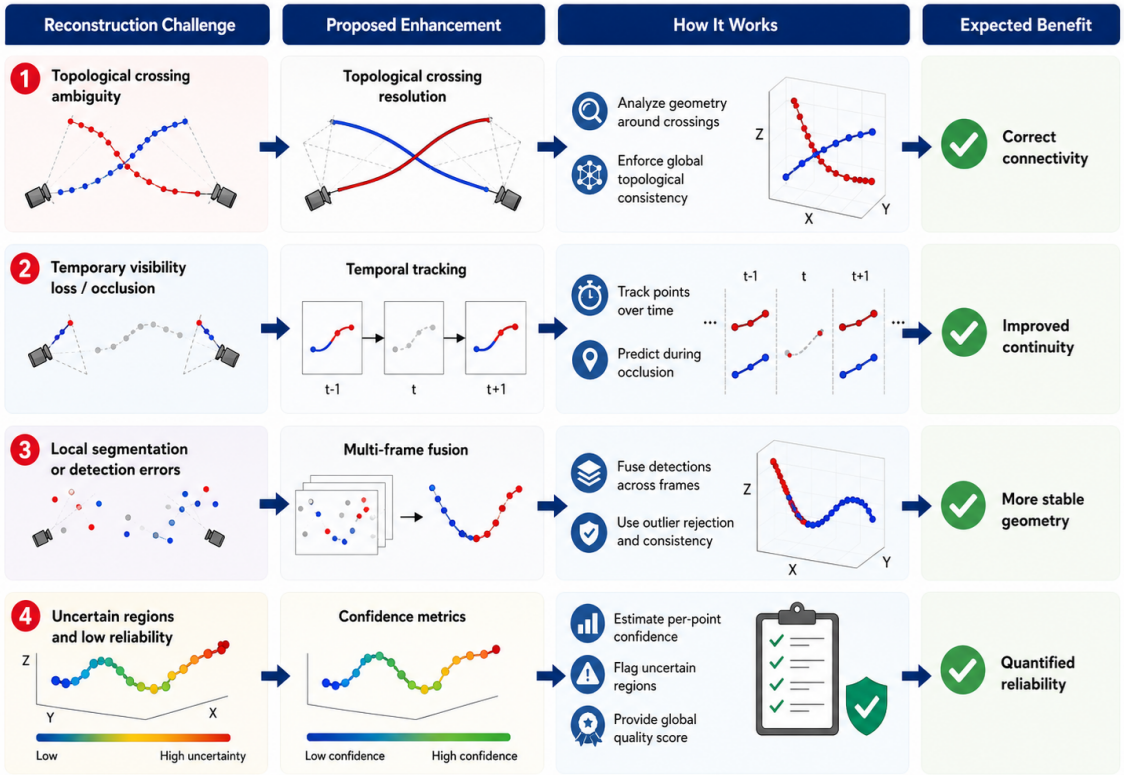


Figure 7.2: Proposed robustness enhancements for the geometry-driven reconstruction framework, including topological crossing resolution, temporal tracking, multi-frame reconstruction, and confidence-based validation.

The development of robust topological reasoning capabilities is expected to improve reconstruction reliability during non-nominal deployment events, including slack tether motion, partial self-intersections, and dynamic oscillatory behavior.

7.1.2 Temporal tracking and multi-frame reconstruction

The current framework processes each stereo pair independently. While this approach simplifies implementation and validation, it does not exploit the temporal continuity naturally present in image sequences.

Future implementations should incorporate temporal tracking mechanisms that propagate geometric information between consecutive frames. By using previously reconstructed tether trajectories as priors, the system could improve correspondence stability, reduce sensitivity to transient image artifacts, and assist in resolving topological ambiguities.

Multi-frame reconstruction techniques may also improve robustness when portions of the tether are temporarily obscured by glare, motion blur, or partial occlusion. The use of temporal consistency constraints would enable the reconstruction process to recover missing information from neighboring observations rather than relying exclusively on a single image pair.

7.1.3 Confidence metrics and reconstruction validation

Future operational implementations will require quantitative indicators to assess reconstruction quality in the absence of external ground-truth references. Such metrics are particularly important for autonomous spaceborne systems, where manual verification of every reconstructed frame is impractical.

A promising approach involves monitoring the consistency of stereo reprojection throughout the reconstruction process. Deviations between observed image locations and reconstructed projections can provide a direct measure of geometric reliability. Additional indicators may include correspondence density, centerline continuity, disparity consistency, and temporal stability across consecutive frames.

The integration of confidence metrics would enable the system to associate each reconstructed tether segment with a quantitative quality estimate. These metrics could be incorporated into telemetry products, providing mission operators with objective indicators of reconstruction reliability and facilitating autonomous decision-making during long-duration tether observation campaigns.

7.2 AI-assisted perception and topology analysis

The exploratory assessment presented in Chapter 6 indicated that current generative AI methods are not suitable as primary reconstruction engines for scientific tether observation. However, the same assessment revealed several perception-oriented tasks in which AI could significantly enhance the robustness of the geometry-driven framework without compromising physical interpretability.

Future AI integration should therefore focus on targeted assistance rather than end-to-end reconstruction. Under this strategy, geometric reconstruction remains responsible for generating scientifically traceable three-dimensional measurements, as specified in Figure 7.3, while AI modules provide support in situations involving ambiguity, image degradation, or complex scene interpretation.

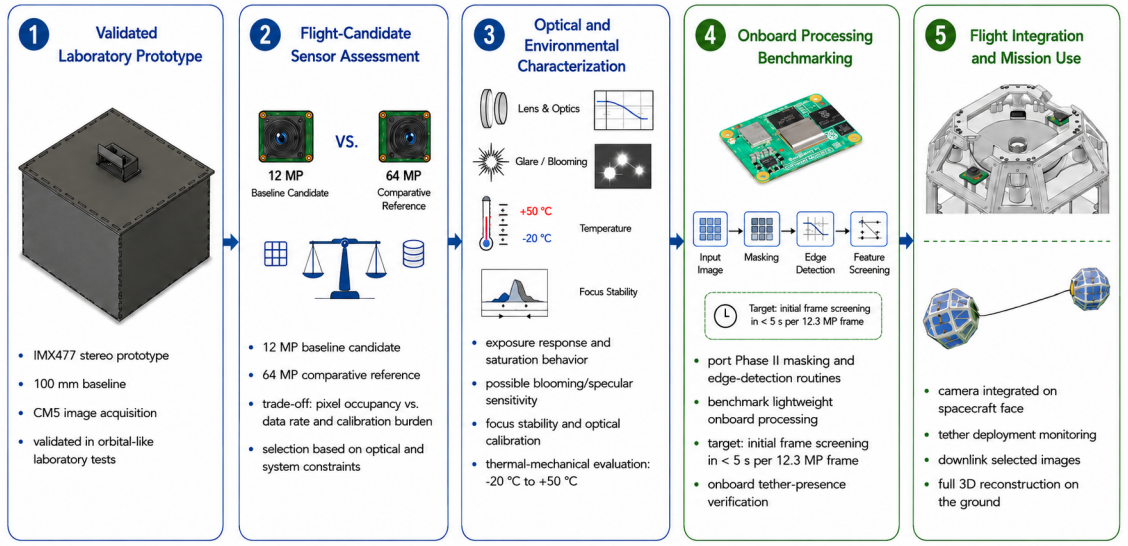


Figure 7.3: Targeted AI assistance within the geometry-driven reconstruction pipeline. AI modules are proposed as support tools for masking, knot solving, and ambiguity resolution, while the final reconstruction remains constrained by stereo geometry.

7.2.1 AI-assisted wire path tracking (including knot solving)

Topological crossings are among the most challenging scenarios for tether reconstruction because multiple physically distinct tether segments may project onto the same image region. Although geometric constraints and stereo correspondence can resolve many of these cases, highly entangled configurations may still produce ambiguous interpretations.

This capability is treated as a single AI component together with ordinary path tracking, not as a separate knot-solving module, because both tasks require exactly the same decision: given a segmented thread image, choose the correct continuation at each point where more than one is geometrically possible. An untangled thread simply has few or no such points, while a knotted one has many; the same trained model is applicable to both. Chapter 4 (Section 4.2, Figure 4.9) proposes a concrete architecture for this component: it reuses the branch-point search already implemented in the classical tracker and replaces only the local scoring rule, which is currently a fixed hand-tuned formula, with one learned from operator-verified paths.

This design offers two advantages over a general-purpose neural reconstruction module. First, it minimizes computational requirements by restricting learned inference to a single scoring decision at each branch point rather than processing the full image. Second, it preserves the interpretability of the overall reconstruction process, since the search structure, backtracking behavior, and downstream stereo matching remain exactly as validated in Chapter 6; only the local weighting of already-physical features (direction continuity, thickness continuity, centeredness) becomes learned instead of hand-tuned.

Reaching this point in practice requires two further steps beyond the current state: growing the path dataset from its present size (1 stereo sample) to the 30–100 samples typically needed for a small classifier of this kind, and building the offline tooling that converts a verified path into labeled (candidate-features, chosen-branch) training examples at each branch point along it.

7.2.2 Neural segmentation and mask generation

Accurate tether extraction remains a critical prerequisite for all subsequent reconstruction stages. A first implementation of this capability already exists: a pixel-level Random Forest classifier (Chapter 4, Section 4.2) trained on 6 manually labeled stereo pairs, selectable as an alternative to manual intensity-based thresholding. While this approach proved effective under the experimental

conditions evaluated in this work, future operational environments may introduce more challenging illumination scenarios.

Examples include severe specular reflections, sensor saturation, non-uniform illumination, partial shadows, and varying background radiometric conditions. Under such circumstances, both the current threshold-based and Random-Forest-based segmentation may become less reliable, and the small training set used so far has not yet been validated on a held-out set.

Future research should therefore evaluate lightweight neural segmentation architectures that can generate robust tether masks under adverse imaging conditions, as a further iteration of the same component. Models derived from U-Net or similar encoder-decoder architectures are promising candidates because they can learn object-specific features while maintaining relatively low computational complexity, and could be trained on the same growing dataset already being collected.

Importantly, the role of these models would be limited to image preprocessing. The resulting binary masks would continue to be processed by the geometry-driven reconstruction pipeline, preserving the physical basis of the three-dimensional reconstruction process.

7.2.3 Monocular depth estimation as auxiliary information

Although stereo vision remains the primary source of depth information in the proposed framework, future systems may benefit from complementary depth estimation techniques that provide additional geometric cues.

Recent advances in foundation models for monocular depth estimation have demonstrated the ability to infer relative depth relationships from single images. While these estimates do not generally provide metric accuracy comparable to stereo triangulation, they can supply useful contextual information regarding foreground-background ordering and local geometric structure.

Future work should investigate whether monocular depth estimation can serve as an auxiliary source of information during reconstruction. Potential applications include resolving depth ambiguities at topological crossings, validating stereo correspondence hypotheses, and improving robustness when stereo matching performance is degraded by glare, motion blur, or partial occlusion.

Under this approach, monocular depth estimation would not replace stereo reconstruction. Instead, it would serve as a secondary information layer that assists the geometry-driven framework when direct geometric measurements become uncertain.

7.3 Hardware maturation and flight implementation

The experimental validation presented in Chapters 5 and 6 demonstrated the feasibility of geometry-driven tether reconstruction using a laboratory-scale prototype based on commercial imaging hardware, as depicted in Figure 7.4. The next stage of development focuses on transitioning from proof-of-concept validation toward a flight-compatible implementation capable of operating under the environmental, computational, and operational constraints of a space mission.

Future work should therefore address three complementary areas: sensor selection and characterization, onboard processing validation, and environmental qualification of the complete imaging system.

7.3.1 Flight-candidate sensor evaluation

The observability analysis presented in Chapter 3 demonstrated that image resolution directly influences tether detectability by increasing the projected tether occupancy in the image plane. However, the analysis also showed that higher pixel counts alone do not eliminate the sub-pixel observation regime encountered at long distances.

Consequently, future sensor selection should not be based exclusively on spatial resolution. Instead, a comprehensive trade-off analysis should consider optical performance, sensitivity, dynamic

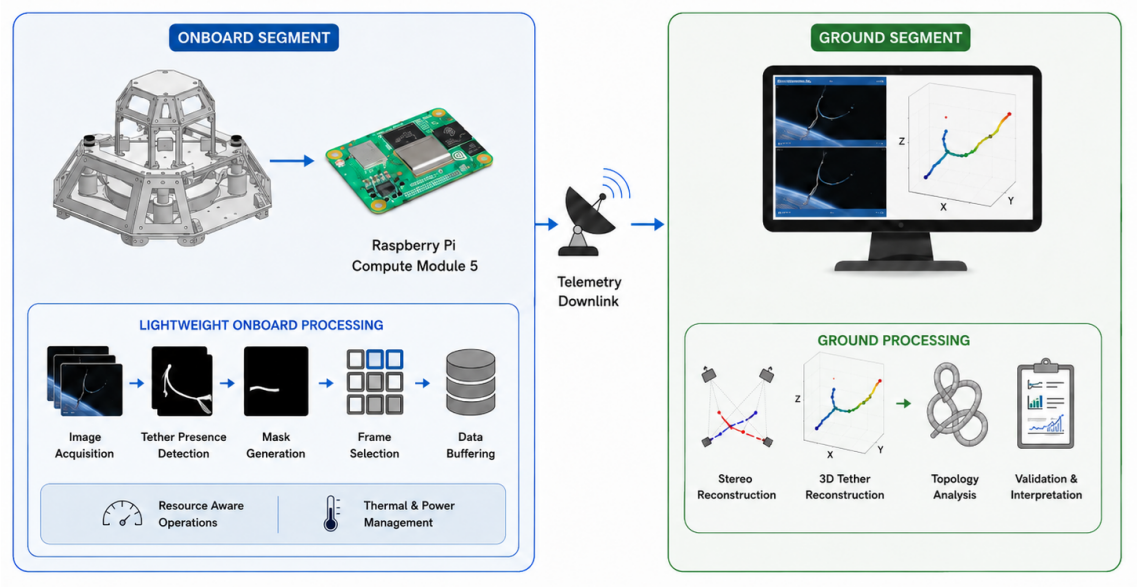


Figure 7.4: Proposed flight implementation architecture, separating lightweight onboard image-processing tasks from full three-dimensional reconstruction and topology analysis performed on the ground segment.

range, data throughput, processing requirements, and environmental robustness.

The IMX477-based architecture remains a suitable baseline for continued development because it supported all experimental validation activities performed in this work and offers interchangeable optics for controlled testing. Alternative high-resolution sensors may provide improved long-range observability but impose additional demands on storage, bandwidth, calibration, and onboard processing.

Future characterization campaigns should evaluate:

- Radiometric sensitivity and signal-to-noise ratio.
- Dynamic range under high-contrast illumination.
- Susceptibility to saturation and blooming effects.
- Optical stability under temperature variations.
- Calibration repeatability and long-term performance.

These measurements will provide the information required to establish a flight-candidate imaging architecture aligned with the observation requirements of future tether missions.

7.3.2 Onboard processing benchmarking

The reconstruction pipeline developed in this work was validated using ground-based processing resources. Future mission implementations will require partial onboard execution to reduce communication bandwidth and support autonomous operation.

A first maturation step consists of porting the preprocessing stages of the geometry-driven framework to a representative flight-computing platform such as the Raspberry Pi Compute Module 5. Particular attention should be given to image acquisition, tether masking, centerline extraction, and preliminary feature validation, as these stages determine whether useful tether information is present within the acquired image.

Benchmarking activities should quantify:

- Processing time per image.

- Memory utilization.
- Processor workload.
- Power consumption.
- Maximum sustainable image acquisition rate.
- Rolling-shutter distortion under spacecraft rotation.

These measurements will establish whether low-level tether detection can be performed onboard before transmitting reduced data products or selected image subsets to the ground segment. Such an approach could significantly reduce downlink requirements while preserving scientifically relevant information.

Longer-term developments may also evaluate the feasibility of performing partial stereo correspondence estimation onboard, reserving full three-dimensional reconstruction for ground-based processing systems.

7.3.3 Environmental qualification and space validation

Although the current prototype successfully demonstrated the proposed reconstruction methodology, its operational performance has not yet been evaluated under representative space environmental conditions.

Future work should therefore include environmental characterization campaigns addressing the principal factors expected during orbital operation. These include thermal variations, vibration exposure, launch-induced mechanical loads, illumination variability, and long-duration operational stability.

Recommended qualification activities include:

- Thermal cycling tests to evaluate focus stability and geometric calibration drift.
- Vibration testing to verify structural integrity of the camera assembly.
- Optical characterization under varying illumination geometries.
- Long-duration operational testing to assess calibration stability over time.
- End-to-end validation using representative deployment sequences and observation distances.

Beyond laboratory qualification, the ultimate maturation objective is the demonstration of tether observation under operationally relevant conditions. Potential intermediate validation platforms include drone-based deployments, high-altitude balloon experiments, and dedicated in-orbit technology demonstrations.

Completion of these activities would increase the Technology Readiness Level (TRL) of the system and provide the experimental evidence required to support integration into future tether deployment, inspection, and monitoring missions.

7.4 Path toward operational tether observation missions

The results presented throughout this work lay the foundation for developing future tether observation systems that support deployment monitoring, structural characterization, and operational diagnostics in space environments. The proposed geometry-driven framework demonstrated that reliable tether reconstruction can be achieved without conventional texture-based photogrammetric techniques, enabling observation of thin, elongated structures that would otherwise be difficult to characterize with traditional computer vision methods.

The next stage of system maturation involves the transition from laboratory validation to operational deployment scenarios. This process requires the progressive integration of the algorithmic, sensing, and processing developments discussed in Sections 7.1 through 7.3. In particular,

Pending: power consumption budget by operational stage (idle, preview, capture, processing). Needs CM5 and camera-module datasheet figures or bench measurement.

Pending: number of images at target resolution obtainable per operational period. Needs onboard storage capacity and/or downlink data rate as inputs.

Pending experiment: validate software-based rolling-shutter correction under controlled rotation rates, to establish the maximum tolerable

improvements in topological ambiguity resolution, temporal tracking, onboard processing, and environmental qualification will increase the robustness and autonomy required for flight operations.

Future mission architectures may employ the proposed framework in several operational roles, including:

- Tether deployment monitoring, providing real-time assessment of deployment geometry, trajectory evolution, and anomaly detection.
- Structural shape reconstruction, enabling characterization of tether curvature, oscillatory motion, and dynamic behavior during mission operations.
- Health and status assessment, supporting the identification of abnormal deployment conditions, entanglements, excessive slack, or unexpected structural deformations.
- Post-deployment analysis that allows reconstruction of tether geometry from stored image sequences for engineering evaluation and mission performance assessment.

Beyond tether-specific applications, the methodology may also be extended to other elongated or low-texture structures encountered in space systems, including deployable booms, flexible appendages, cables, and lightweight structural elements. The underlying principles of geometric continuity, radiometric localization, and constrained stereo reconstruction are not limited to tethered observations and may therefore support a broader class of spaceborne inspection and monitoring tasks.

From a mission perspective, the proposed architecture enables a distributed processing strategy in which computationally efficient detection and validation functions are executed onboard, while high-fidelity three-dimensional reconstruction is performed on the ground segment when required. This approach balances onboard resource constraints with the need for accurate geometric characterization and is consistent with the operational concepts commonly employed in modern small satellite missions.

Ultimately, the maturation pathway described in this chapter provides a practical roadmap for evolving the current proof-of-concept implementation into a flight-capable tether observation instrument. By combining physics-based reconstruction, targeted AI assistance, and flight-oriented hardware validation, the framework offers a scalable approach for future orbital tether missions and contributes to the broader objective of improving autonomous monitoring capabilities for next-generation space systems.

Chapter 8

Conclusions

This work presented the conceptual design, optical characterization, experimental validation, and maturation roadmap of the LINKU stereo vision system for orbital tether observation. The study addressed the challenge of reconstructing the three-dimensional geometry of a thin, low-texture tether under visual conditions representative of Low Earth Orbit, including high radiometric contrast, directional illumination, limited background texture, and sub-pixel target visibility.

The results demonstrate that orbital tether observation cannot be treated as a conventional texture-based photogrammetry problem. Instead, the tether must be reconstructed through a methodology that combines geometric continuity, radiometric localization, centerline extraction, and constrained stereo correspondence. This conclusion is central to the proposed LINKU camera system architecture.

8.1 Research objectives and contributions

The principal objective of this work was to evaluate whether a stereo vision system could provide useful geometric information for monitoring and characterizing a tethered satellite configuration. In particular, the system was intended to support deployment verification, three-dimensional tether reconstruction, oscillation analysis, and complementary interpretation of GNSS, inertial, and tether-tension measurements.

This objective was addressed through four main contributions.

1. The work established an optical observability framework for tether detection under orbital-like conditions. The analysis quantified the trade-off between field of view, pixel density, depth of field, geometric resolution, and radiometric detectability. It also showed that, at long observation distances, a millimeter-scale tether enters the sub-pixel regime, requiring radiometric detection rather than purely geometric edge measurement.

2. The study demonstrated the limitations of conventional photogrammetry for thin tether reconstruction. Standard feature-based methods have been shown to depend heavily on background texture and visual keypoints, making them unsuitable for isolated, low-texture tethered observation.

3. The work developed and validated a geometry-driven stereo reconstruction methodology. Unlike conventional approaches, the proposed framework treats the tether as a continuous geometric structure and reconstructs its spatial trajectory by extracting the centerline, establishing stereo correspondence, and performing triangulation.

4. The study clarified the role of AI within the LINKU architecture. Rather than adopting AI as the primary reconstruction engine, the results support its use as a targeted assistance layer for two components: cable segmentation, already implemented and trained on laboratory data, and AI-assisted wire path tracking, which unifies ordinary path tracing and topological

knot/crossing resolution into a single proposed component and is currently at the dataset-collection stage.

8.2 Principal findings

The principal findings of this work are summarized as follows:

- Conventional photogrammetry is not suitable as the primary reconstruction strategy for orbital tether observation. Its dependence on texture, stable visual features, and dense correspondence makes it unreliable for thin, low-texture targets observed against dark backgrounds.
- The tether observation problem contains two distinct detection regimes. In the near field, where the tether occupies at least one pixel, geometric edge and centerline extraction remain viable. In the far field, where the tether becomes sub-pixel, radiometric localization becomes the dominant detection mechanism.
- Increasing sensor resolution improves long-range pixel occupancy but does not eliminate the sub-pixel regime at mission-relevant distances. Therefore, optical design must be combined with radiometric and geometry-driven processing.
- The proposed geometry-driven reconstruction framework successfully recovered physically plausible tether geometries under both planar and volumetric configurations. This validates the use of constrained stereo geometry for thin tether reconstruction under low-texture conditions.
- Topological crossings and self-intersections remain challenging localized cases. These ambiguities do not invalidate the reconstruction framework, but they require additional temporal tracking or topology-aware support for fully autonomous operation.
- End-to-end generative AI reconstruction is not currently appropriate as the primary source of scientific tether geometry because of computational cost, limited physical traceability, and the risk of hallucinated geometry.
- Targeted AI assistance remains promising for two well-bounded tasks: robust mask generation (already implemented) and wire path tracking under both simple and topologically ambiguous configurations (proposed, not yet implemented), provided that the final geometric reconstruction remains constrained by physically interpretable stereo methods.

8.3 Validation of the proposed framework

The proposed framework was validated through a progressive experimental strategy. The first stage established the failure modes of conventional photogrammetry. The second stage demonstrated the feasibility of geometry-driven reconstruction using controlled stereo datasets. The third stage assessed AI-based methods as exploratory support tools rather than operational reconstruction methods.

The experimental results confirm that the LINKU stereo vision system can extract and reconstruct tether-like geometries under controlled space-emulated visual conditions. The use of a dark enclosure, directional illumination, stereo acquisition, and a 1 mm tether proxy provided a representative laboratory environment for evaluating the principal optical constraints expected in orbit.

The validation does not imply that the current prototype is flight-ready. However, it demonstrates that the core observation principle is technically feasible. The system can detect and reconstruct thin tether geometries when the target is visible, sufficiently contrasted, and not severely affected by saturation, glare, or unresolved topological ambiguity.

The resulting architecture is therefore suitable as a proof-of-concept baseline for future maturation. Lightweight onboard processing can be used for image selection, tether presence verification, and

data prioritization, while full three-dimensional reconstruction and advanced topology analysis can be performed on the ground segment.

8.4 Scientific and engineering contributions

This work contributes to the development of optical monitoring systems for tethered satellite missions in three main ways.

From a scientific perspective, it demonstrates that tether observation in orbit should be modeled as a geometric and radiometric detection problem rather than as a conventional photogrammetric reconstruction problem. This distinction is essential for thin, low-texture structures whose apparent image footprint may fall below one pixel at long range.

From a methodological perspective, the work proposes a reconstruction strategy that replaces texture matching with centerline extraction, geometric continuity, epipolar constraints, and stereo triangulation. This provides a physically interpretable alternative for reconstructing slender structures under sparse visual conditions.

From an engineering perspective, the work defines a practical development path to transform a laboratory prototype into a flight-compatible observation instrument. The proposed roadmap includes robustness improvements, topology-aware processing, targeted AI support, onboard processing benchmarks, and environmental qualification of candidate hardware.

Together, these contributions provide a foundation for future tether monitoring systems capable of supporting deployment verification, anomaly identification, shape reconstruction, and post-deployment dynamic analysis.

8.5 Final remarks

The LINKU stereo vision system has progressed from an optical observation concept to an experimentally validated reconstruction approach for thin tether-like structures. The work showed that conventional photogrammetry is insufficient for this application, while geometry-driven stereo reconstruction provides a more appropriate and scientifically interpretable solution.

The proposed framework does not eliminate all challenges associated with orbital tether observation. Specular saturation, intermittent visibility, self-intersections, and flight hardware constraints remain important areas for future development. Nevertheless, the results establish a clear technical basis for continued maturation.

The most important conclusion is that reliable tether reconstruction is achievable when the problem is approached by leveraging the target's physical characteristics: thin geometry, radiometric contrast, geometric continuity, and stereo constraints. This principle defines the core contribution of the LINKU camera system and provides a practical path toward autonomous visual monitoring of tethered satellite systems in future orbital missions.

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